

**OVERVIEW OF
THE SKELETAL SYSTEM
OSTEOLOGY**

**Department of Anatomy
School of Medical Sciences
Access course 2026**

Graduate Entry Medicine Programme



- **Bone, or osseous tissue,** is a hard, dense connective tissue that forms most of the adult skeleton.

 - The skeletal system includes
 - all of the bones
 - Cartilages
 - ligaments
- support and give shape to
the body and body
structures.
-
- The adult skeleton has 206 bones

- 
- Bone tissue differs greatly from other tissues in the body.
 - Bone is hard and many of its functions depend on that characteristic hardness.
 - Bone is also dynamic in that its shape adjusts to accommodate stresses



Functions of Skeletal System

- **SUPPORT:** Hard framework that supports and anchors the soft organs of the body.
- **PROTECTION:** Surrounds organs such as the brain and spinal cord.
- **MOVEMENT:** Allows for muscle attachment therefore the bones are used as levers.
- **STORAGE:** Minerals (calcium) and lipids are stored within bone material.



Functions of Skeletal System

- **BLOOD CELL FORMATION:** The bone marrow is responsible for blood cell production.
- **TRIGLYCERIDE (FAT) STORAGE:** It is stored in bone cavities.
- **HORMONE PRODUCTION:** Bones produce osteocalcin which regulate bone formation.

Classification of bones based on shape

- The adult skeleton (206 bones)
 - divided into five categories based on shape.
 - Long bones
 - Short bones
 - Flat bones
 - Sesamoid bones
 - Irregular bones
- Each categorical shape of bone has a distinct function.

- **Long bones**

- Typically longer than wide (tubular)
- Have a shaft with heads at both ends
- Contain mostly compact bone
- Long bones function as levers; they move when muscles contract.
 - Examples: Femur, humerus, radius, ulna, Tibia, Fibula

- **Short bones**

- Generally cube-shape
- Contain mostly spongy bone
- Short bones provide stability and support as well as some limited motion.
 - Examples: Carpals, tarsals

- **Flat bones**

- Thin and flattened, usually curved
- Thin layers of compact bone around a layer of spongy bone
- Flat bones serve as points of attachment for muscles and often protect internal organs.
 - Examples: Skull, ribs, sternum

- **Irregular bones**

- Irregular in shape
- Do not fit into other bone classification categories
 - Example: Vertebrae and hip bone

- **Sesamoid bone**

- a small, round bone that, as the name suggests, is shaped like a sesame seed.
- form in tendons where a great deal of pressure is generated in a joint.
- protect tendons by helping them overcome compressive forces.
 - Example: patella



Structure of a Typical Long Bone

- **Epiphysis**—ends of a bone, contain red marrow, mostly spongy (or cancellous, trabecular) bone.
- **Diaphysis**—“shaft” of a bone, made of *compact bone*; has a medullary cavity containing yellow marrow (storage site for triglycerides)
- **Metaphysis**—the growing area of a long bone where diaphysis and epiphysis meet. Has the cartilaginous epiphyseal (growth) plate.
- **Articular cartilage**—a thin layer of hyaline cartilage covering each epiphysis

Commonest site of bone marrow aspiration – manubrium sterni in adults, iliac crest in children



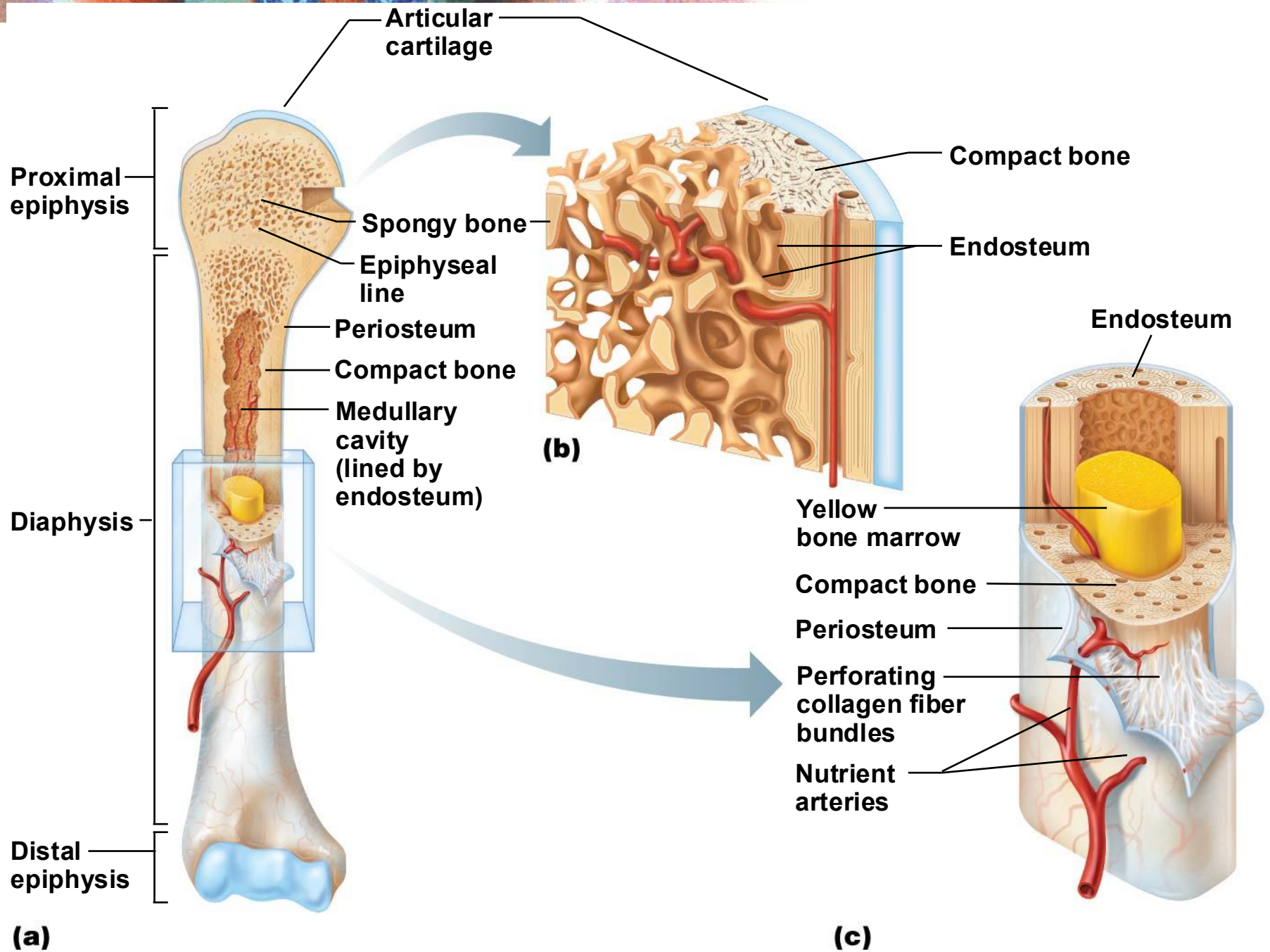
Structure of a Typical Long Bone

■ **Periosteum**

- connective tissue **covering outer surfaces of bones**
- Are osteogenic. Important in the repair of fractures, hence bone stripped of its periosteum will not survive
- Has numerous sensory nerve fibers so is very sensitive to pain

■ **Endosteum**

- Single layered and **lines inner bone surfaces;**
- Osteogenic and contributes to new bone formation.





Bone (Osseous tissue)

- **Highly vascular**
- **Consists of**
 - **Organic components (about 35% of the dry weight)**
 - cells, collagen fibers (95% mainly type I), and ground substance
 - **Inorganic (calcified) components (about 65% of the dry weight)**
 - composed of calcium, phosphate, bicarbonate, citrate, magnesium, potassium, and sodium; primarily of hydroxyapatite crystals, $\text{Ca}_{10}(\text{PO}_4)_6(\text{OH})_2$

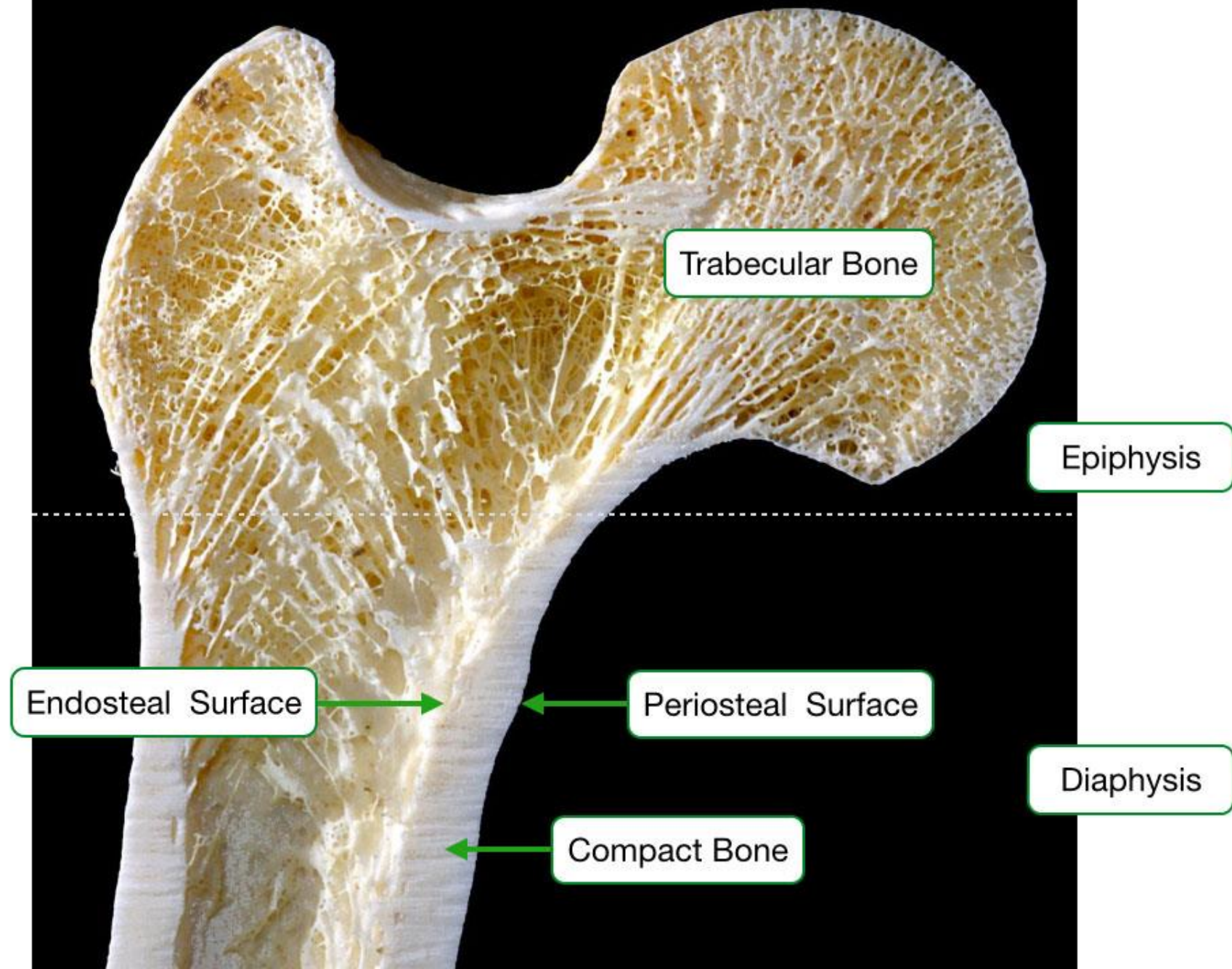


Bone Cells

- **Osteoprogenitor (osteogenic) cells**—located in the periosteum and the endosteum; persist throughout life as stem cells; can be activated later for bone repair of fractures or other repair.
- **Osteoblasts**— immature bone cells that secrete bone matrix.
- **Osteocytes**— mature bone cells housed in a *lacunae*; maintain bone matrix; cytoplasmic extensions pass through canaliculi in lamellae to connect to other osteocytes
- **Osteoclasts**—large, motile, multinucleated cells that resorb or break down bone

Types of bones

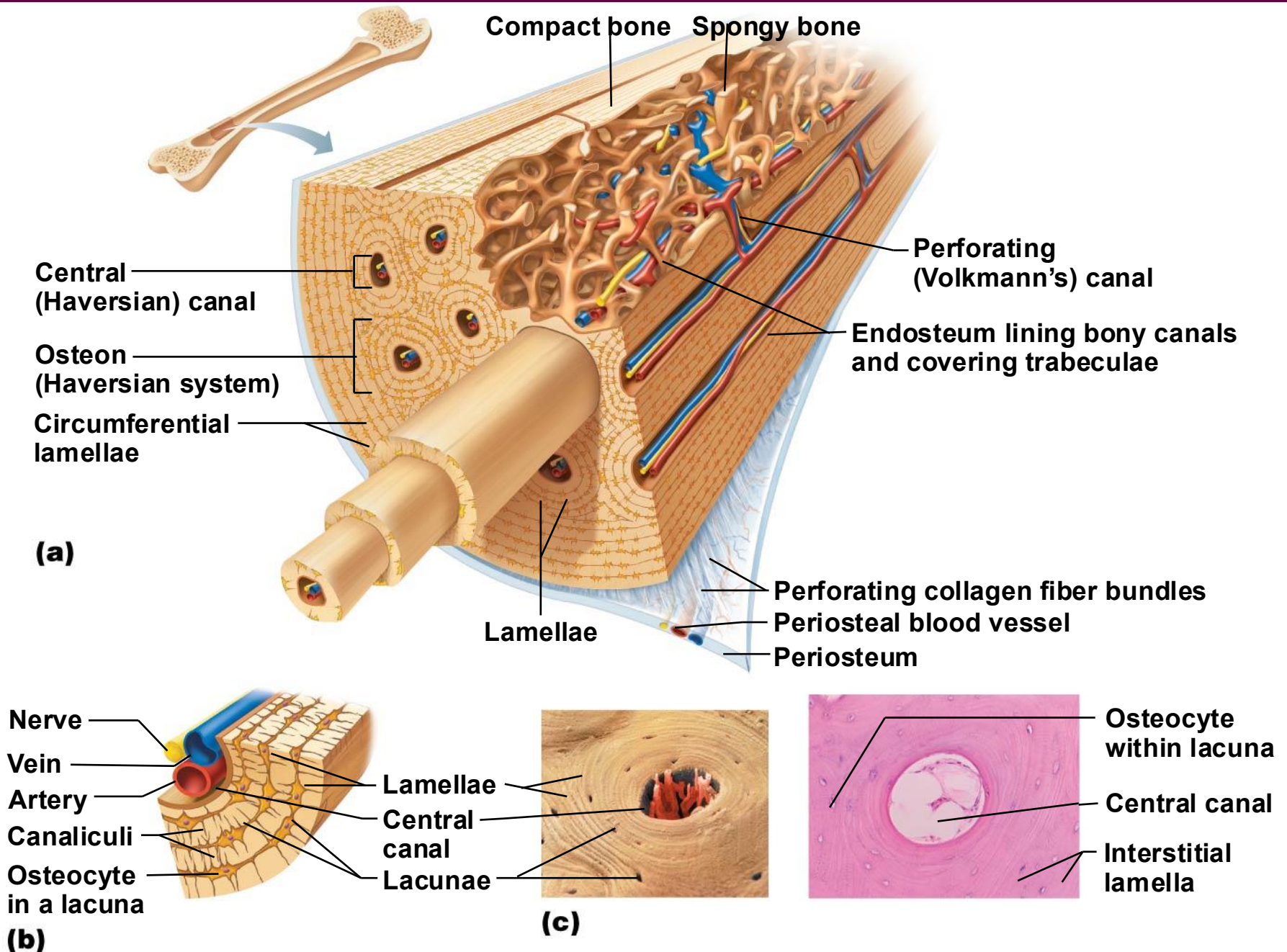
- **Compact (dense) bone** - 80% of bone
 - Forms the outside of the bone. Made of lamellar (mature) bone.
 - Consists of **osteons (haversian systems)**
 - basic structural units of lamellar/mature bone
- **Spongy (cancellous or trabeculae) bone**- 20% of bone
 - forms the inside of the bone
 - lacks osteons.
 - consists of columns of bone called **trabeculae** (little beams), which contain lamellae, osteocytes, lacunae, and canaliculi. Spaces between trabeculae are filled with red bone marrow.



Osteons (haversian systems)

- Each **osteon** is composed of calcified matrix arranged in concentric **lamella** around a **central (Haversian or osteonal) canal**. The canal contains blood vessels, nerves and loose connective tissue.
- **Lacunae**- small spaces between lamellae that contain mature bone cells called **osteocytes**.
- Projecting from the lacunae are **canaliculi**, networks of minute canals containing the processes of osteocytes. Canaliculi provide routes for nutrients to reach osteocytes and for wastes to leave them
- Perforating (Volkmann's) canals lie at right angles to the central canals and connect the blood vessels and nerves from the periosteal and endosteal surfaces to the central canals and the marrow cavity

Microscopic structure of compact bone.

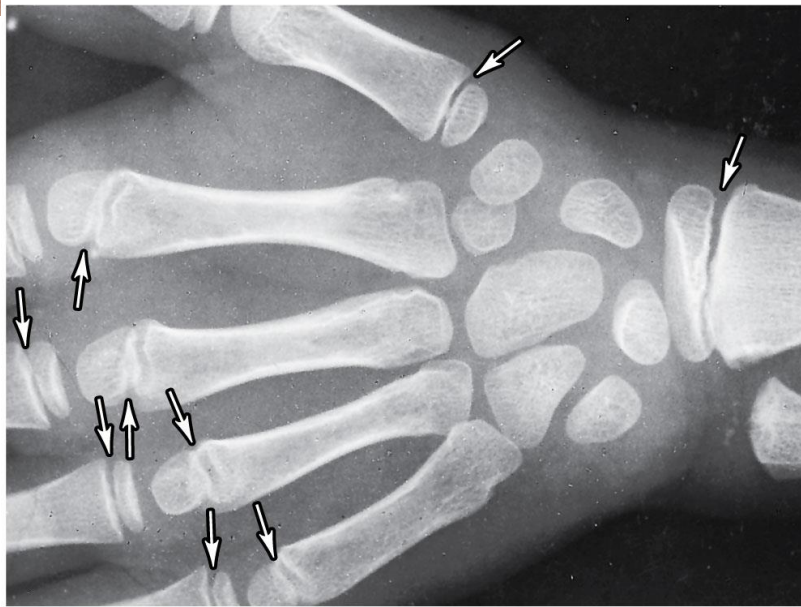


Osteogenesis (or ossification) - bone formation

- Begins during the 6-8th week of embryonic development.
- All bones are derived from mesenchyme by 2 different processes:
 - **Intramembranous ossification**
 - direct formation of bone from mesenchyme (embryonic connective tissue).
 - forms most flat bones of skull, mandible, sternum and clavicle.
 - **Endochondral ossification**
 - formation of bone from (hyaline) cartilage model
- Stages of bone development
 - Formation of unmineralised matrix (**osteoid**)
 - Mineralisation of matrix (calcification).

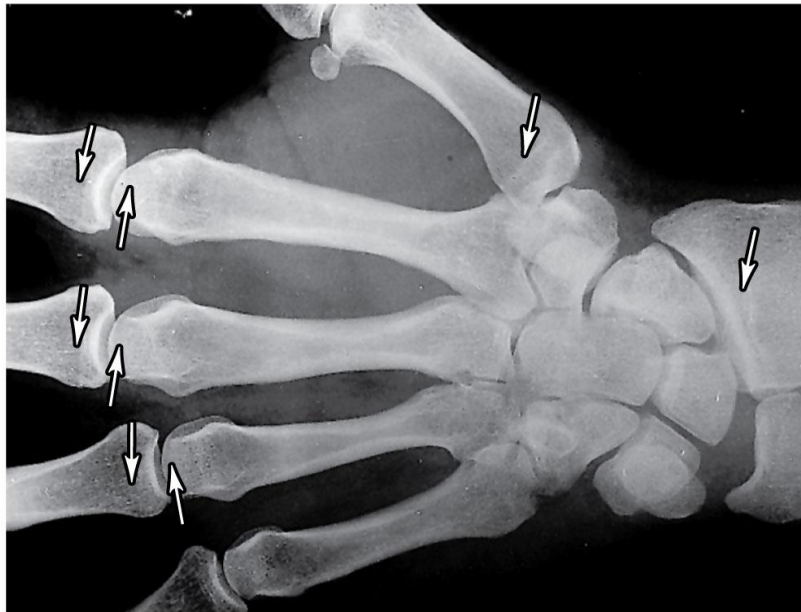
Endochondral ossification

- The site where bone first forms is the **primary centre of ossification** (mesenchymal tissue, osteoblasts, blood-forming cells, and blood vessels).
 - Appears 7-12th week in the middle of the diaphysis.
- **Secondary ossification centres appear in the epiphyses** much later, usually after birth.
- **The thickness of the epiphyseal plate does not change during childhood. As the long bone grows, bone is added to the diaphyseal end of the epiphyseal plate.**
- Damage to the epiphyseal plate, following fracture of a long bone in child may lead to shortening of the limb.
- Estrogen influences early closure of the epiphyseal plate of bones (about 18 years in females and 21 years in males)



An x-ray showing **epiphyseal plate** in a child (arrows)

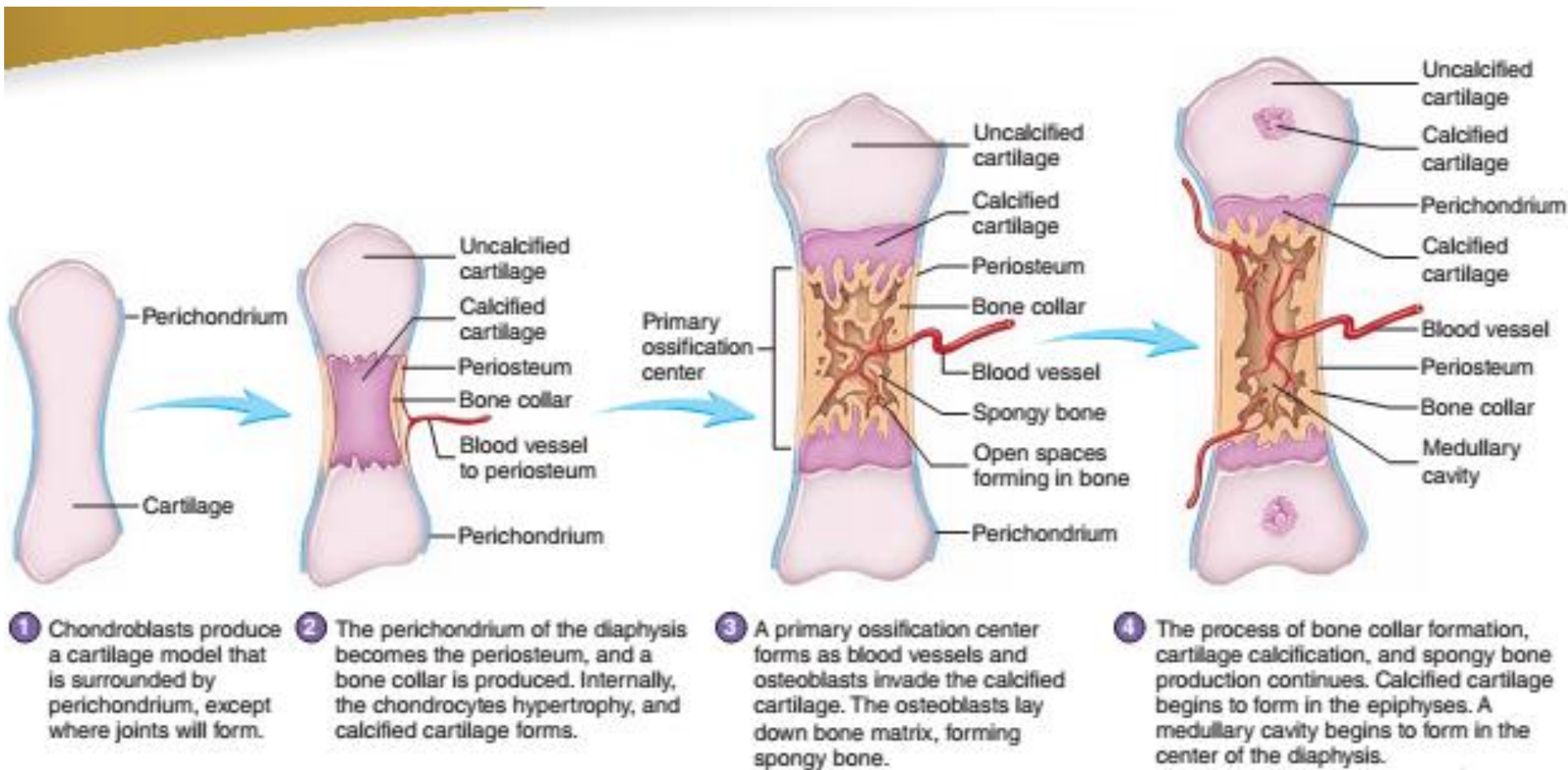
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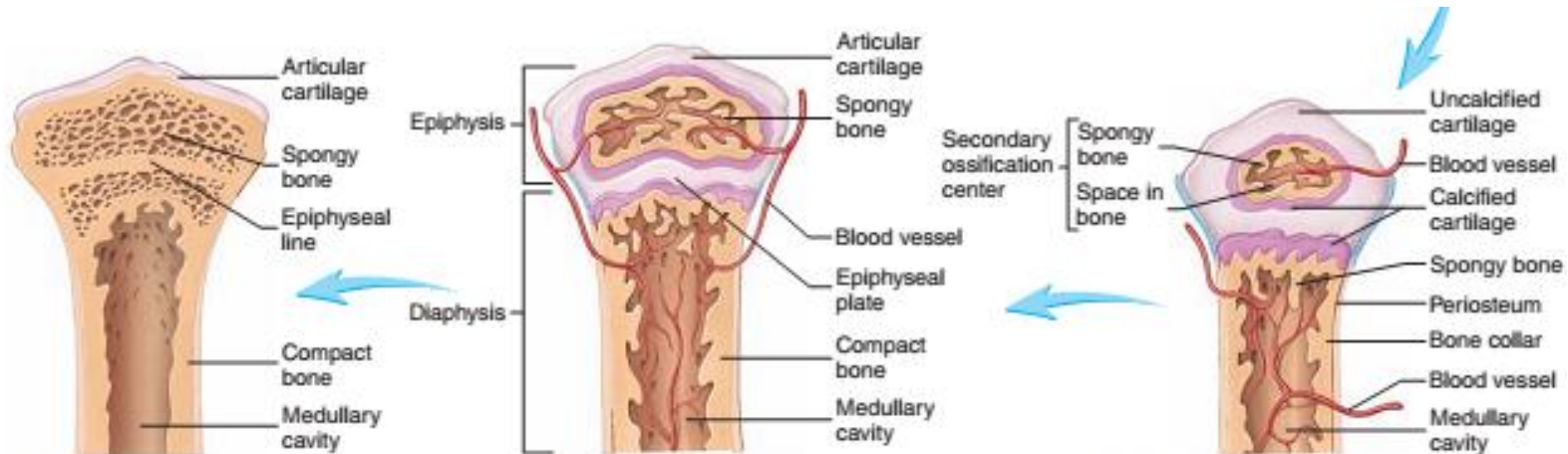
An x-ray showing **epiphyseal lines** in an adult (arrows)

b

Endochondral Ossification



Bone Development

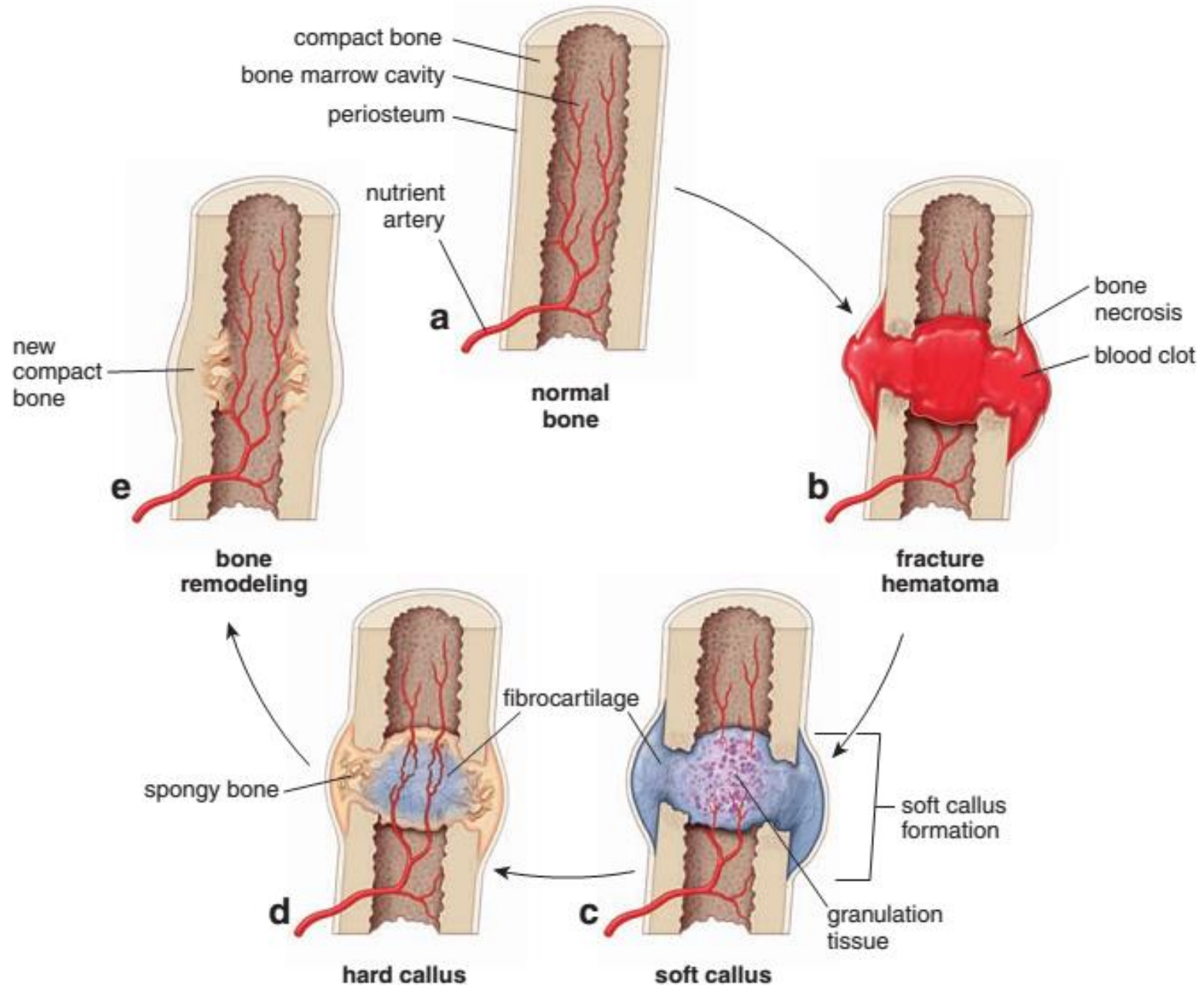


7 In a mature bone, the epiphyseal plate has become the epiphyseal line, and all the cartilage in the epiphysis, except the articular cartilage, has become bone.

6 The original cartilage model is almost completely ossified. Unossified cartilage becomes the epiphyseal plate and the articular cartilage.

5 Secondary ossification centers form in the epiphyses of long bones.

Bone fracture and stages of bone healing process





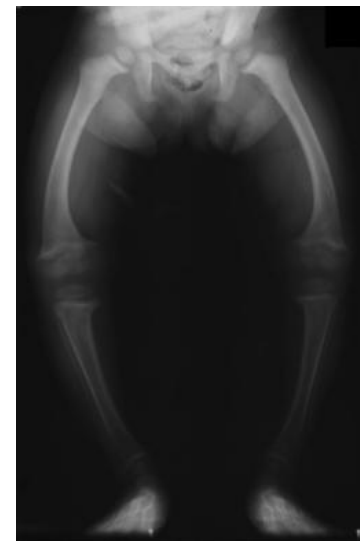
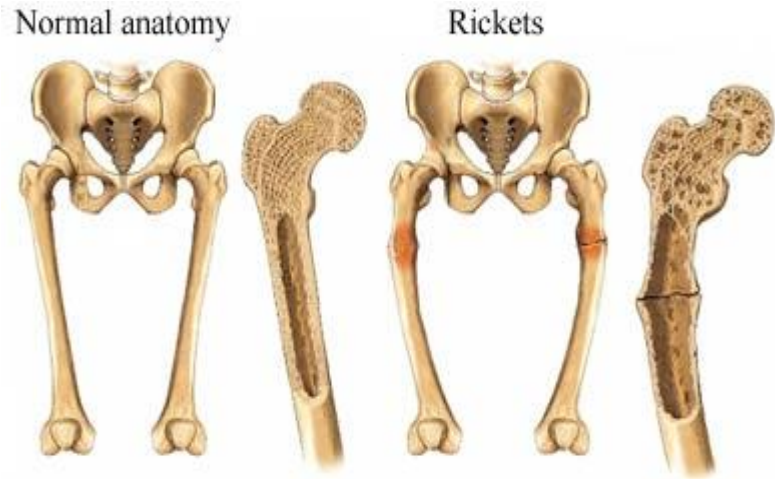
Role of hormones in bone formation

- **Parathyroid hormone** (produced by the parathyroid gland) increases blood Ca^{2+} level by stimulating osteoclast activity
- **Calcitonin** (produced by the thyroid gland) decreases blood Ca^{2+} level by inhibiting osteoclast activity.
- Estrogen decline in postmenopausal women increases their risk of bone loss. Estrogen replacement therapy is associated with increased risk of heart attack, stroke, and breast cancer.

Weight-bearing activities stimulate osteoblasts and, consequently, help build thicker, stronger bones and retard loss of bone mass that occurs as people age.

Vitamin D Deficiencies

- Osteomalacia and rickets result from inadequate mineralization of bone.
- **Rickets:** insufficient amounts of dietary calcium or Vitamin D in children causing bowing of the skeleton. In adults the condition is called **Osteomalacia**.



Osteoporosis

- Abnormal loss of bony tissue resulting in fragile porous bones attributable to a lack of calcium; most common in postmenopausal women
- Bone cells become less active due to estrogen loss, bone resorption outpaces bone formation.
- Increased susceptibility to fractures of the hip and wrist, vertebral compression, loss of body height and development of kyphosis



Normal bone



Close-up view



Bone with osteoporosis



Close-up view



- **Osteopenia** is a decreased calcification of bone or a reduced bone mass due to an inadequate osteoid synthesis. Women lose 8% bone mass per decade, men-3%
- **Albers-Schonberg disease (Osteopetrosis):** An inherited disorder characterized by an *increase in bone density*; in severe forms the bone marrow cavity may be obliterated. Occurs due to defective resorption of immature bone.
- **Osteophyte (Bony spur):** Small abnormal bony outgrowth.
- **Osteomyelitis:** inflammation of the bone caused by pus forming bacteria.
- **Osteitis:** inflammation of bone tissue.



Pituitary growth failure

- Inadequate growth hormone production in childhood
- Reduced epiphyseal cartilage activity; abnormally short bones leading to shorter stature
- Children treated with synthetic human growth hormone



Gigantism and Acromegaly

Gigantism

Overproduction of *growth hormone* before puberty, often due to pituitary tumor – leads to excessive growth at the epiphyseal plates

Produces height over 2.7 m (8 ft. 11 in.)

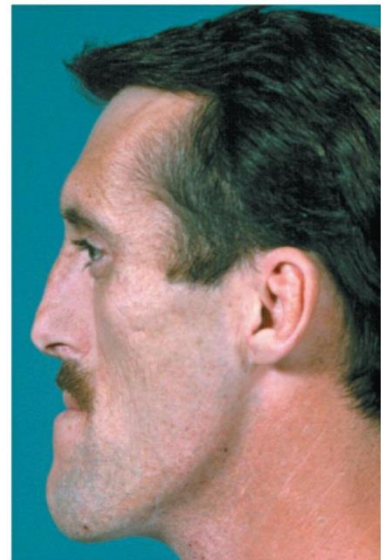
Puberty often delayed



Acromegaly

Overproduction of *growth hormone* after epiphyseal cartilages close

Bones get thicker, not longer, especially affects face, jaw, and hands



Achondroplasia

- Common form of dwarfism
- Defective endochondral ossification
- Dominant disorder: Mutation of FGFR3 gene which inhibits cartilage growth at epiphyseal plate
- Epiphyseal cartilage of long bones replaced by bone early in life
- Results in short limbs
- Trunk normal size



Marfan's syndrome

- Inherited metabolic condition that results in abnormal connective tissue structure
- Excessive *cartilage* formation at epiphyseal plate
- Results in very tall person with long, slender limbs and cardiovascular problems.



Skeletal System

Learning Outcomes

After you have completed this topic, you should be able to:

- Identify the two major subdivisions of the skeleton and list the bones found in each area.
- List at least three functions of the skeletal system.
- Name the five main classifications of bones.
- Describe how the skull of a newborn infant (or fetus) differs from that of an adult, and explain the function of fontanelles
- Discuss the importance of the intervertebral discs and spinal curvatures.
- Describe important differences between a male and a female pelvis
- Describe the two systems for classifying joints.

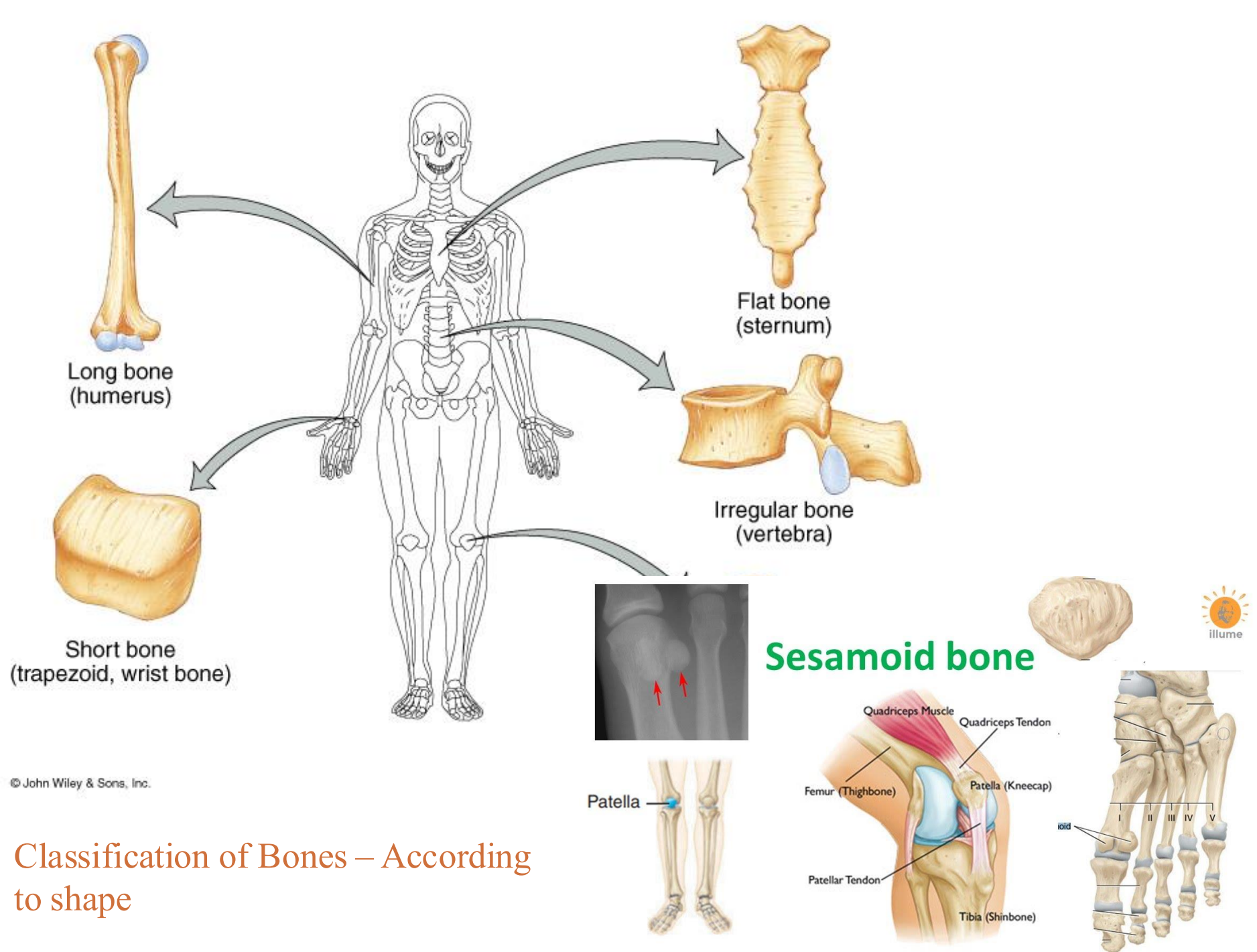
Skeletal System

- Explain the structure of a fibrous joint, list the three types, and give an example of each type.
- Contrast the two types of cartilaginous joints and give examples of each type.
- Illustrate the structure of a synovial joint and explain the roles of the components of a synovial joint.
- Classify synovial joints based on the shape of the bones in the joint and give an example of each type.
- Distinguish among uniaxial, biaxial, and multiaxial synovial joints
- Describe the major joints of the body



Skeletal System

- The skeleton is composed of cartilages and bones
- Bone is a specialized connective tissue characterized by a mineralized extracellular matrix
- Bones are the organs of the skeletal system; bone tissue is the structural component of bones.
- **Functions of bones**
 - ✓ Support and shape
 - ✓ Mechanical basis for movement
 - ✓ Protection for vital structures
 - ✓ Mineral storage – calcium and phosphate
 - ✓ Blood cell formation (Haemopoiesis)
 - ✓ Energy metabolism



Classification of Bones – According to shape

Bone Markings

Table 6.1

Bone Markings

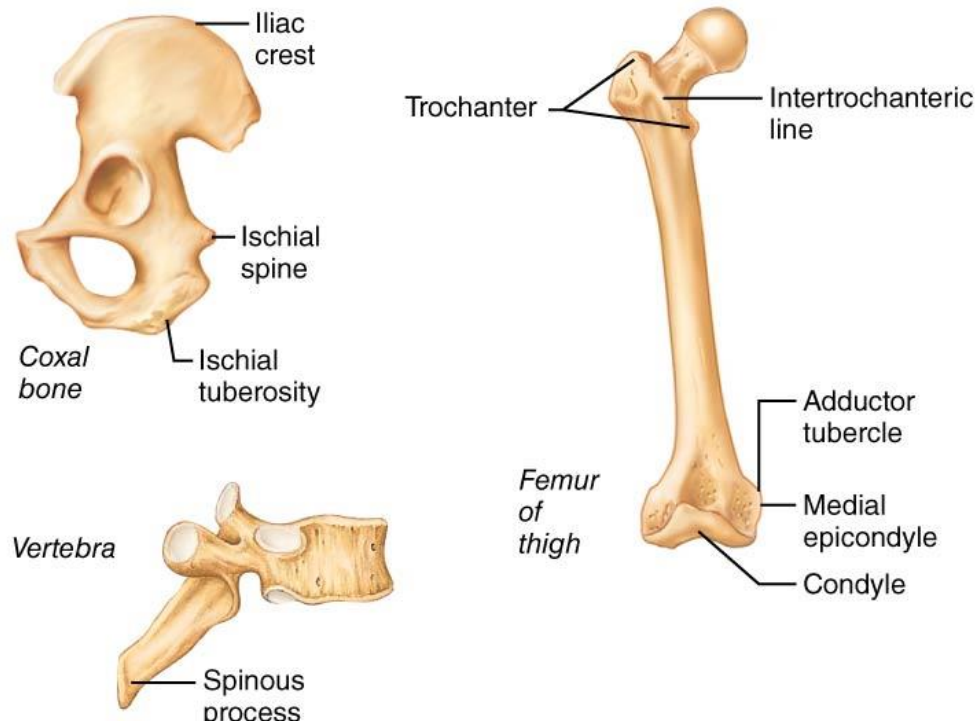
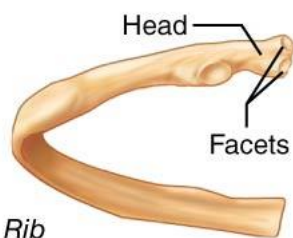
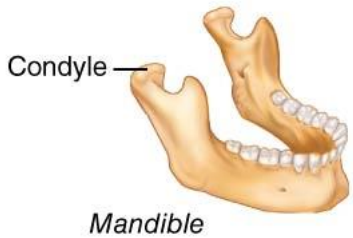
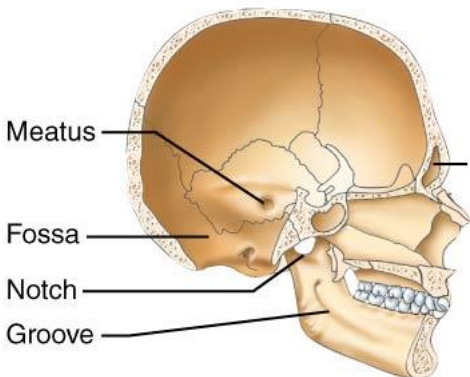
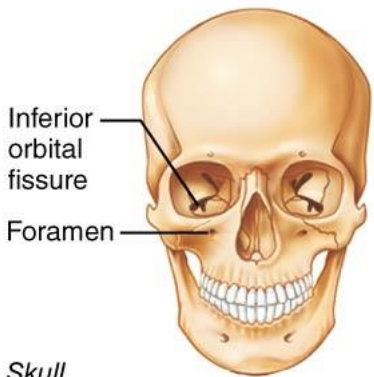
Name of Bone Marking	Description	Illustration
PROJECTIONS THAT ARE SITES OF MUSCLE AND LIGAMENT ATTACHMENT		
Tuberosity (too"be-ros'ŷ-te)	Large rounded projection; may be roughened	 The illustration block contains three anatomical drawings. The top drawing shows a lateral view of the coxal bone (hip bone) with labels for the Iliac crest, Ischial spine, and Ischial tuberosity. The middle drawing shows a vertebra with a label for the Spinous process. The right drawing shows a femur (thigh bone) with labels for the Trochanter, Intertrochanteric line, Adductor tubercle, Medial epicondyle, and Condyle. The label 'Femur of thigh' is placed next to the bone.
Crest	Narrow ridge of bone; usually prominent	
Trochanter (tro-kan'ter)	Very large, blunt, irregularly shaped process (the only examples are on the femur)	
Line	Narrow ridge of bone; less prominent than a crest	
Tubercle (too'ber-kl)	Small rounded projection or process	
Epicondyle (ep"ŷ-kon'dĭl)	Raised area on or above a condyle	
Spine	Sharp, slender, often pointed projection	
Process	Any bony prominence	

Table 6.1 Bone Markings (continued)

Name of Bone Marking	Description	Illustration	
SURFACES THAT FORM JOINTS			
Head	Bony expansion carried on a narrow neck	 <i>Rib</i>	 <i>Mandible</i>
Facet	Smooth, nearly flat articular surface		
Condyle (kon'dīl)	Rounded articular projection, often articulates with a corresponding fossa		
DEPRESSIONS AND OPENINGS			
For Passage of Vessels and Nerves			
Foramen (fo-ra'men)	Round or oval opening through a bone	 <i>Meatus</i> <i>Fossa</i> <i>Notch</i> <i>Groove</i> <i>Sinus</i>	 <i>Inferior orbital fissure</i> <i>Foramen</i> <i>Skull</i>
Groove	Furrow		
Fissure	Narrow, slitlike opening		
Notch	Indentation at the edge of a structure		
Others			
Fossa (fos'ah)	Shallow basinlike depression in a bone, often serving as an articular surface		
Meatus (me-a'tus)	Canal-like passageway		
Sinus	Cavity within a bone, filled with air and lined with mucous membrane		



The Skeletal System

- Adult human skeleton is composed of 206 named bones grouped into two divisions
 - ✓ Axial skeleton (80 bones)
 - ✓ Appendicular skeleton (126 bones)
- At birth, they are about 270 bones. The number decreases due to gradual fusion of separate bones.

The Axial Skeleton – 80 bones

Axial Skeleton

a. Skull

8	cranial bones		
	frontal 1	temporal 2	
	parietal 2	sphenoid 1	
	occipital 1	ethmoid 1	
14	facial bones		
	maxilla 2	lacrimal 2	
	zygomatic 2	nasal 2	
	palatine 2	vomer 1	
	inferior nasal concha 2		
	mandible 1		22 bones

b. Middle ear bones

	malleus 2	
	incus 2	
	stapes 2	6 bones

c. Hyoid

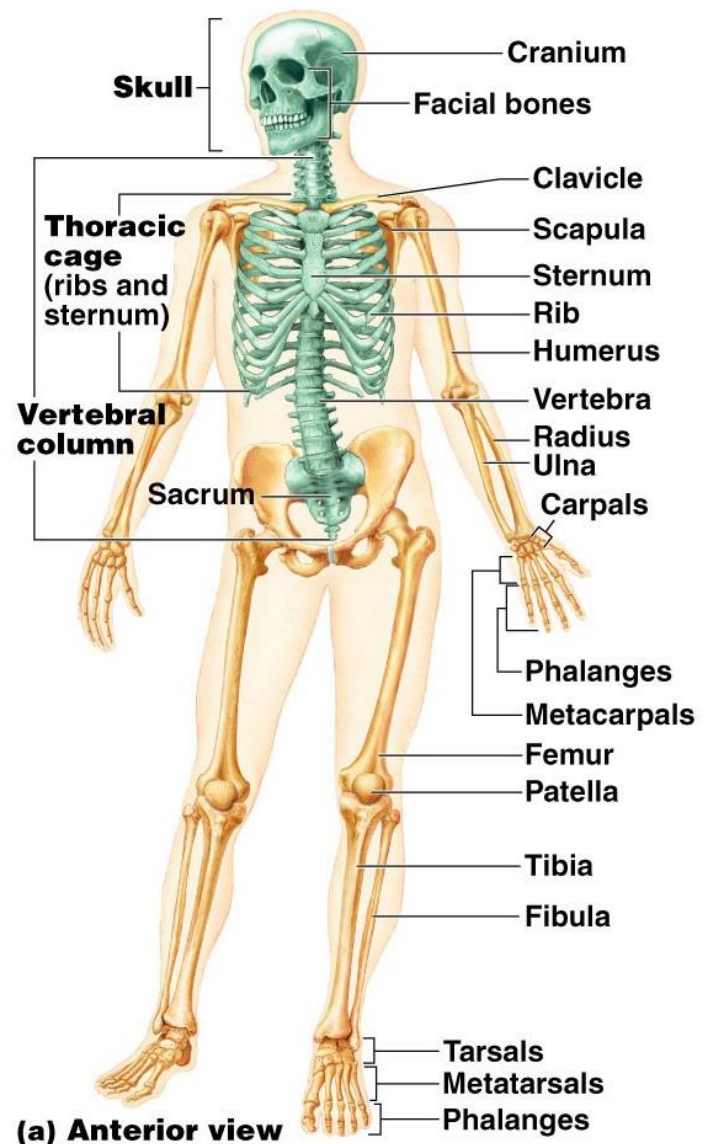
	hyoid bone 1	1 bone
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d. Vertebral column

	cervical vertebra 7	
	thoracic vertebra 12	
	lumbar vertebra 5	
	sacrum 1	
	coccyx 1	26 bones

e. Thoracic cage

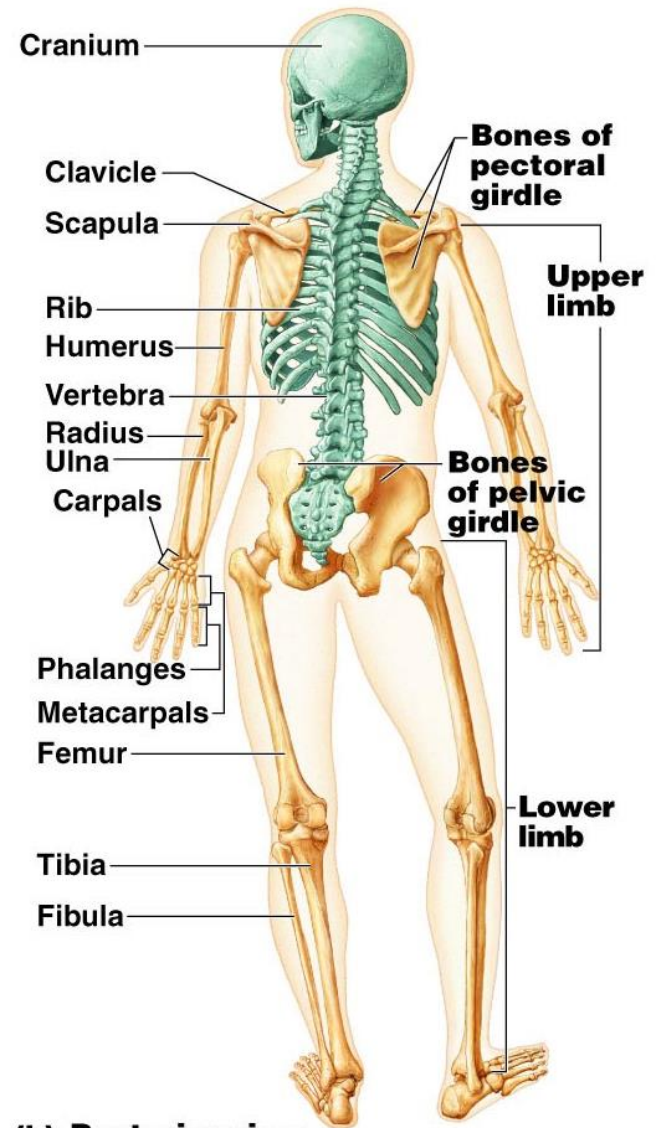
	rib 24	
	sternum 1	25 bones



The Appendicular Skeleton – 126 bones

Appendicular Skeleton

a. Pectoral girdle	
scapula 2	
clavicle 2	4 bones
b. Upper limbs	
humerus 2	
radius 2	
ulna 2	
carpal 16	
metacarpal 10	
phalanx 28	60 bones
c. Pelvic girdle	
hip bone 2	2 bones
d. Lower limbs	
femur 2	
tibia 2	
fibula 2	
patella 2	
tarsal 14	
metatarsal 10	
phalanx 28	60 bones



(b) Posterior view



The Axial Skeleton

Cranium

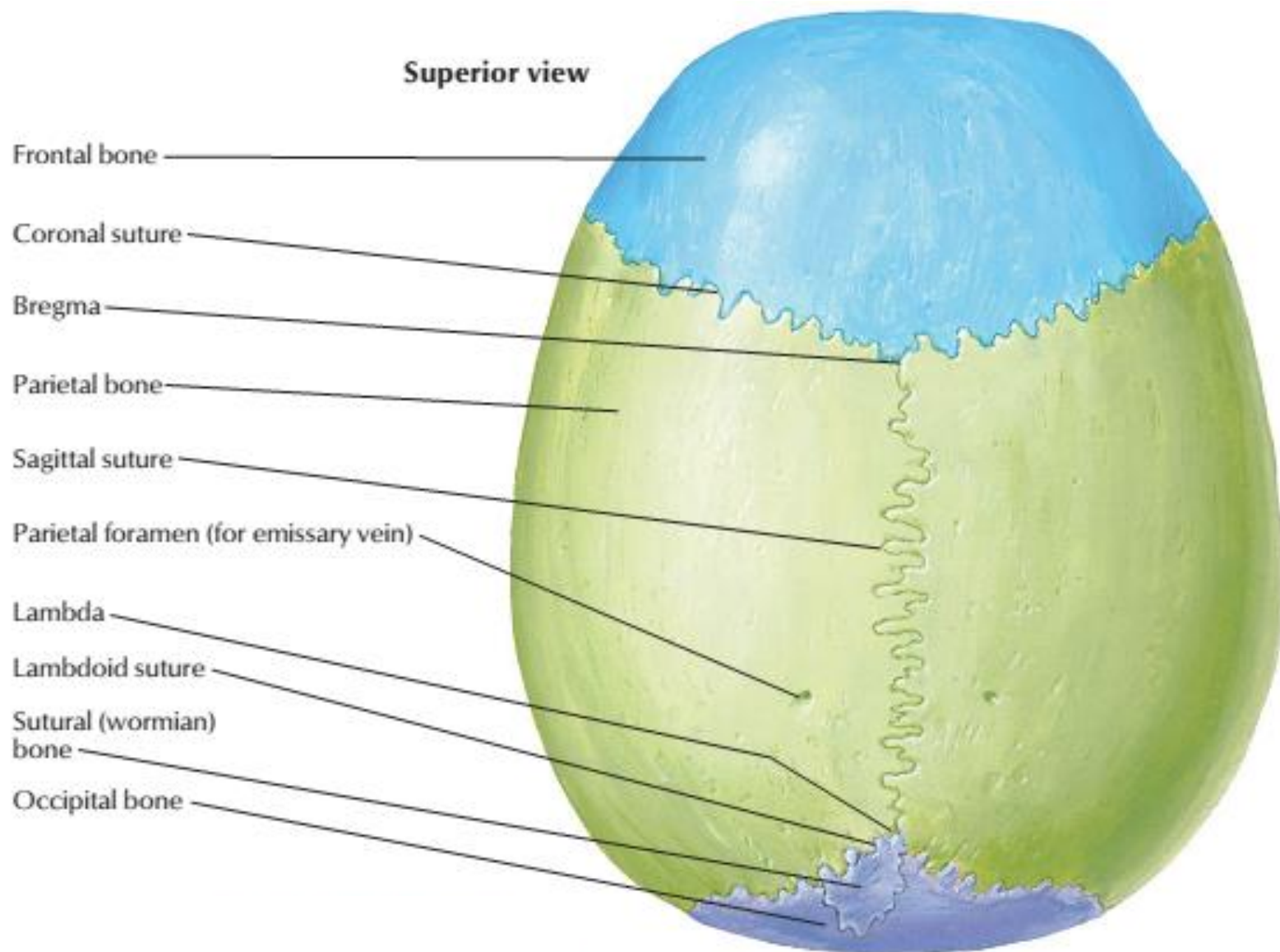
- The cranium has a
 - **vault/calvaria/ skullcap** and
 - **base/floor**
 - Internally, prominent bony ridges divide the cranial base into 3 distinct fossae: Anterior, middle and posterior cranial fossae
- 8 cranial bones
 - **Paired bones:** Temporal bones, Parietal bones
 - **Unpaired bones:** Frontal bone, Occipital bone, Sphenoid bone, Ethmoid bone

“Old People From Timbuktu Eat Spiders.”

Occipital, Parietal, Frontal, Temporal, Ethmoid, Sphenoid

PEST OF 6 (cranial bones)

Calvarium





Parietal Bones and Sutures

- Parietal bones form superior and lateral parts of skull
- Form sutures with other bones
 - **Coronal suture** with frontal bone
 - **Squamous suture** with temporal bone
 - **Sagittal suture** with each other (right and left parietal bones)
 - **Lambdoid suture** with occipital bone

Bregma – the point where sagittal and coronal sutures meet.

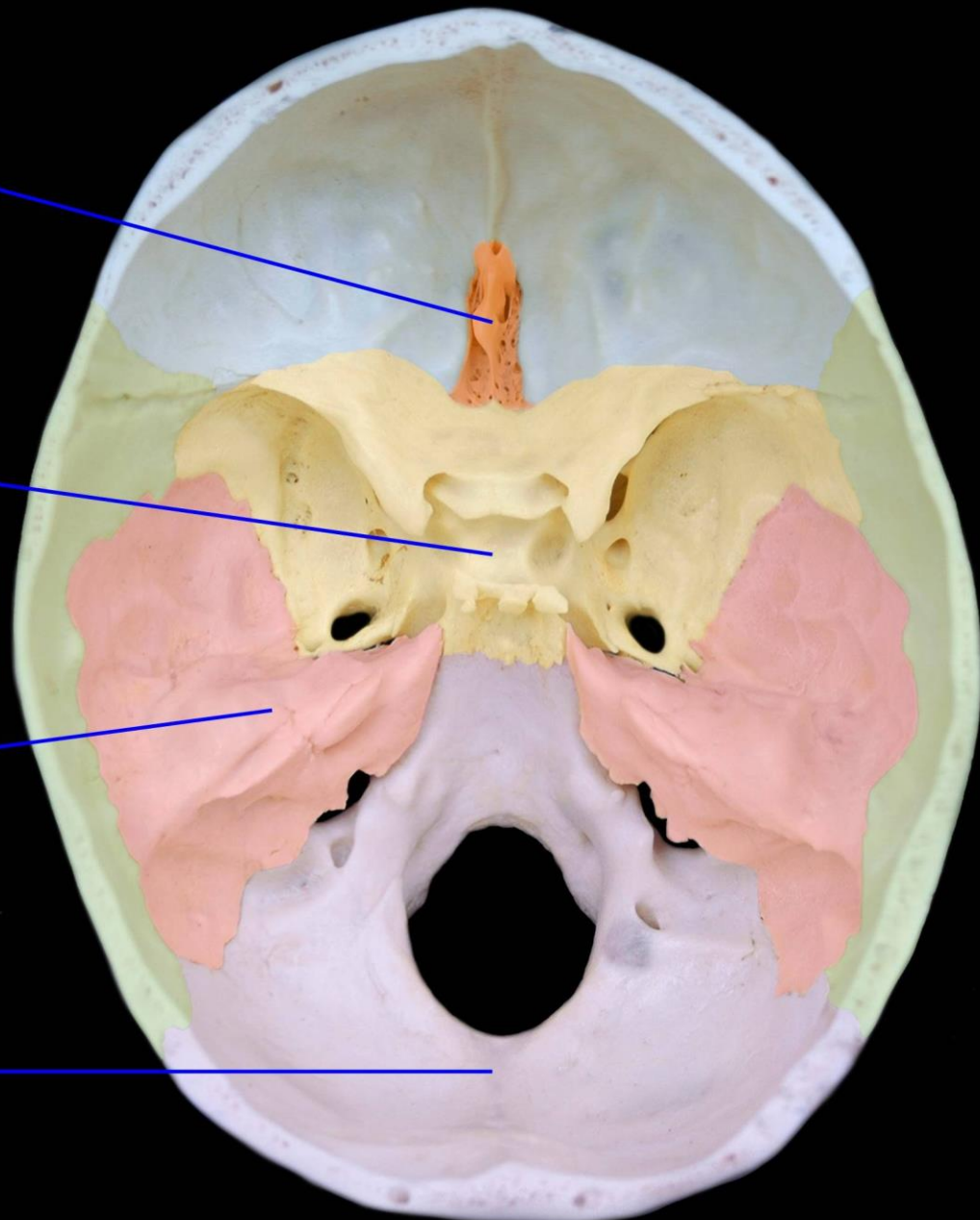
Lambda – the point where sagittal and lambdoid sutures meet.

Ethmoid bone

Sphenoid bone

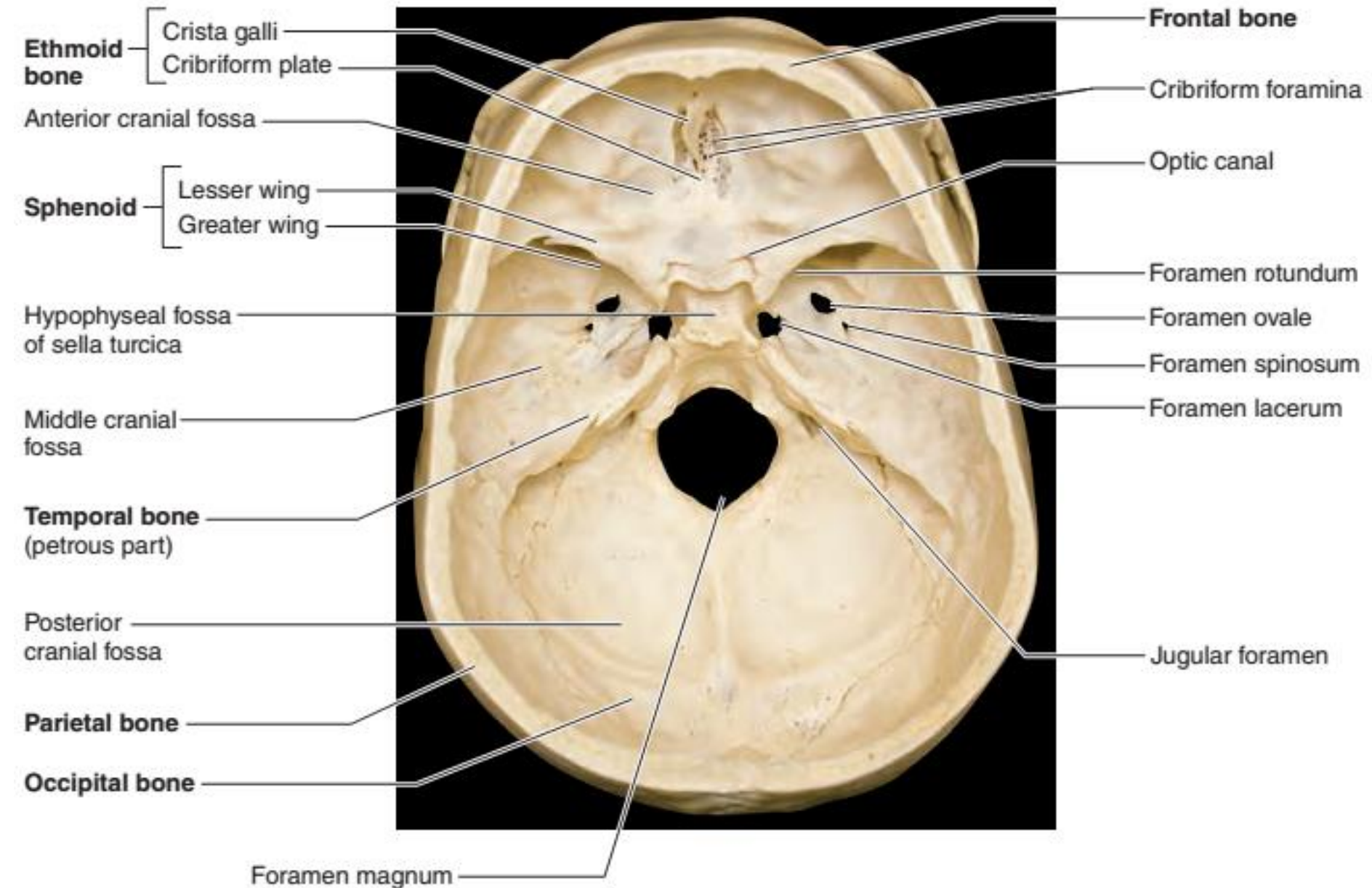
Temporal bone

Occipital bone



Cranial base, superior

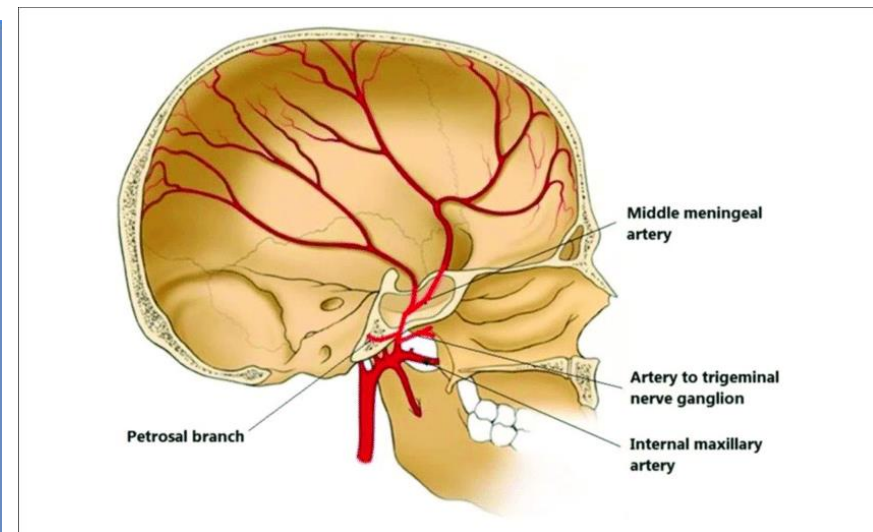
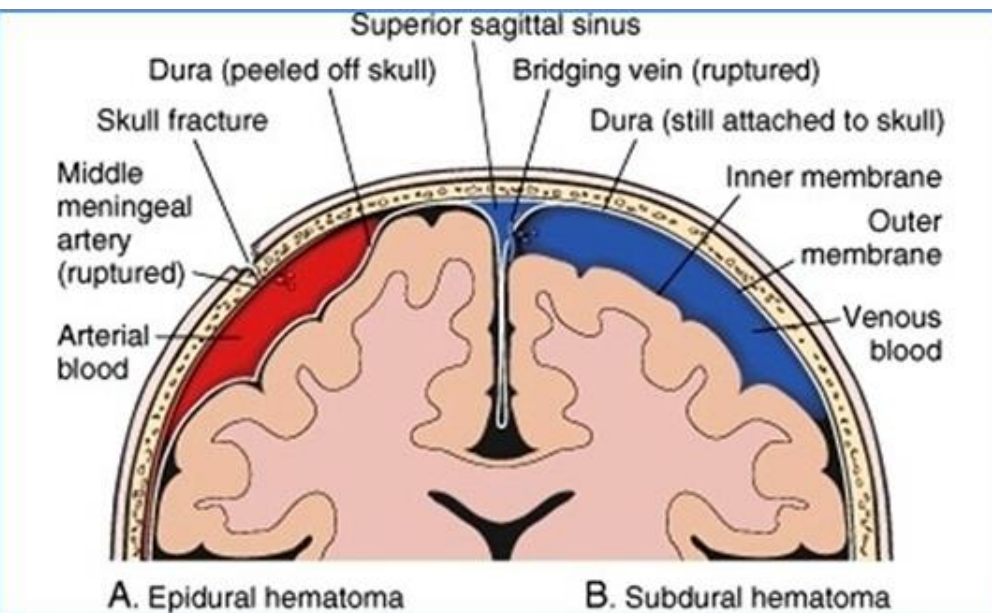
Floor of cranium



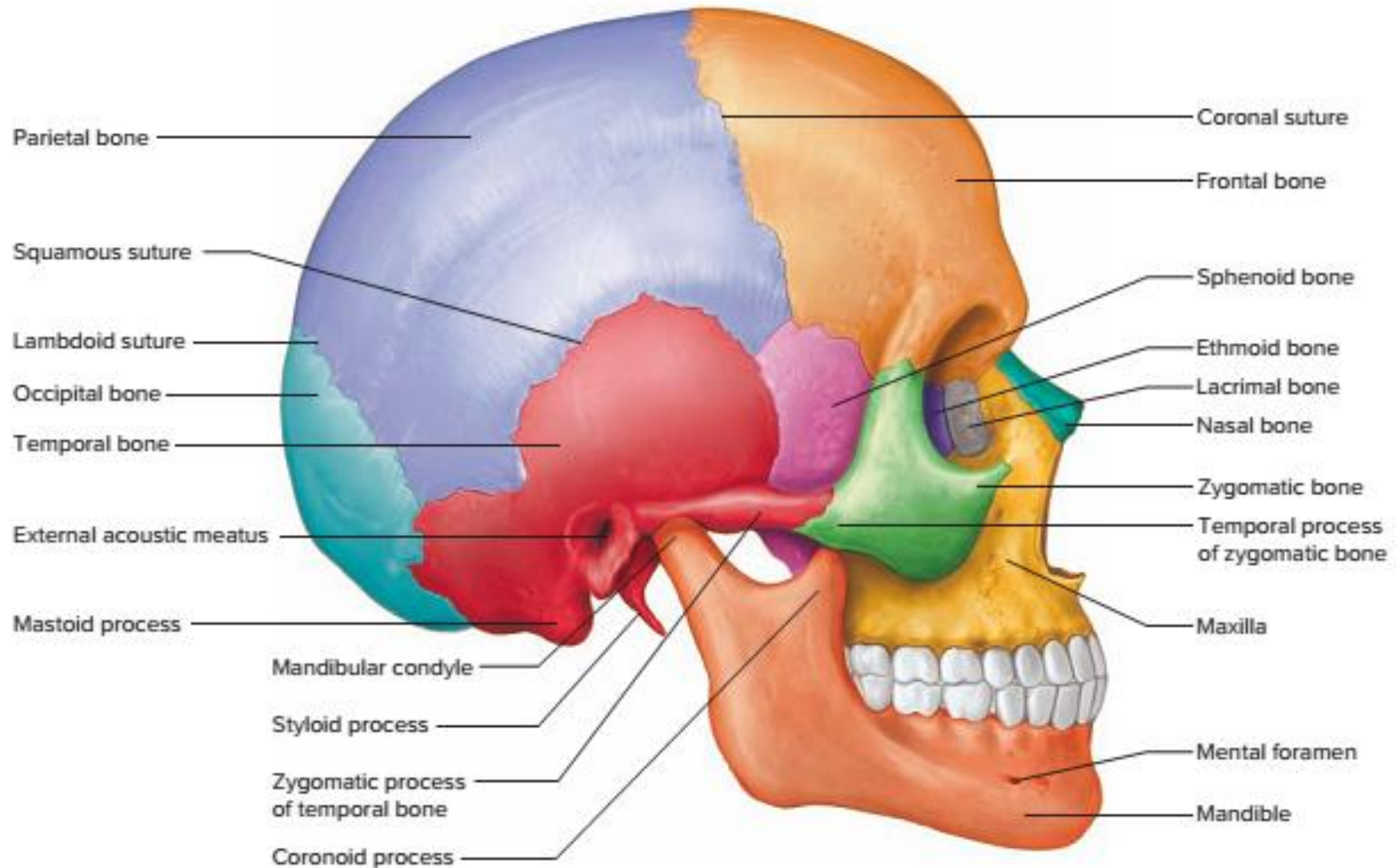
Parietal Bones and Sutures

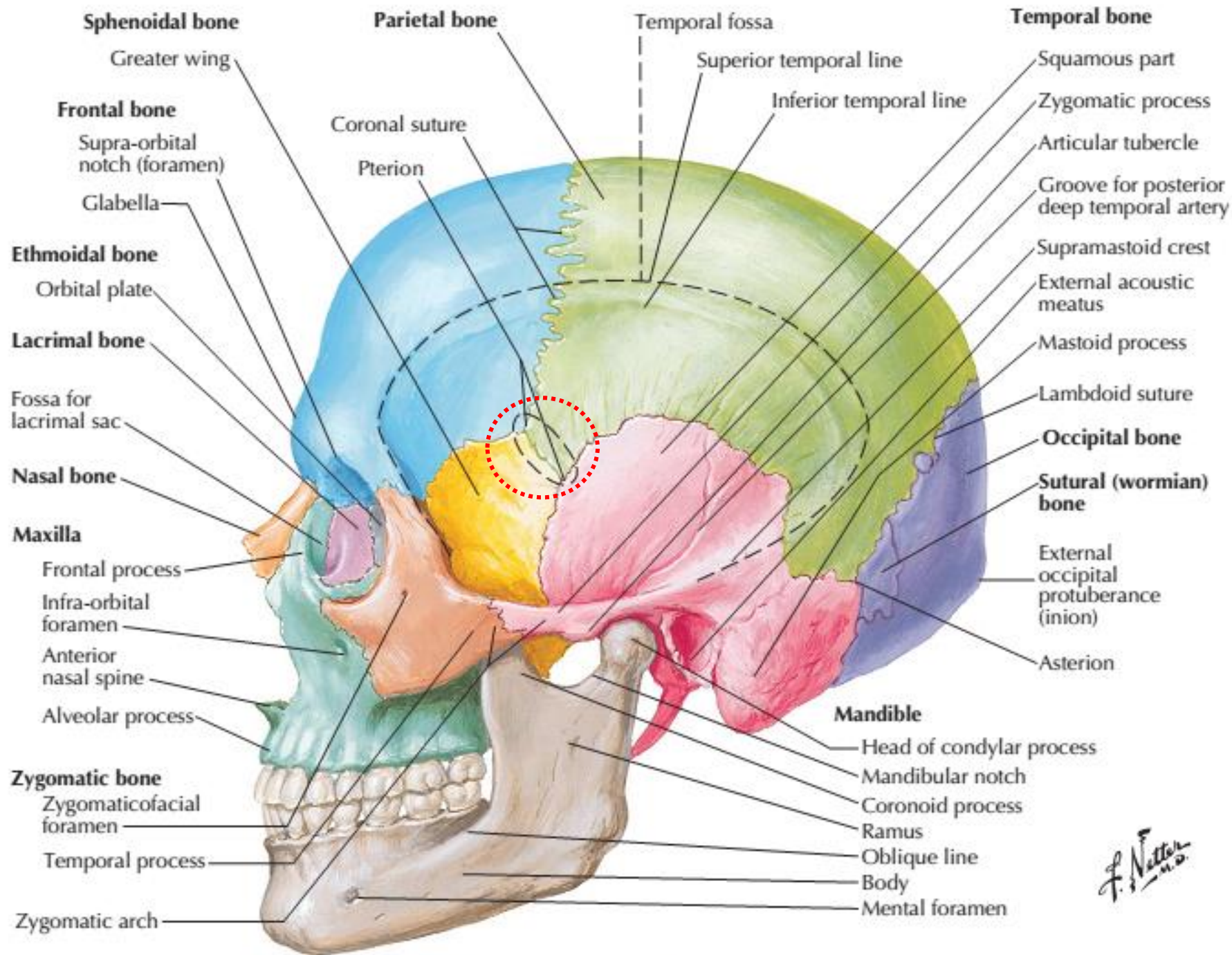
Pterion – region on the lateral aspect of skull where frontal, sphenoid, temporal and parietal bones meet; forms an H-shaped sutures. Is 3–4 cm superior to the midpoint of the zygomatic arch

*A blow to the pterion may damage the **middle meningeal artery** (or one of its branches), leading to **epidural (extradural) hematoma**.*

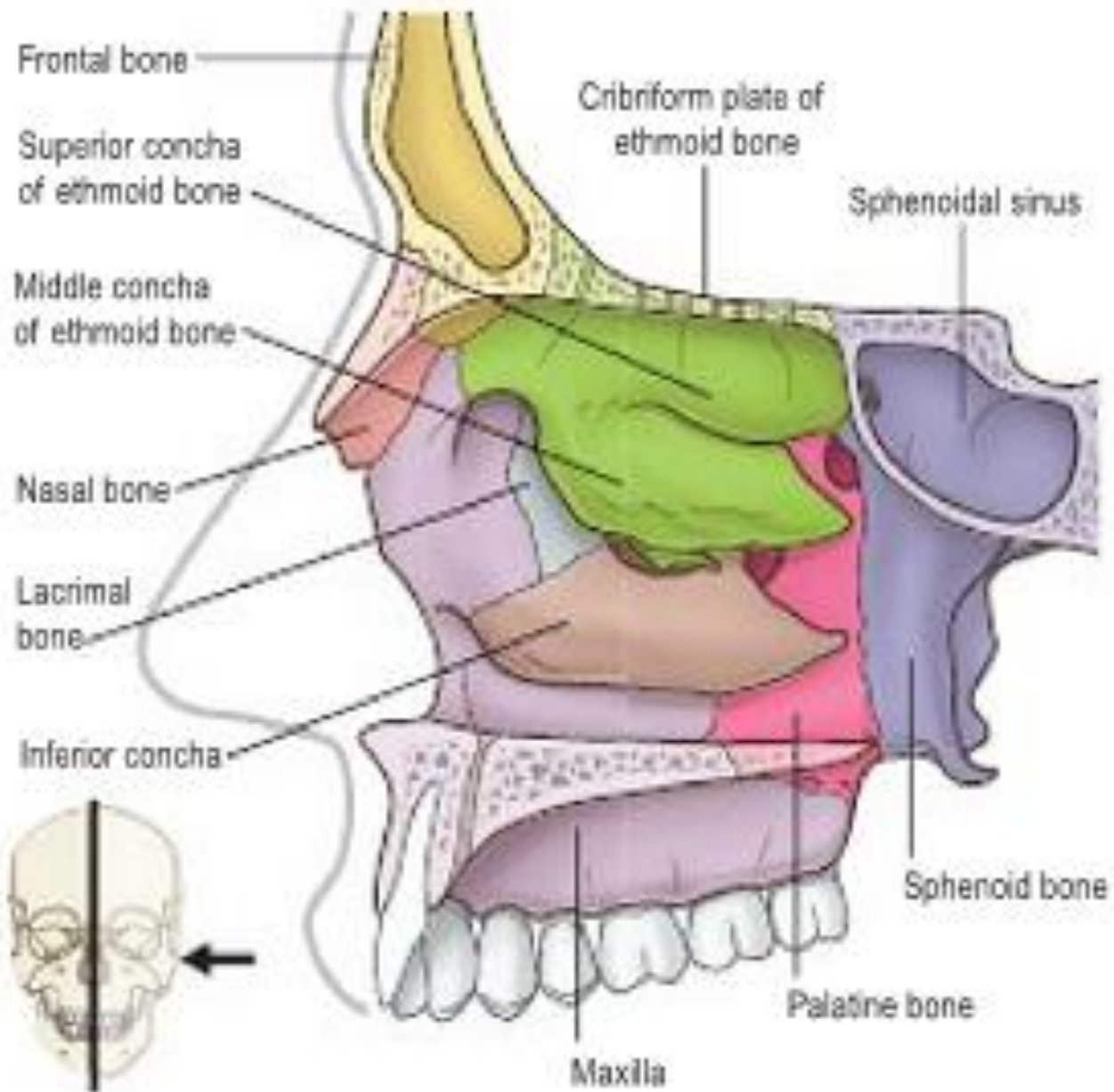


Lateral View of the Skull

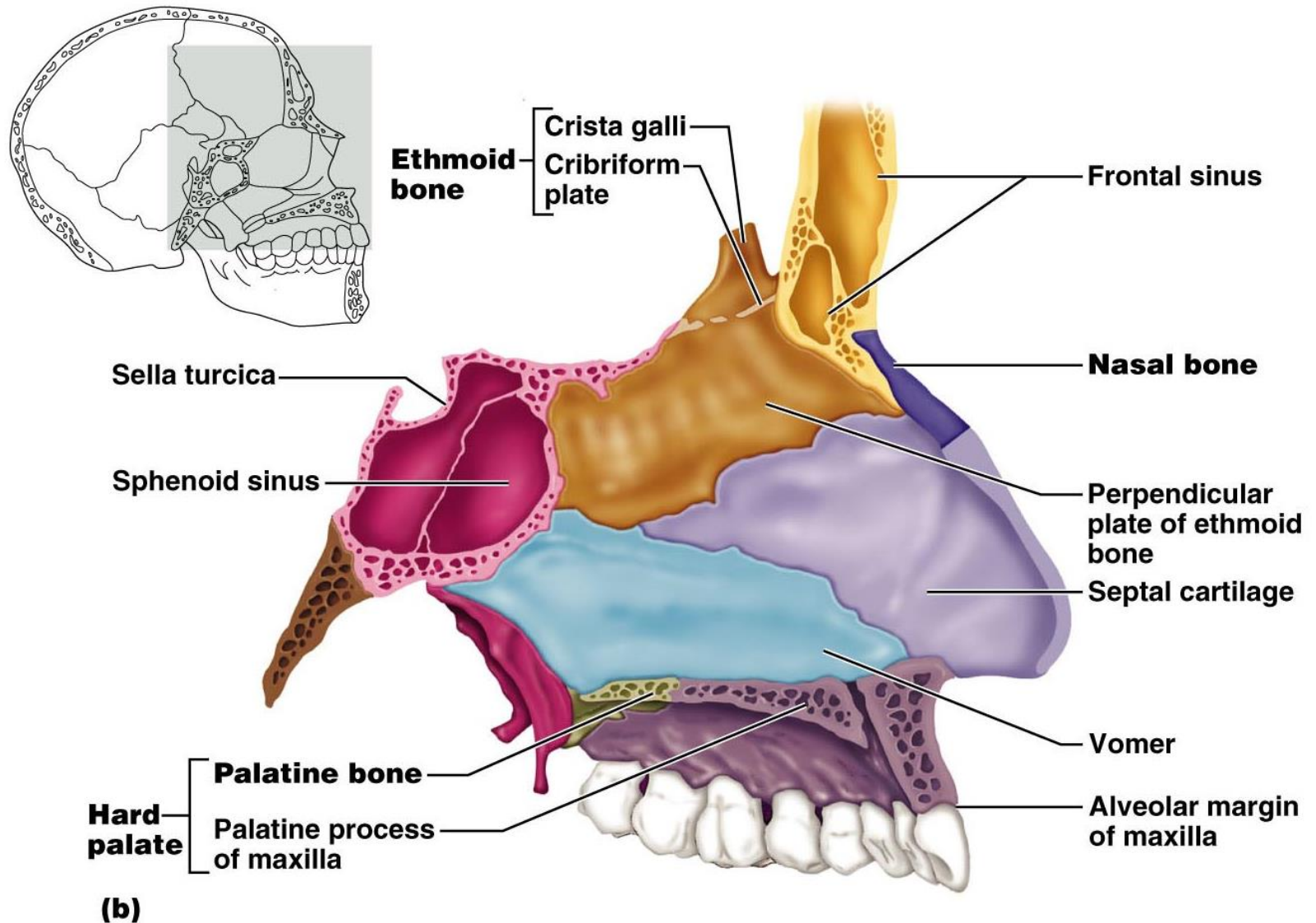




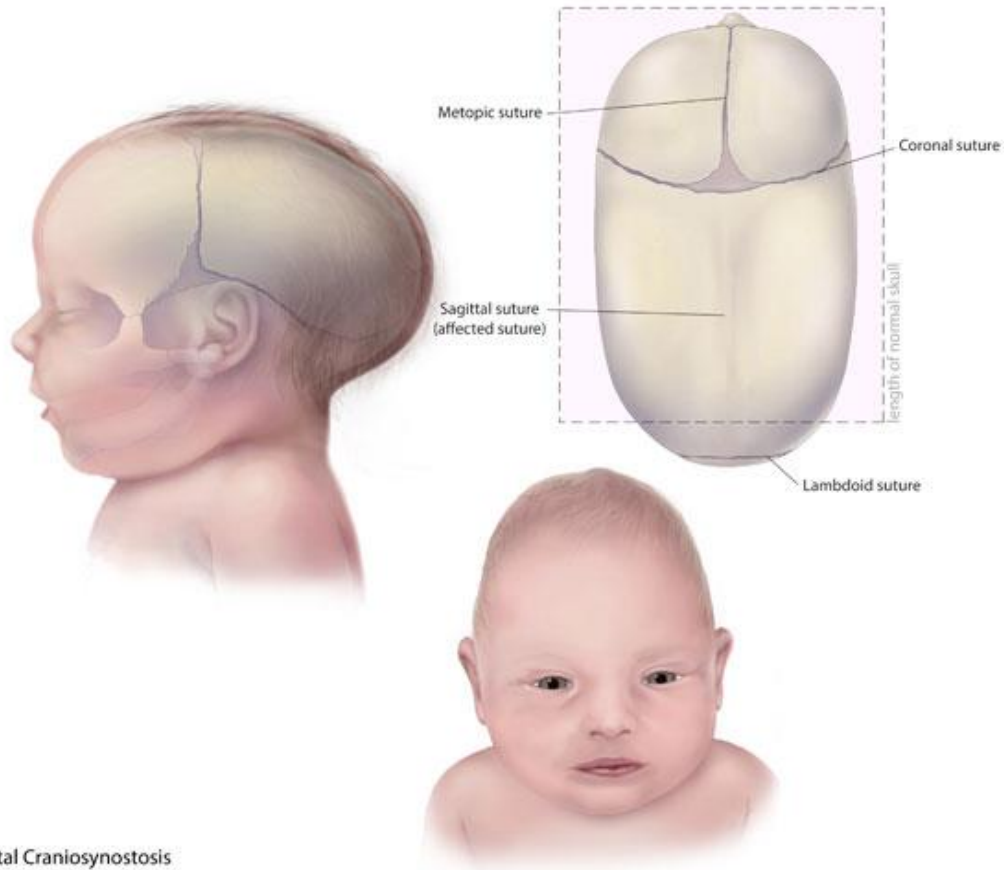
The Ethmoid Bone- Lateral wall of nasal cavity



Nasal septum



The Skull



Sagittal Craniosynostosis

Scaphocephaly (premature closure of sagittal suture)



Plagiocephally (asymmetry in the shape of the head)

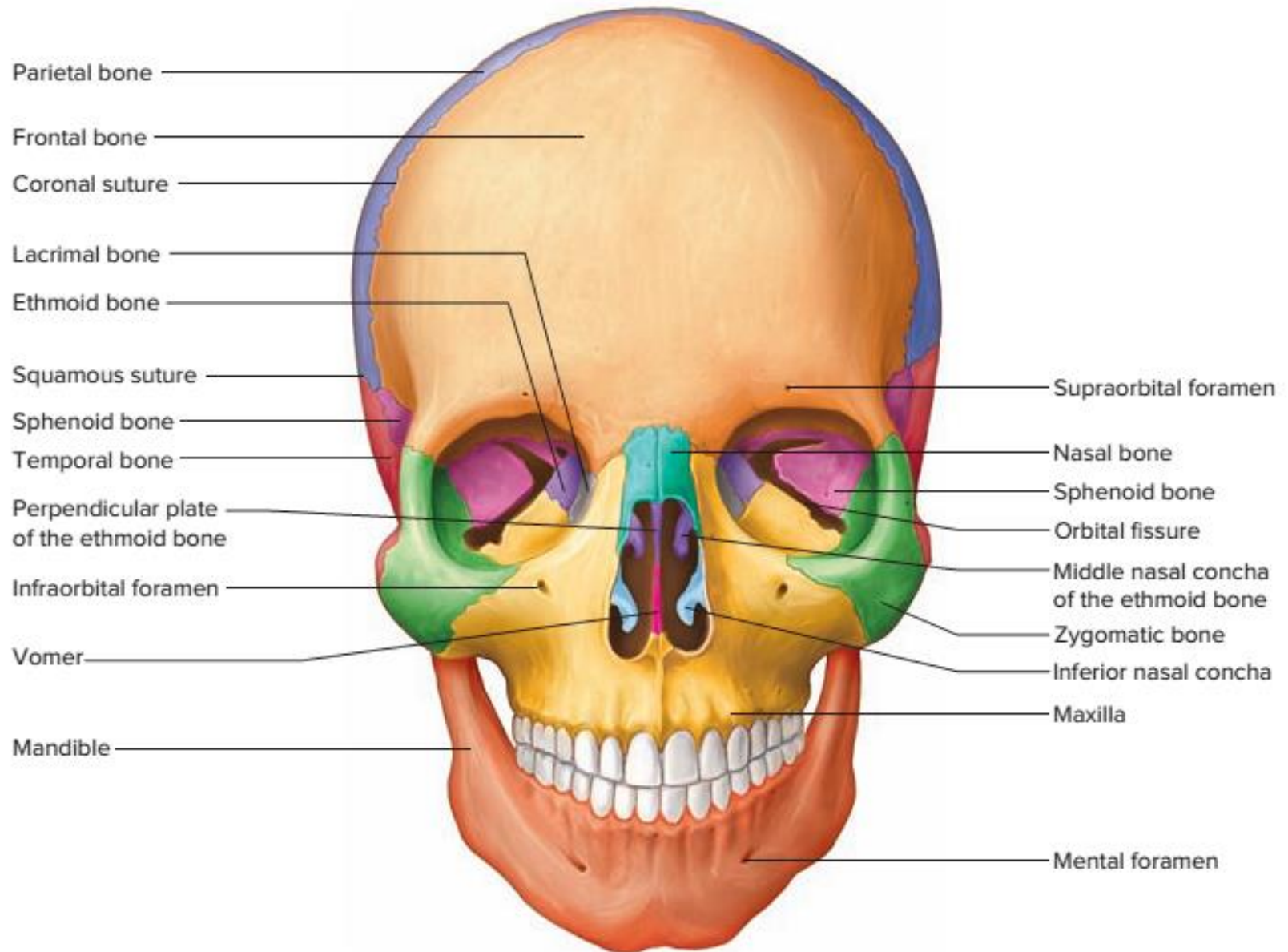
Facial Bones (14)

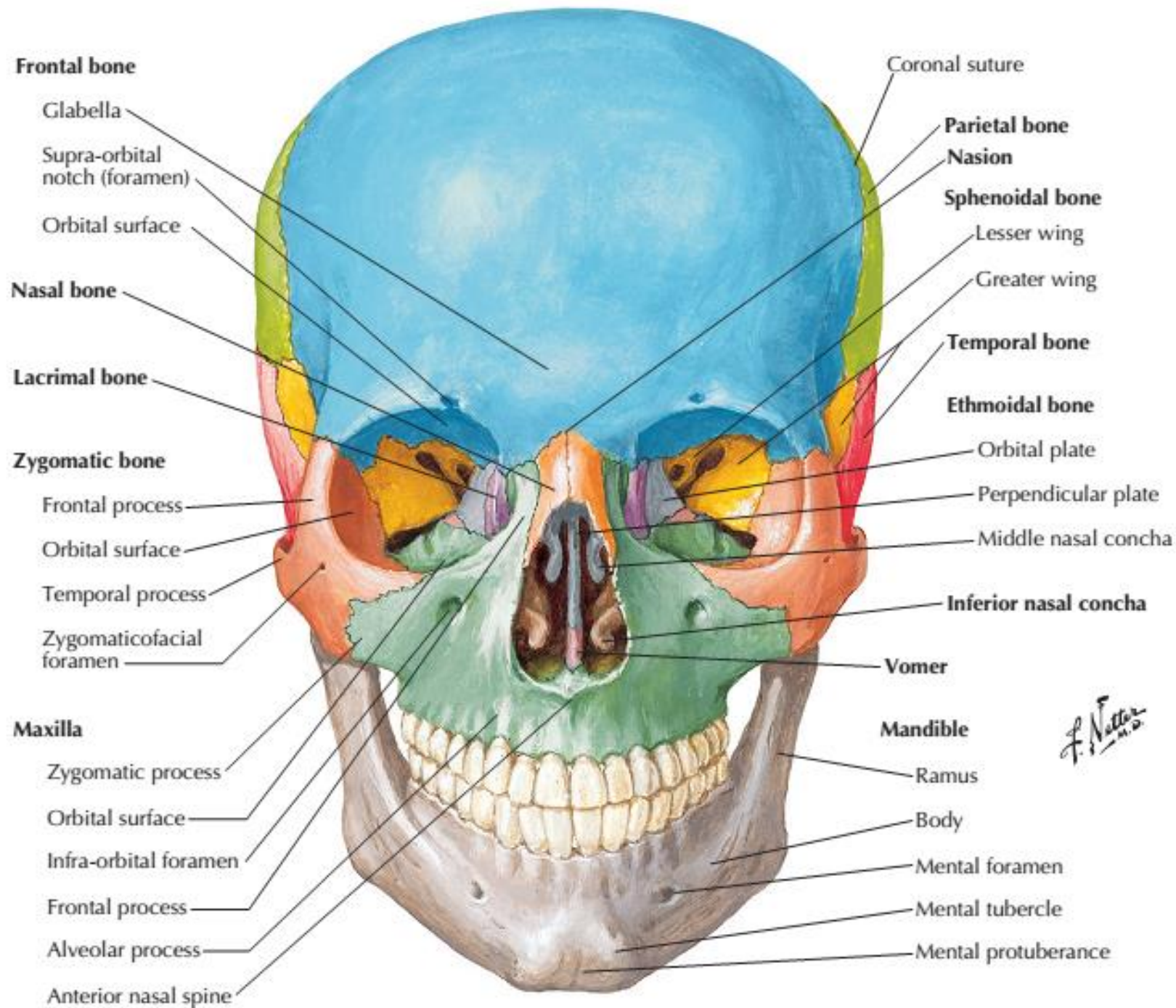
- Unpaired bones
 - ✓ Mandible (lower jaw)
 - ✓ Vomer

- Paired bones
 - ✓ Maxillae (upper jaw)
 - ✓ Zygomatic bones (cheek bones)
 - ✓ Nasal bones
 - ✓ Lacrimal bones
 - ✓ Palatine bones
 - ✓ Inferior nasal conchae

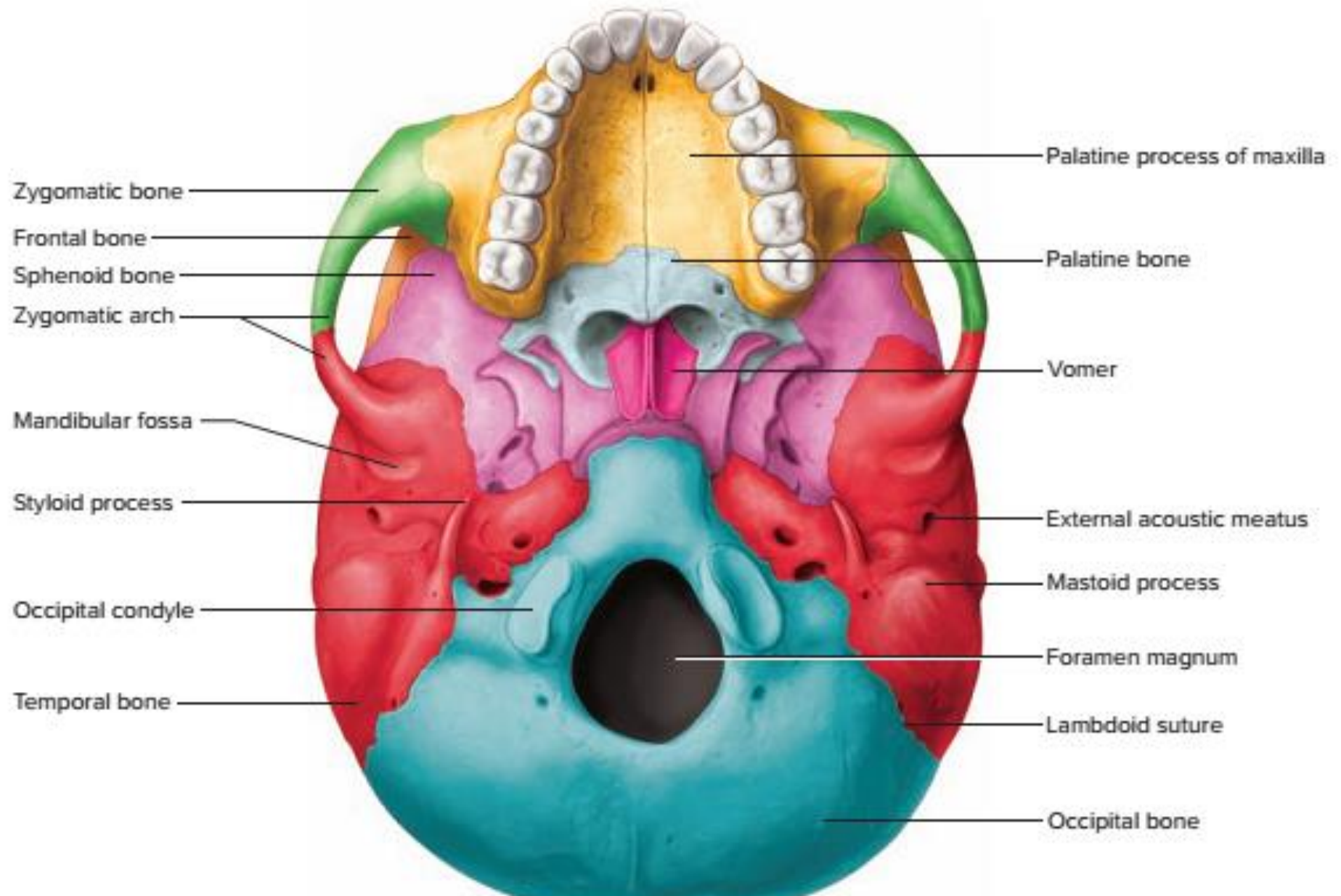
Virgil Is Now Making My Pet Zebra Laugh

Anterior View of the Skull

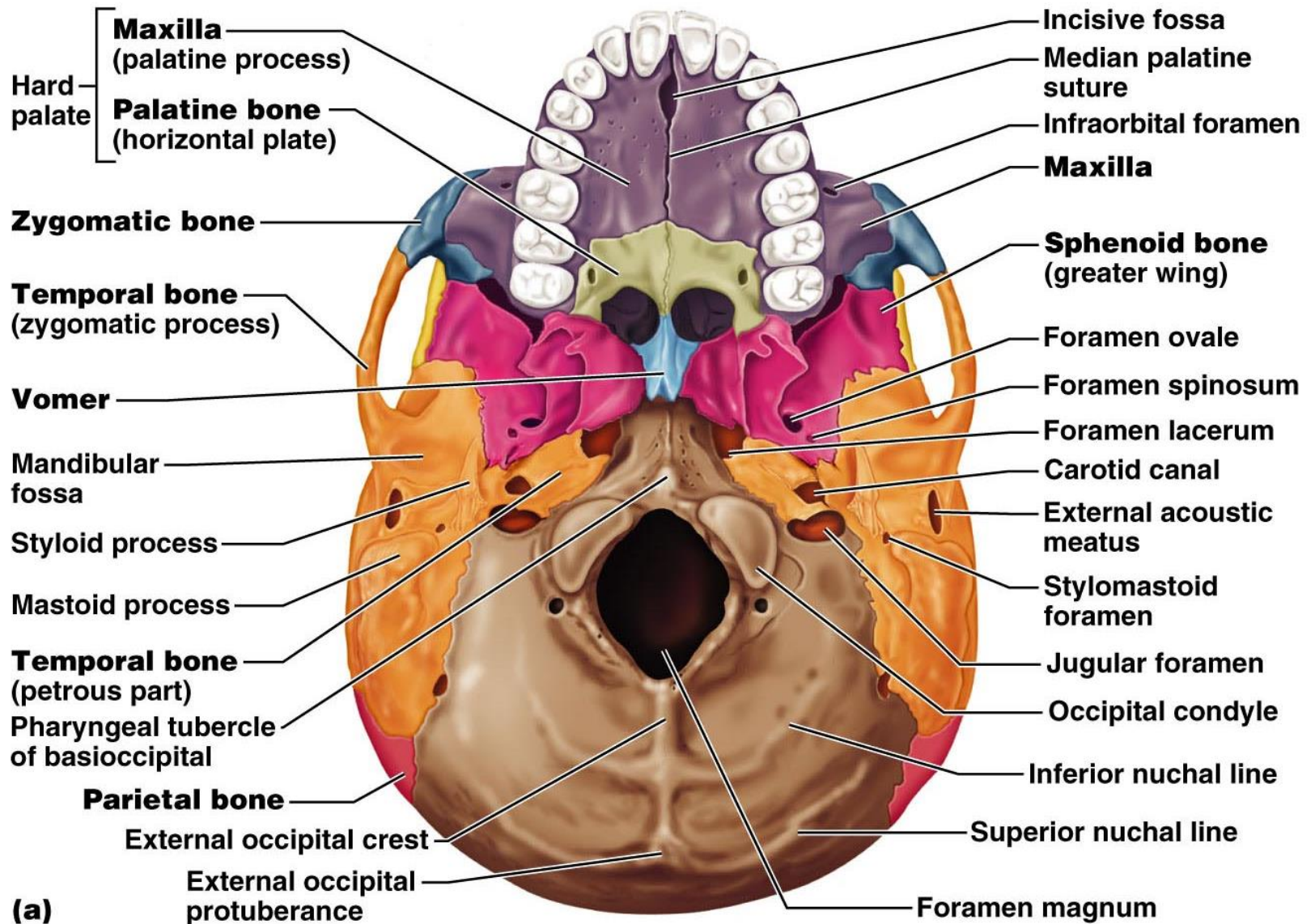




Inferior View of the Skull



Maxilla



(a)

Cleft lip and/or cleft palate

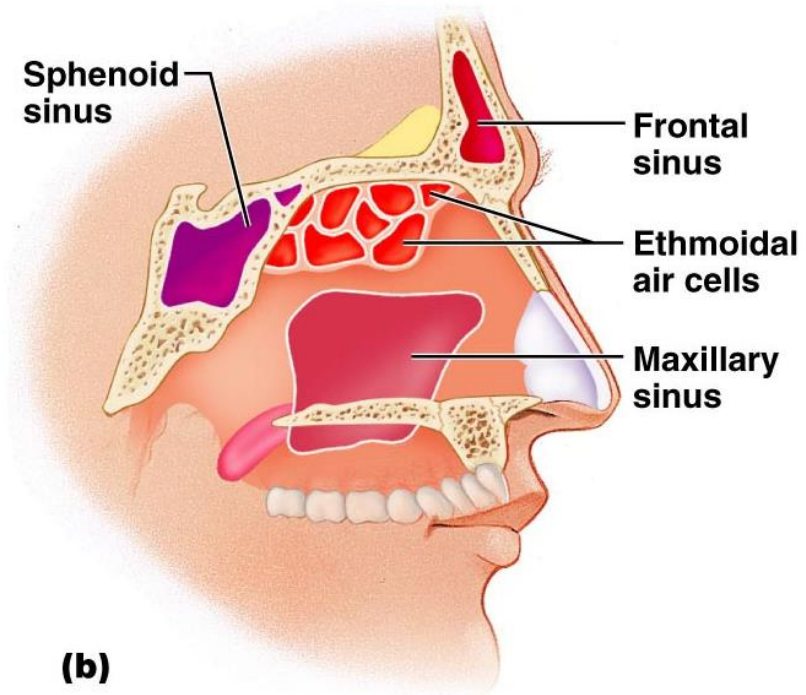
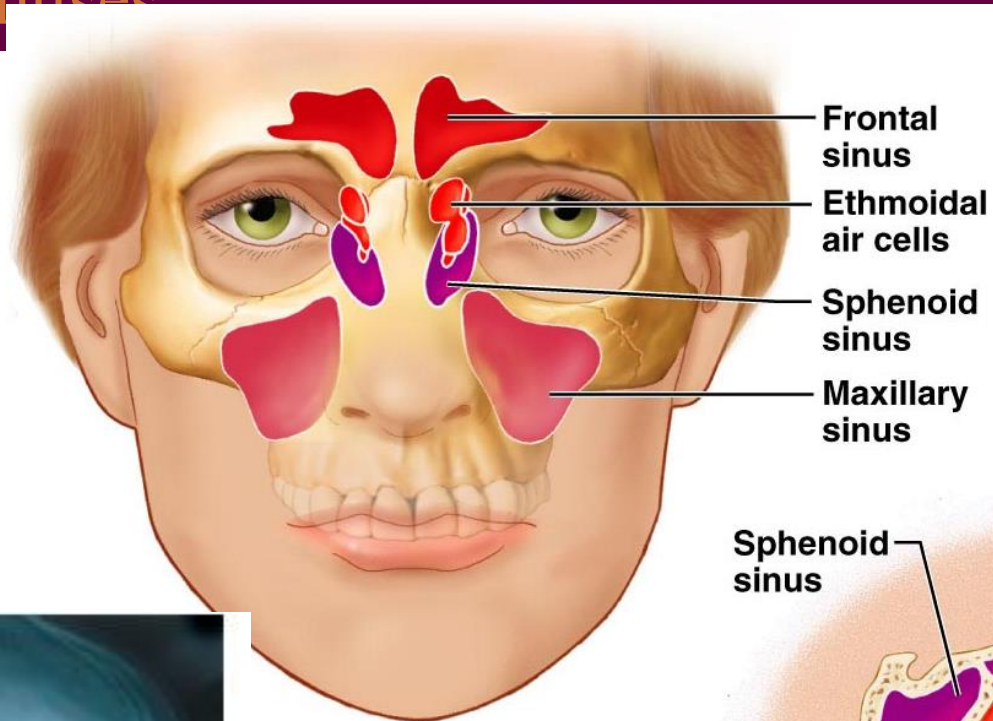
- A **cleft lip** results if the maxillae do not form normally, and a **cleft palate** occurs when the palatine processes of the maxillae do not fuse with one another.
- Can lead to aspiration (inhalation) of food into the nasal cavity and lungs, resulting in pneumonia.
- Can be repaired surgically.
- Likelihood of cleft palate minimized by taking **folic acid** during pregnancy



Paranasal Sinuses

- Air-filled sinuses located within
 - ✓ Frontal bone
 - ✓ Ethmoid bone
 - ✓ Sphenoid bone
 - ✓ Maxillary bones – largest
- Sinuses reduce the skull's weight and increase voice intensity by resonance
- **Sinusitis** -inflammation of the paranasal sinuses, especially the ethmoidal and maxillary sinuses

Paranasal Sinuses



Fontanelles (“little fountain”)

- Membranous areas (soft spots) in infant skull; represents areas where intramembranous ossification is incomplete at birth
- Eventually fuse to form the immovable joints called *sutures* before a baby is 2 years old
- There are **6 fontanelles**: anterior, posterior, paired mastoid (posterolateral) and paired sphenoidal (anterolateral)
 - Anterior fontanelle is the last to fuse (between 12 and 18 months old). Point of fusion is the **bregma**.
 - Posterior fontanelle ossifies 2–3 months after birth. Point of fusion is the **lambda**.
 - Anterolateral fontanelle— closes 3 months after birth,
 - Posterolateral fontanelle – closes 1-2 months after birth

Fontanelles (“little fountain”)

■ Functions of fontanelle

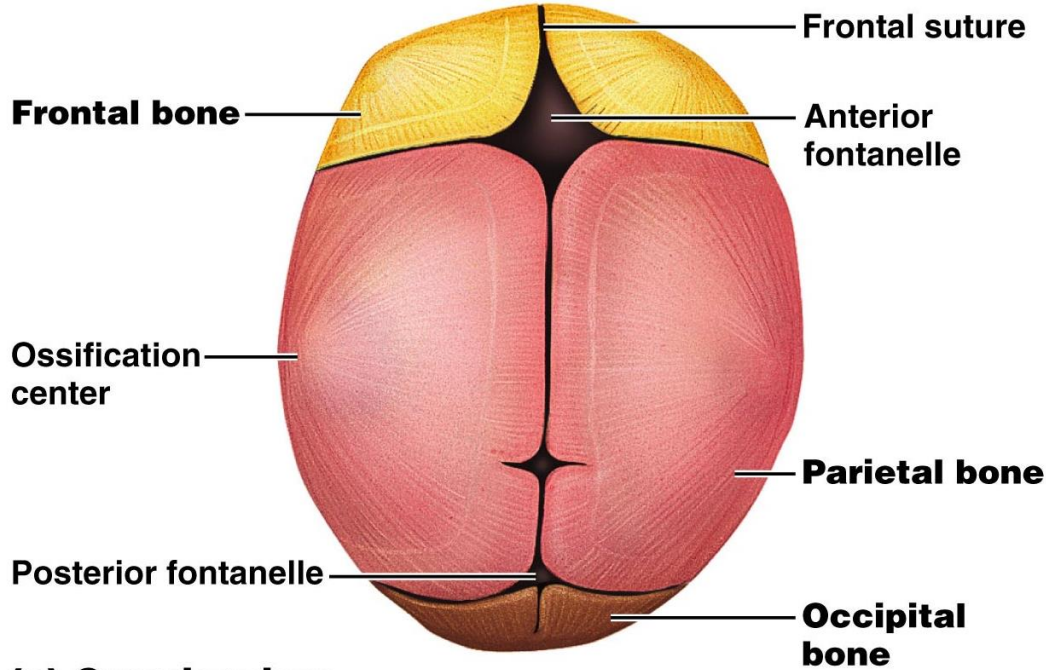
- ✓ Permits some overlapping of the skull bones (moulding) during childbirth.
- ✓ Permits growth of the brain
- ✓ Permits early rapid growth of an infant’s skull

■ Palpation of fontanelles during infancy enables physicians to determine:

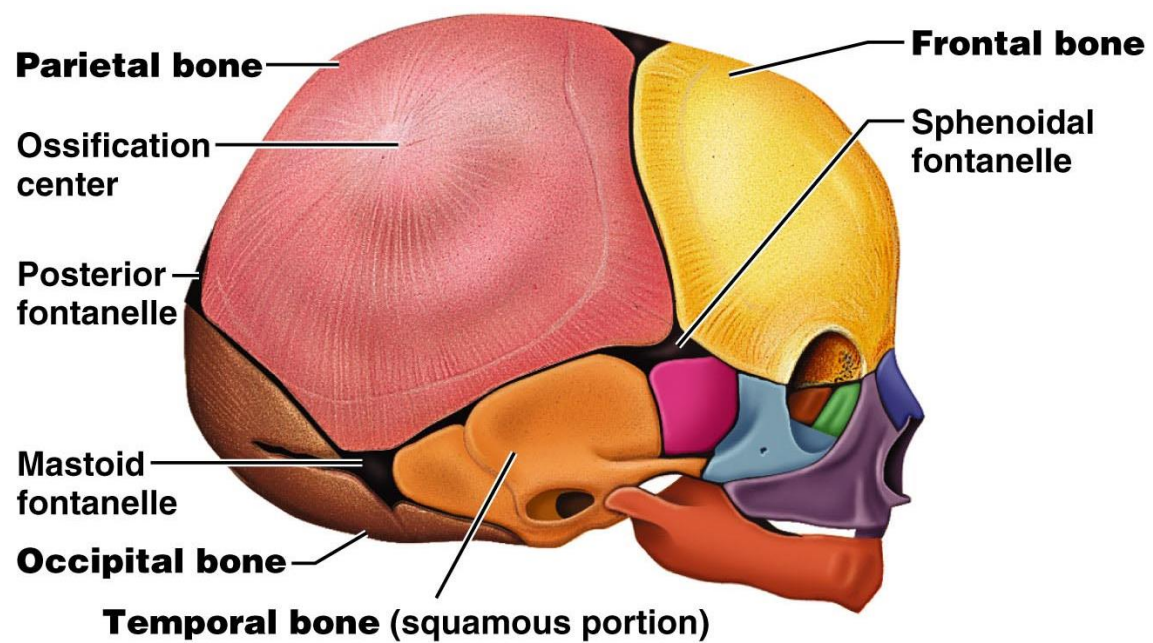
- *Growth progress of the frontal & parietal bones*
- *Degree of hydration of the infant (a depressed fontanelle indicates dehydration)*
- *Level of intracranial pressure (bulging fontanelle indicates increased pressure on the brain)*



Fontanelles



(a) Superior view



(b) Lateral view

Fontanelles

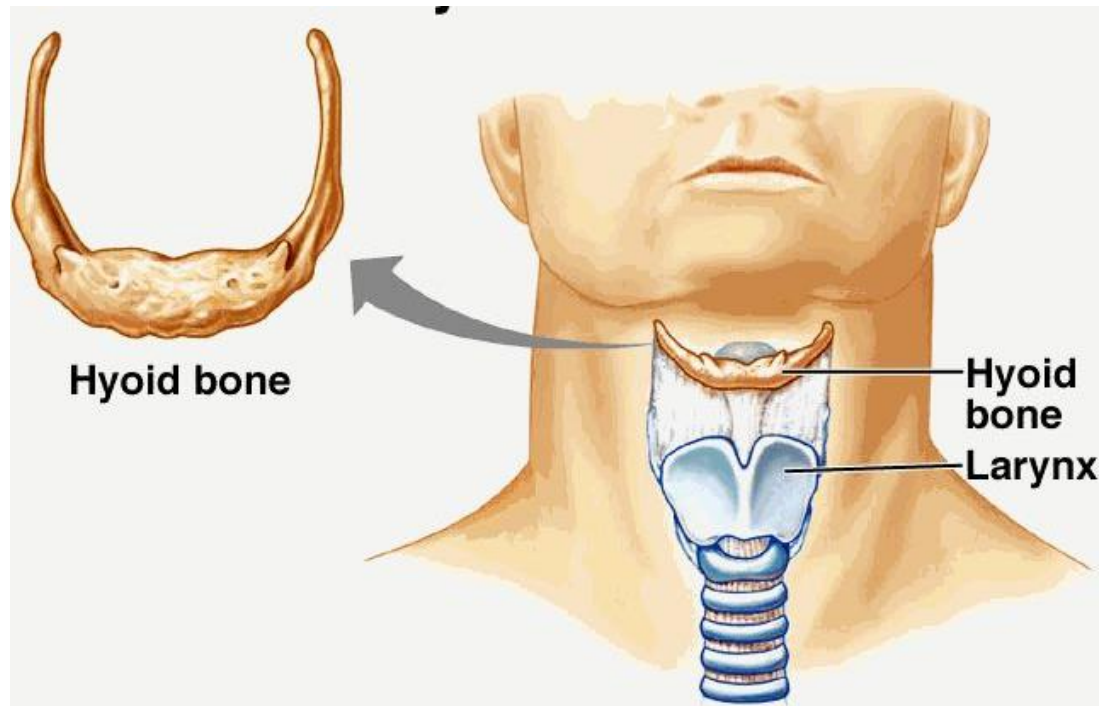
Other characteristics of an infantile skull

- relatively small face
- prominent forehead and large orbits.
- Small jaw and nasal cavity
- sinuses are incompletely formed
- skull bones are thin, but somewhat flexible and thus are less easily fractured than adult skull bones.



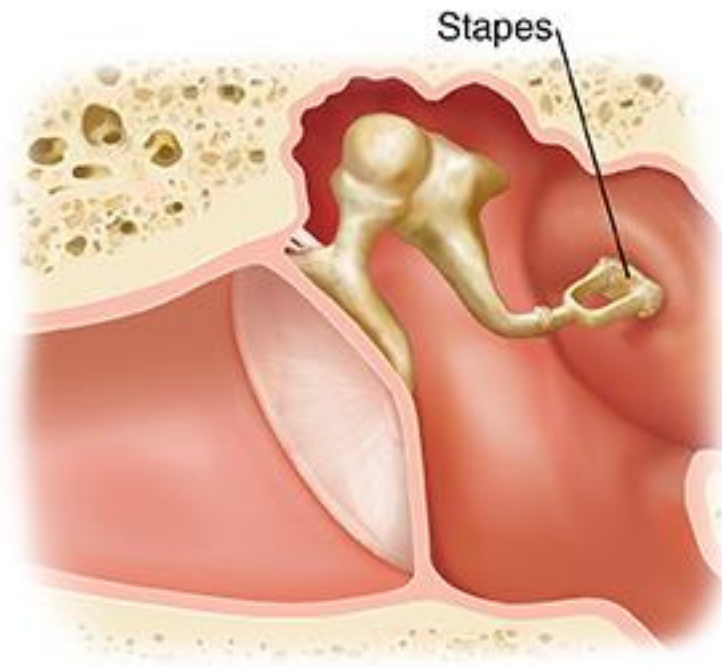
The Hyoid Bone

- It does not articulate with any other bone,
- Acts as a movable base for the tongue. Provides attachment for neck muscles that raise and lower the larynx during swallowing
- Fractured in one-third of all homicides by strangulation



Auditory ossicles

- Auditory ossicles: 3 small bones (malleus, incus, and stapes) present in each middle ear cavity



The Vertebral Column (Spine)

- Consists of a series of separate bones (26), or **vertebrae** in the adult
- Vertebral column is about 70 cm (28 inches)

Functions

- Protect the spinal cord
- Provide opening for exit of spinal nerves
- Allow movements of the spine
- Support the skull
- Intervertebral discs act as shock absorbers
- Gives attachment to the ribs, shoulder girdle and upper limbs, and the pelvic girdle and lower limbs



Regions and Normal Curvatures

- **Cervical vertebrae**
 - 7 vertebrae in the neck
- **Thoracic vertebrae**
 - 12 vertebrae in the thorax
- **Lumbar vertebrae**
 - 5 vertebrae of the lower back
- **Sacrum**
 - Inferior to lumbar vertebrae
 - 5 fused vertebrae. In children, they are separate vertebrae
- **Coccyx**
 - Most inferior region of the vertebral column
 - 3-5 fused vertebrae. In children, they are separate vertebrae

Development of Secondary Curvature

- At birth, the spine of a newborn is C-shaped, with one curve
- As the infant lifts his or her head, after about 3 months, the neck (**cervical**) curve and its muscles develop
- As the infant learns to crawl and stand, the lower back (**lumbar**) curve and its muscles develop (after 12–15 months). Strong back muscles help give the child the strength and balance to walk and run.



Regions and Normal Curvatures

- 4 distinct curvatures give vertebral column an S-shape
- Curvatures increase the resilience of the spine
- **Thoracic and sacral curvatures**
 - Convex
 - **Primary curvatures** - present during foetal developmental curvatures
 - **accommodation curves** (accommodate internal organs)

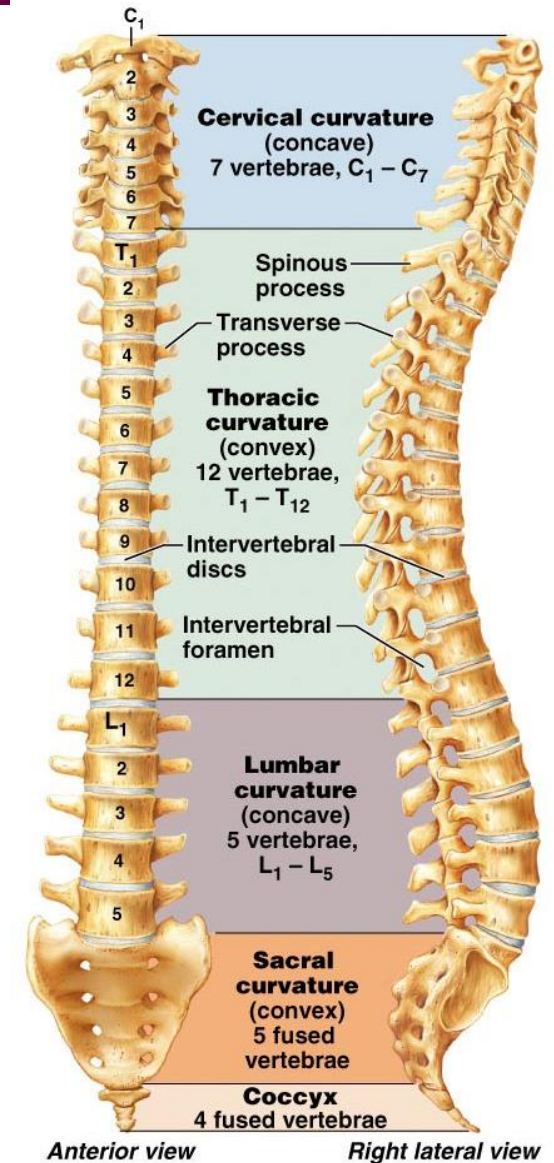
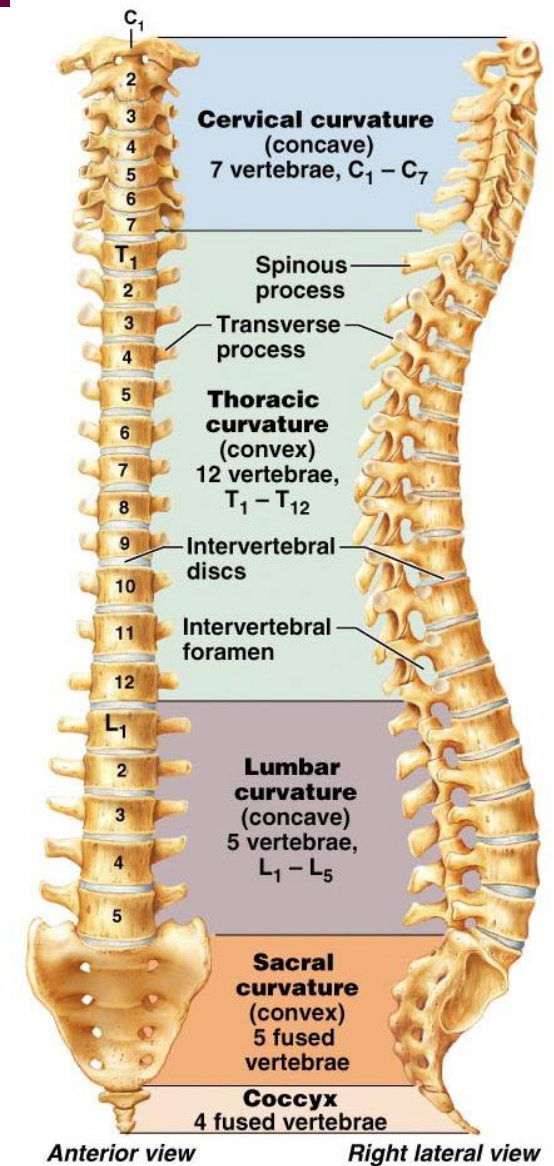


Figure 7.13

Regions and Normal Curvatures

- **Cervical and lumbar curvatures**
 - Concave
 - **Secondary curvatures** - appear after birth
 - Develop when a baby begins to walk
 - **Compensation curves** (shift body weight for upright posture)
 - Redistributes weight of the upper body over the lower limbs



Abnormal spinal curvatures

- ✓ **Scoliosis** – an abnormal lateral curvature mostly in the thoracic region; Causes: congenital, polio, weak muscles
- ✓ **Kyphosis** (“hunchback”) – an exaggerated thoracic curvature, common in aged women, tuberculosis of spine, osteomalacia
- ✓ **Lordosis** (“swayback”) – an exaggerated lumbar curvature, tuberculosis of spine, osteomalacia. Temporary lordosis: obese men, pregnant women



(a) Scoliosis



(b) Kyphosis



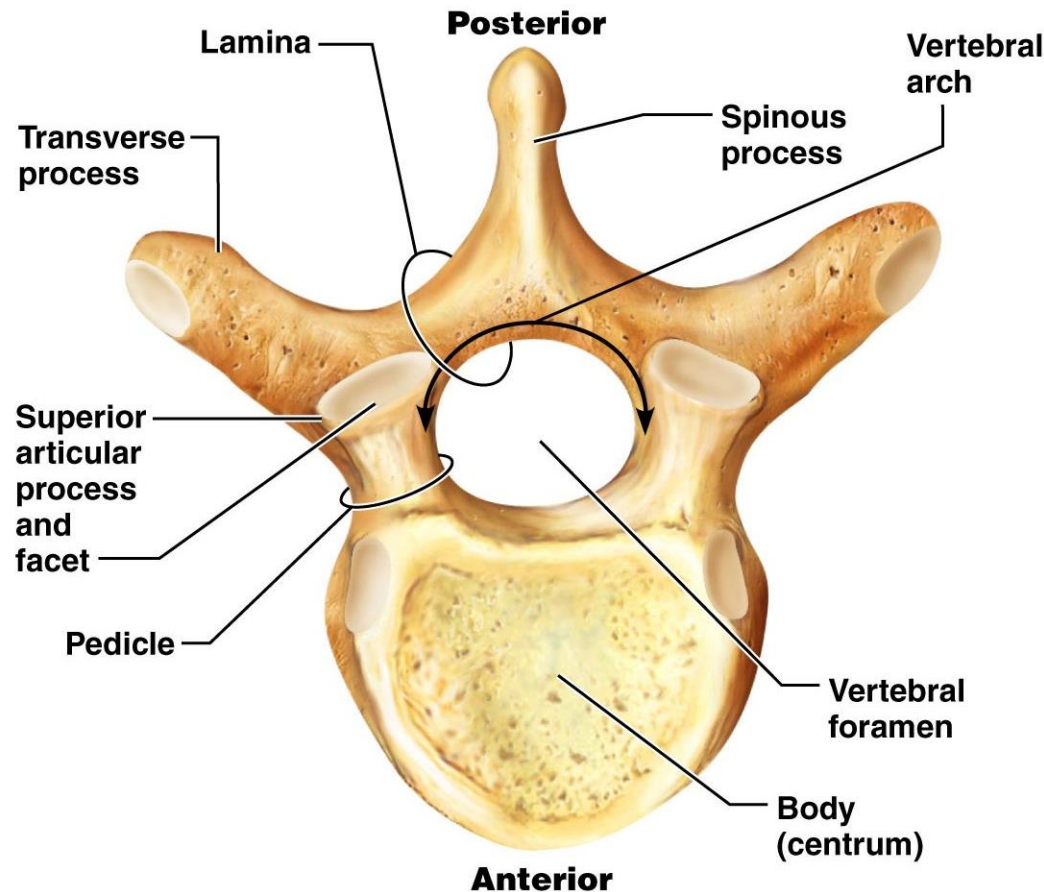
(c) Lordosis

Figure 7.27 Abnormal spinal curvatures.



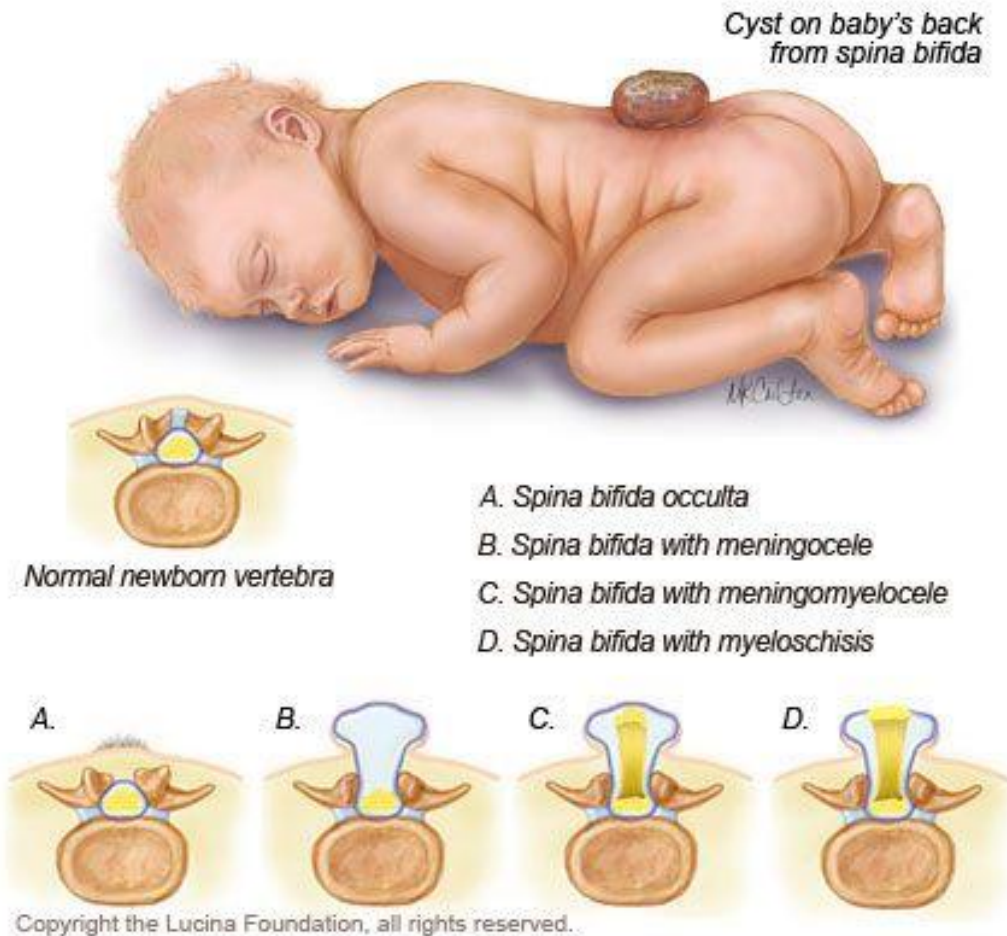
Typical Vertebra – By convention thoracic

- Consists of a **body** and **vertebral arch**
- Vertebral (neural) arch
 - encloses a vertebral foramen
 - consists of
 - **Pedicles** – projects from body
 - **Laminae** – unite to form spinous process
- Transverse process project laterally
- Facets for articulation with adjacent vertebrae



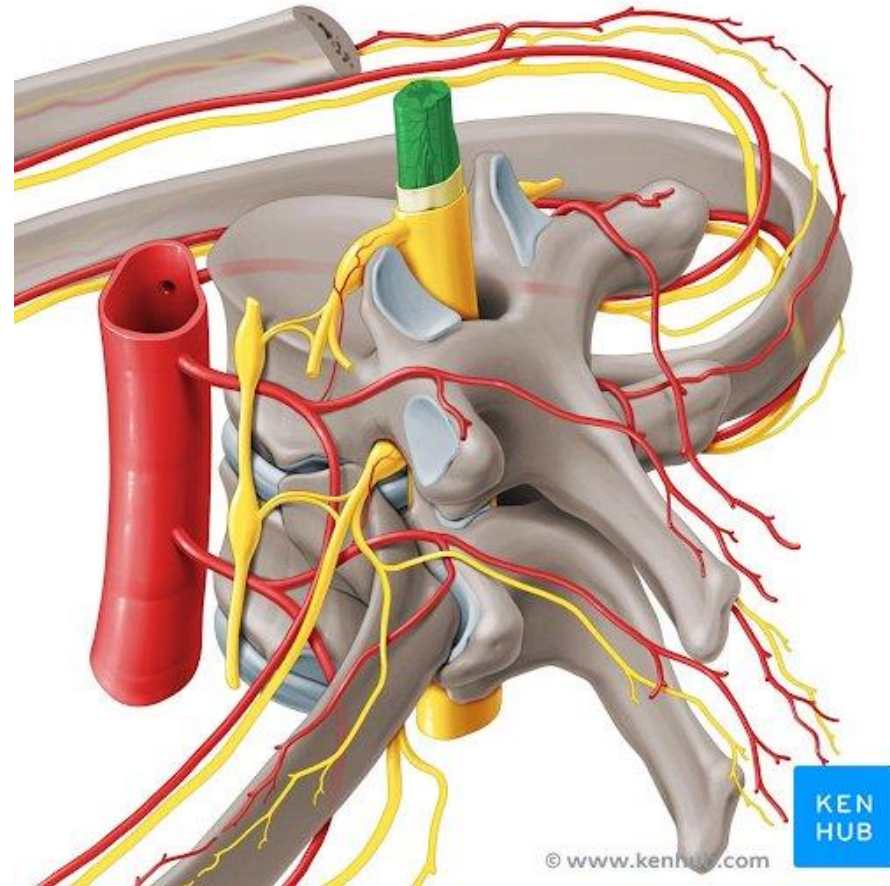
Spina Bifida

- Congenital disorder where the laminae of the vertebrae do not unite
- Bones can't support weight
- occurs most frequently in the lumbosacral region.
- associated with folic acid deficiency in certain genetically susceptible individuals
- Treatment is surgery

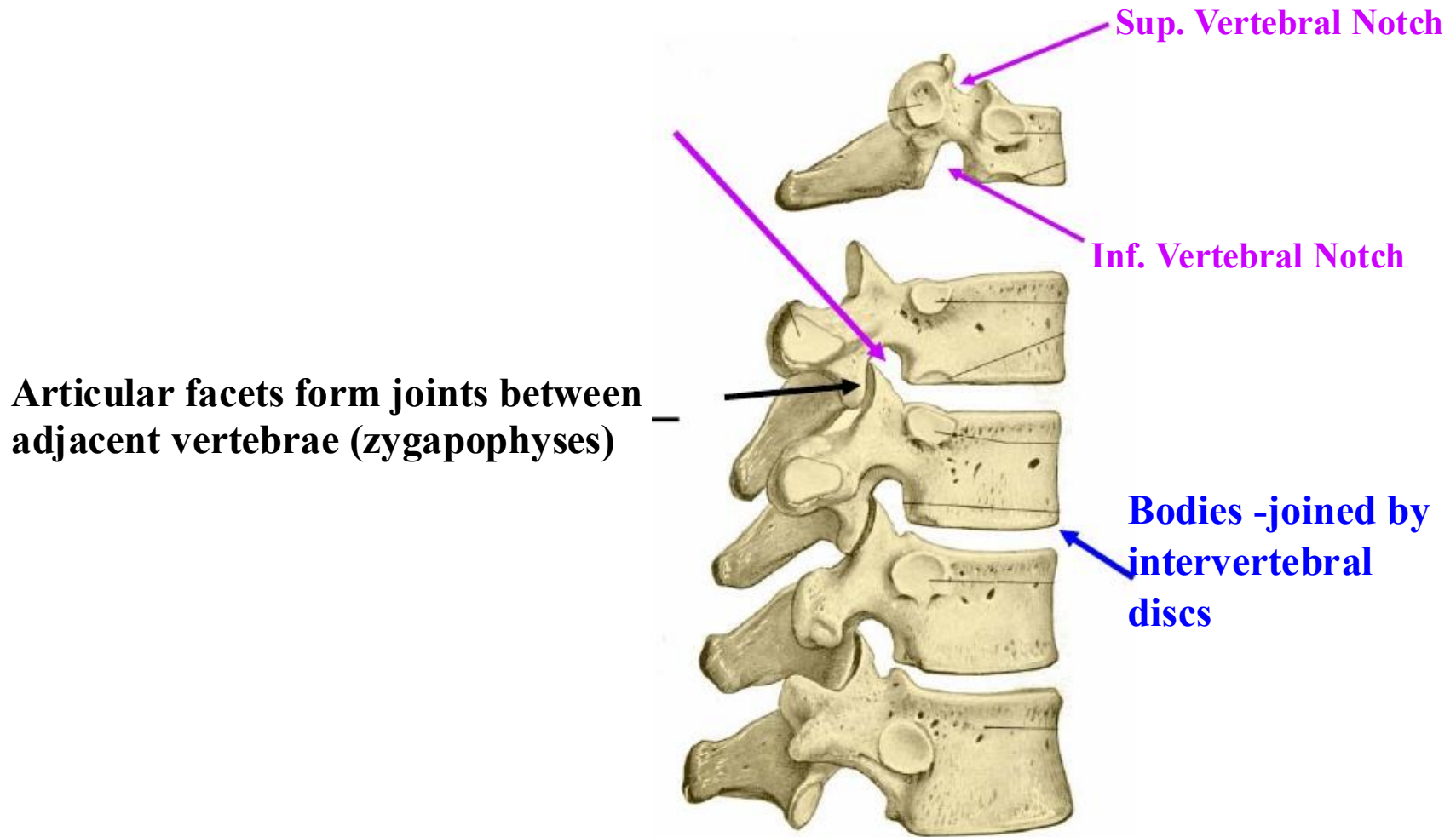


Vertebrae

- Vertebral foramina of 2 or more vertebrae form a **neural canal** where the spinal cord passes.
- Spinal nerves form and exist at the **Intervertebral foramina**

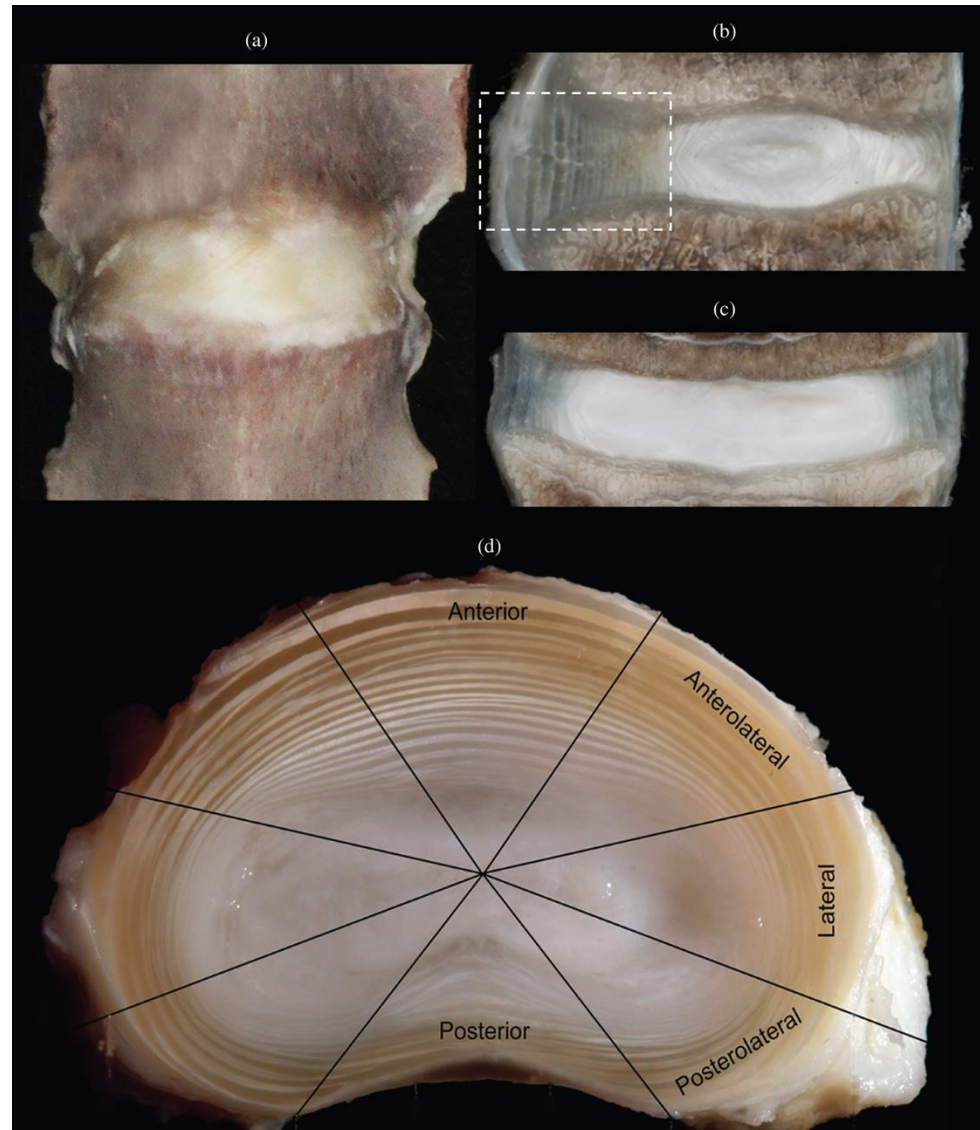


LATERAL VIEW OF VERTEBRA



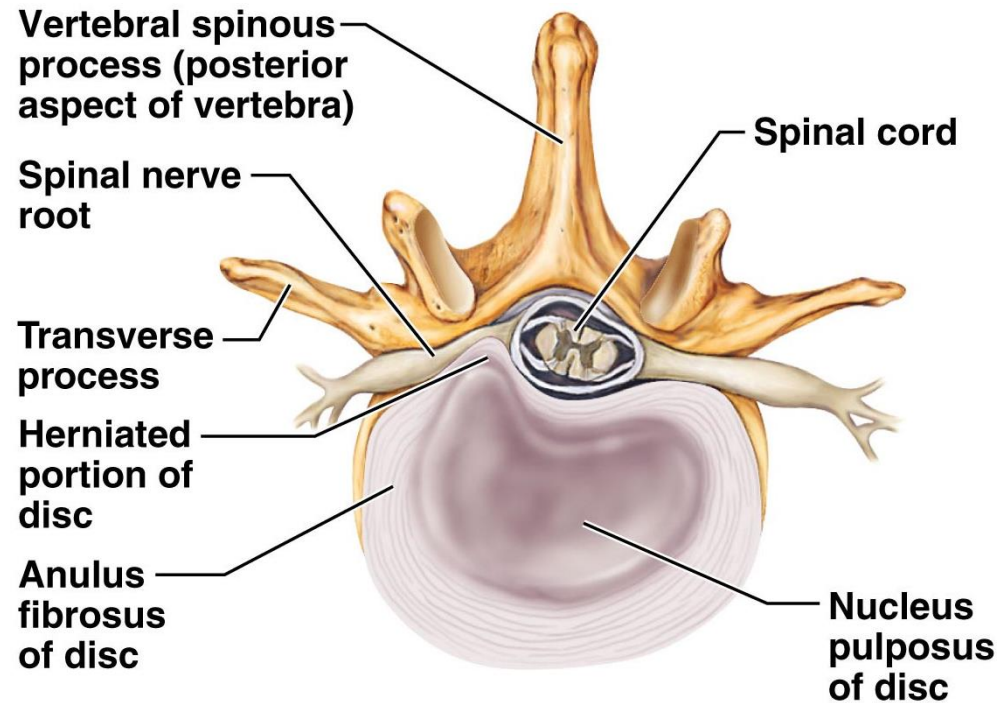
Intervertebral Discs

- Cushion-like pads between vertebrae
 - Is a fibrocartilage
- Act as shock absorbers
- Account for 20-25% of the height of the vertebral column
- Absent between C1 & C2, fused sacrum and coccyx
- Composed of inner **Nucleus pulposus** and outer **annulus fibrosus**



Herniated Disc/Prolapsed disc/Slipped disc

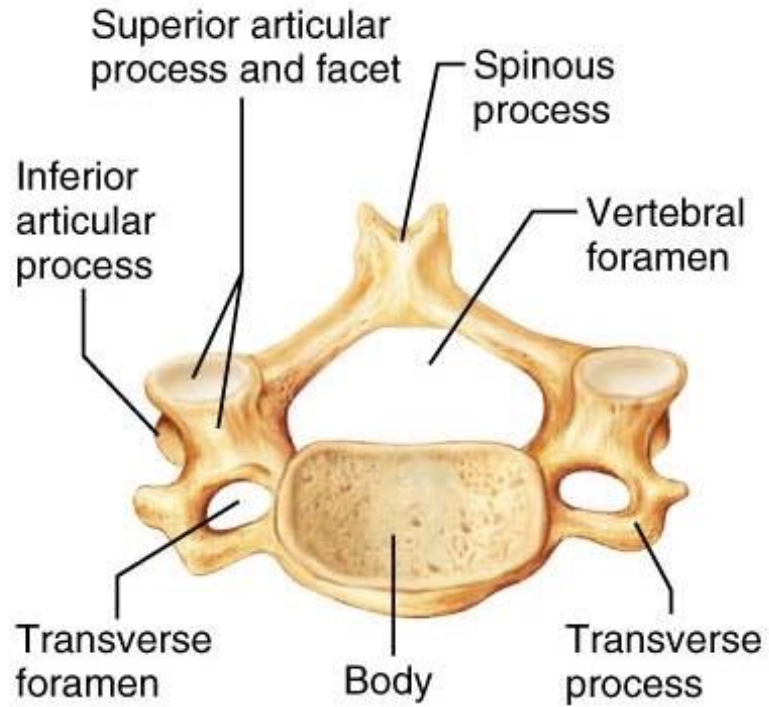
- Involves rupture of the annulus fibrosus followed by protrusion of the nucleus pulposus.
- The herniated part of the disk may push against and compress the spinal cord, cauda equina, or spinal nerves.
- **Occurs posterolaterally**—the posterior longitudinal ligament prevents the herniation from proceeding directly posteriorly.
- **Usually occurs in L4- L5 or L5-S1; may result in sciatica**



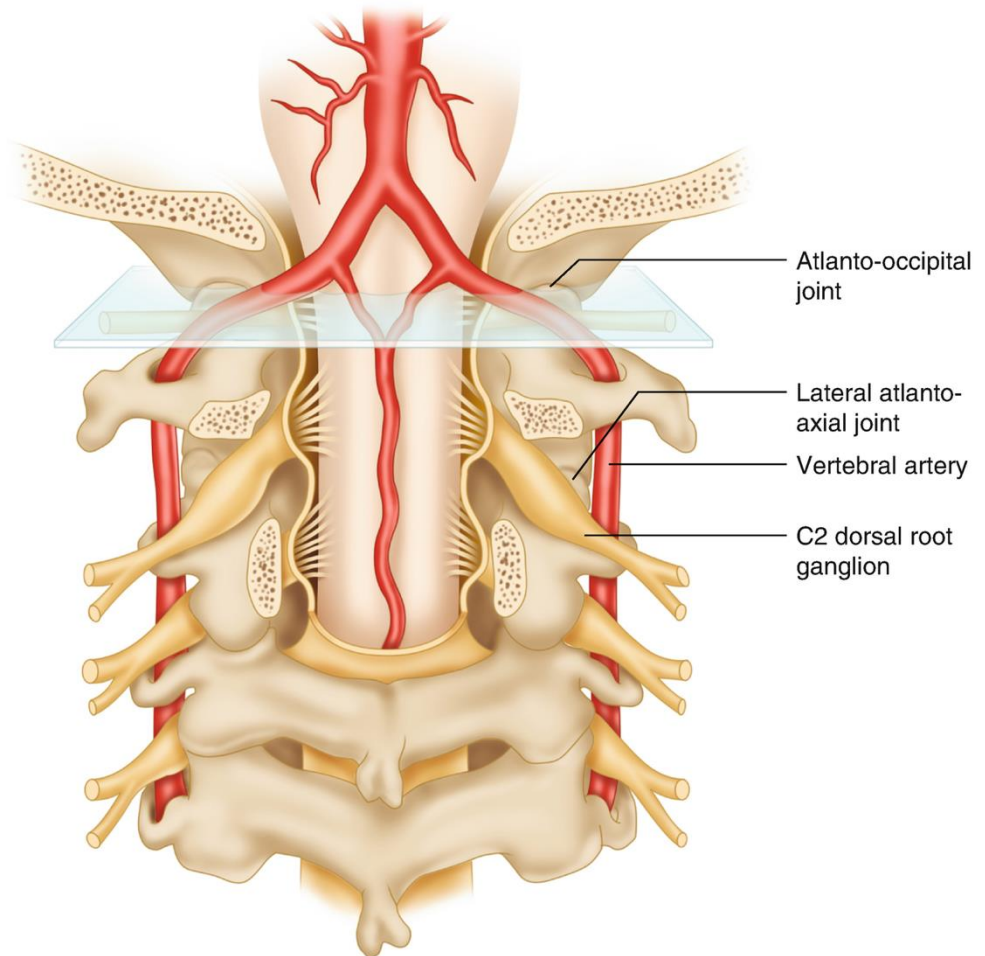
(c)

- May be caused by trauma to the spine. Aging is also a contributing factor: nucleus pulposus loses cushioning properties, annulus fibrosus weakens

Cervical Vertebrae



(a) Cervical





Cervical vertebrae

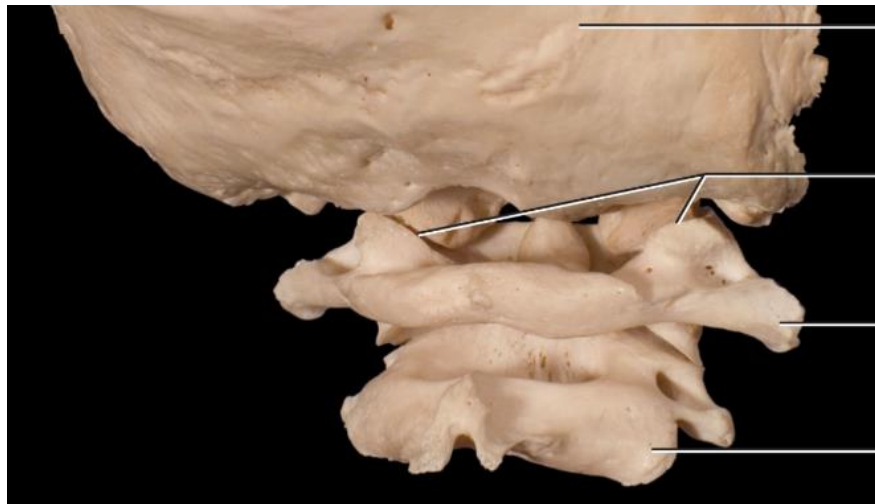
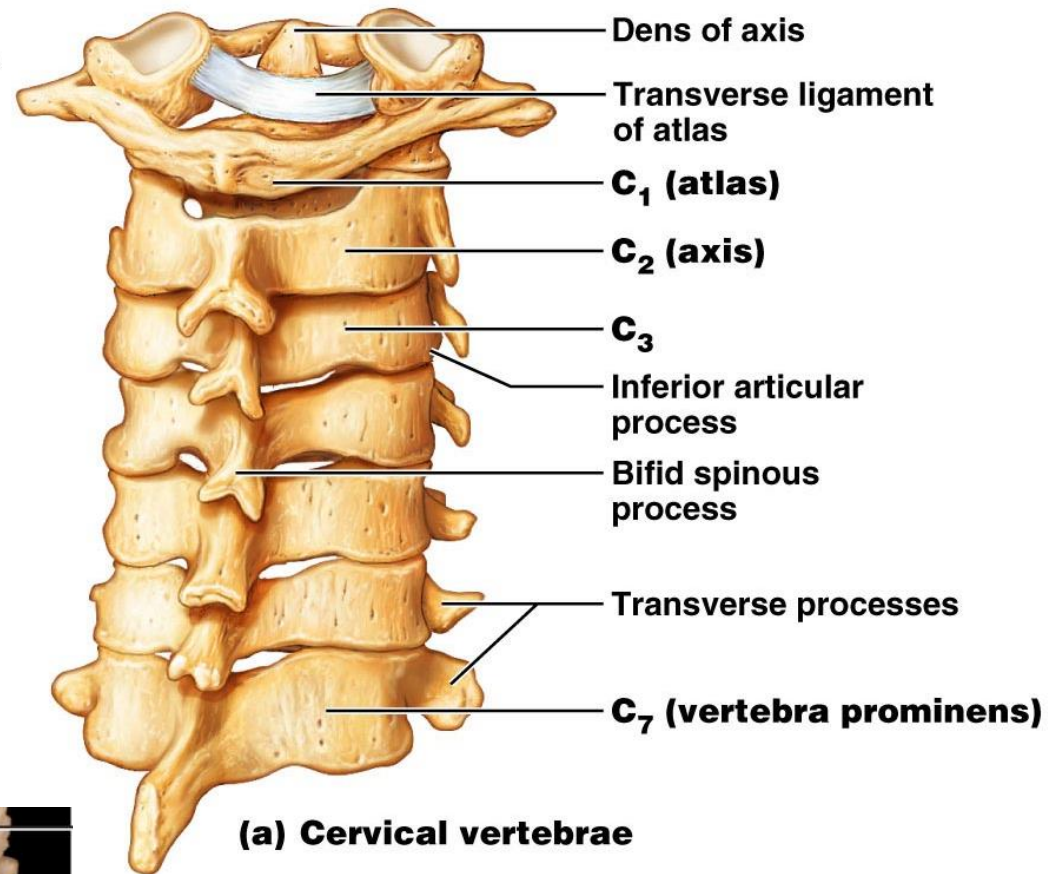
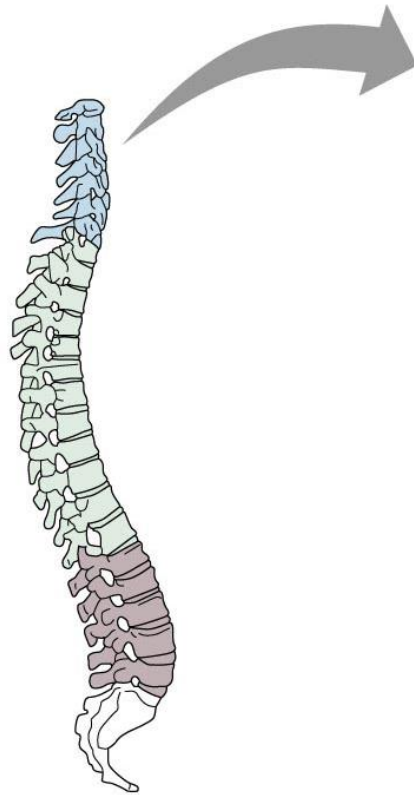
Atlas

- Lacks a body and spinous process
- Superior articular facets articulate with the occipital condyles to form atlanto-occipital joint (Condylod joint) that permits **nodding of the head “yes”**

Axis

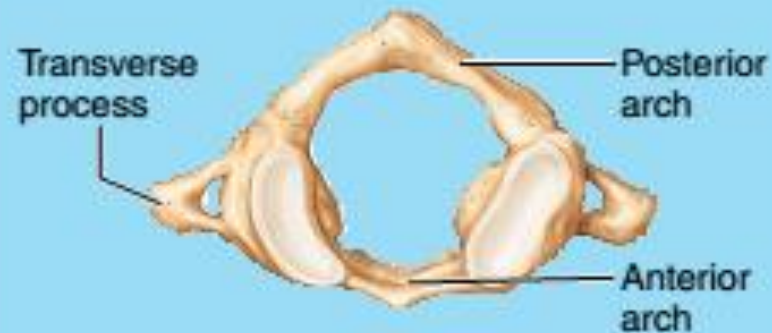
- Has a **dens or odontoid process**. The den articulates with atlas (pivot joint) to help **rotate the head**. The transverse ligament secures the dens
- **C₇, vertebra prominens**, has a long spinous process. Is used as landmark in counting vertebra

Cervical Vertebrae

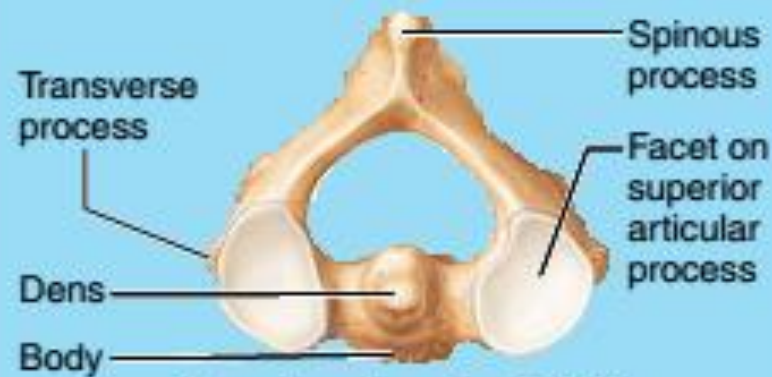


Atlanto-occipital joint

(a) ATLAS AND AXIS

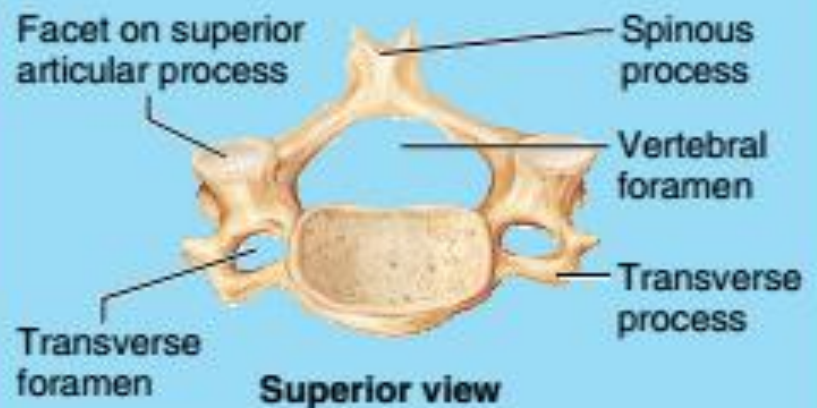


Superior view of atlas (C₁)

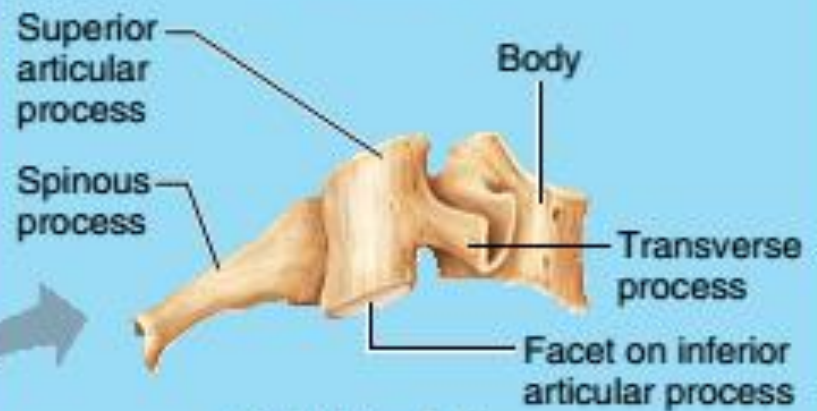


Superior view of axis (C₂)

(b) TYPICAL CERVICAL VERTEBRAE

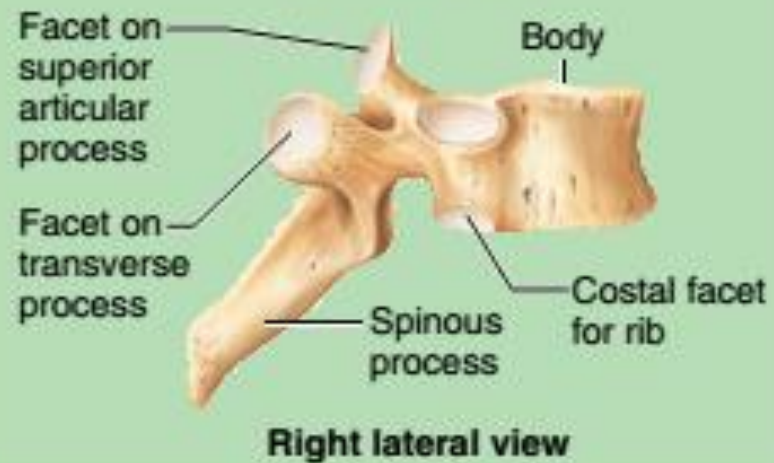
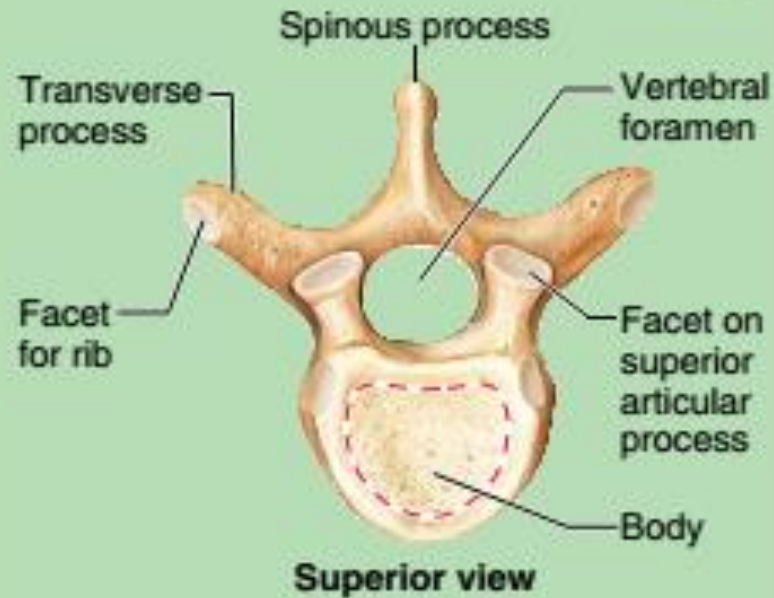


Superior view

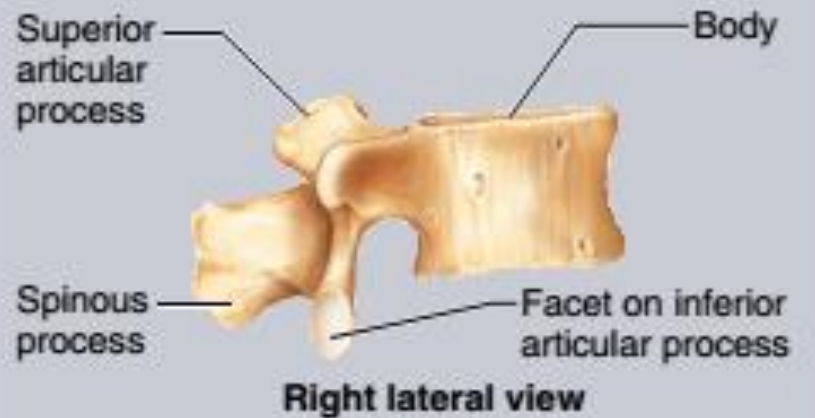
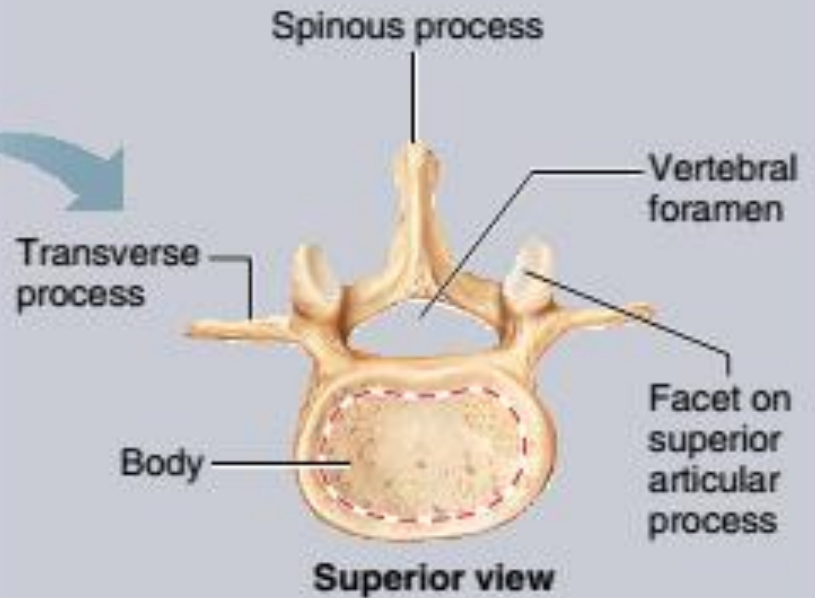


Right lateral view

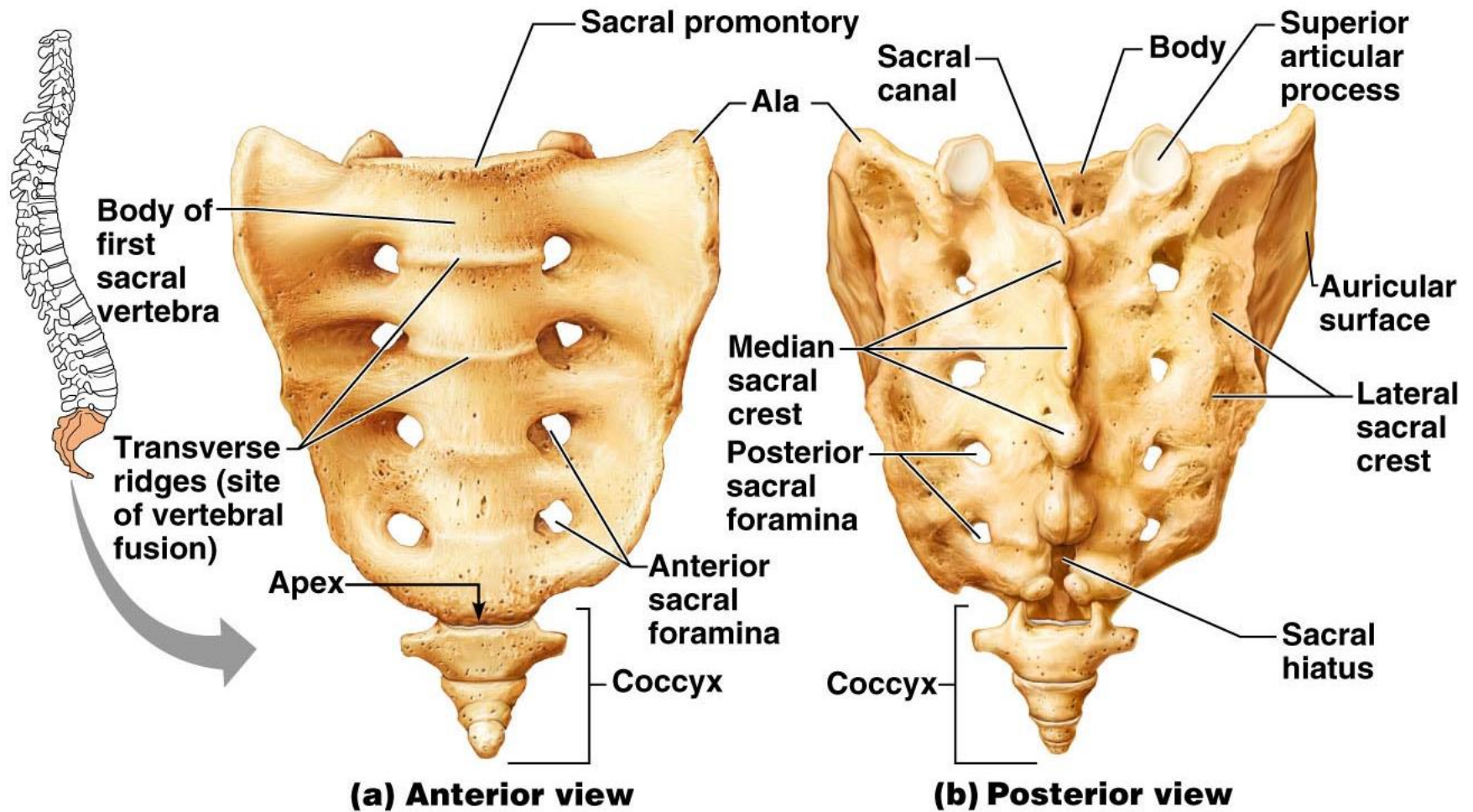
(c) THORACIC VERTEBRAE



(d) LUMBAR VERTEBRAE



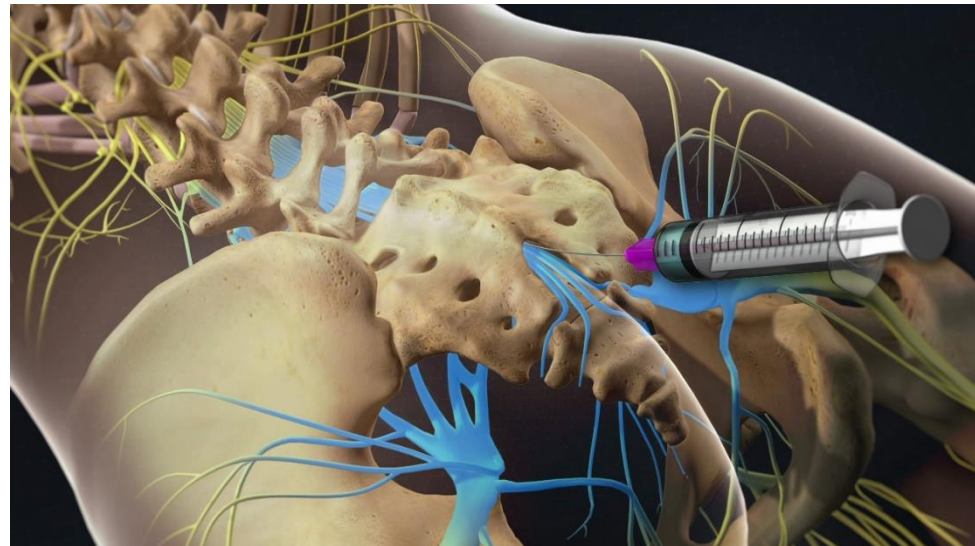
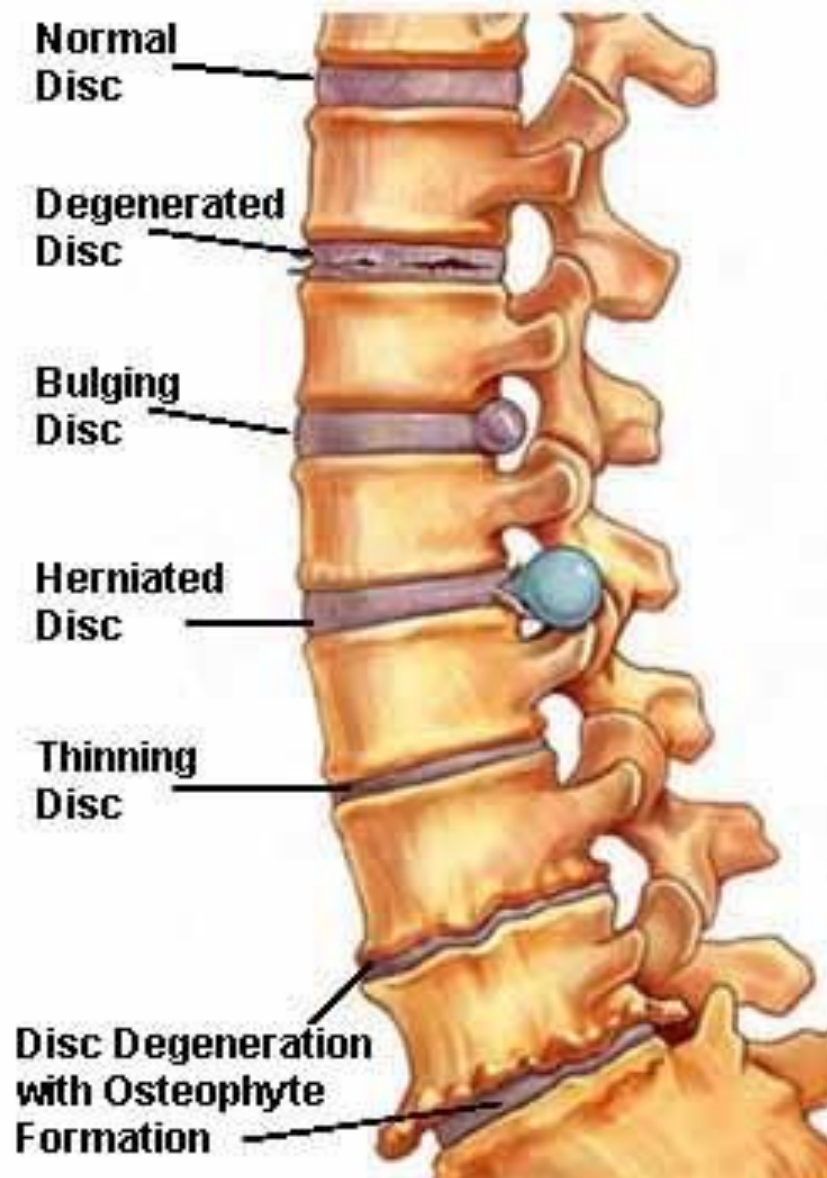
Sacrum and coccyx



Comparison of Cervical, Thoracic, and Lumbar Vertebrae

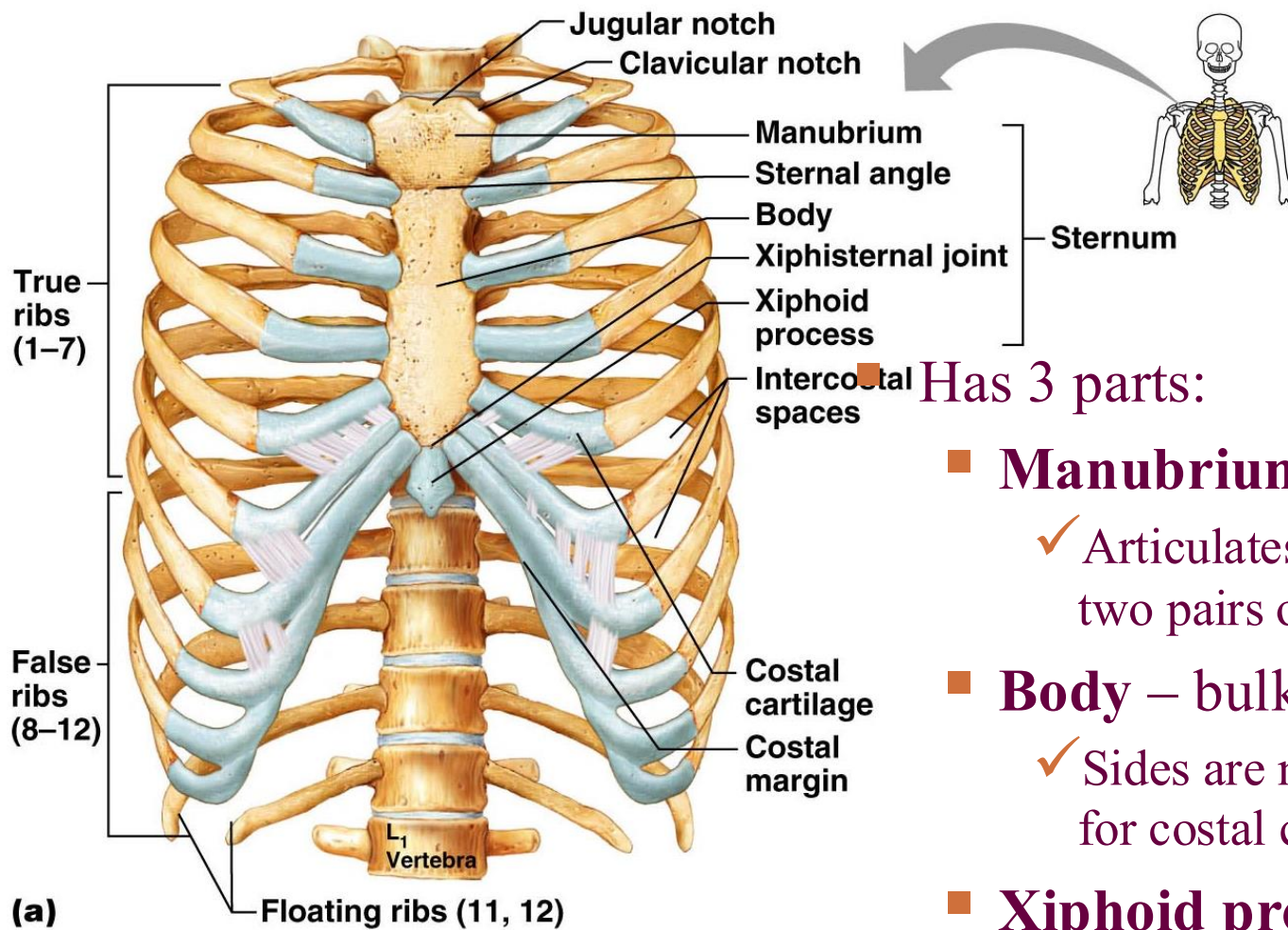
Characteristic	Cervical vertebrae	Thoracic vertebrae	Lumbar vertebrae
Body shape and size	Small and oval; C ₁ lacks a body; C ₂ has the dens on the superior surface of its body	Larger and heart-shaped; contain costal facets	Largest and kidney-shaped
Vertebral foramen shape	Triangular	Circular	Flattened triangular
Transverse processes	Contain transverse foramina	Long; contain articular facets for ribs	Short with no facets or foramina
Spinous processes	Most are fork-shaped; C ₁ lacks a spinous process	Long; point inferiorly	Thick; point posteriorly
Appearance (superior view)			

Examples of Disc Problems



Epidural block

Sternum and Ribs



Has 3 parts:

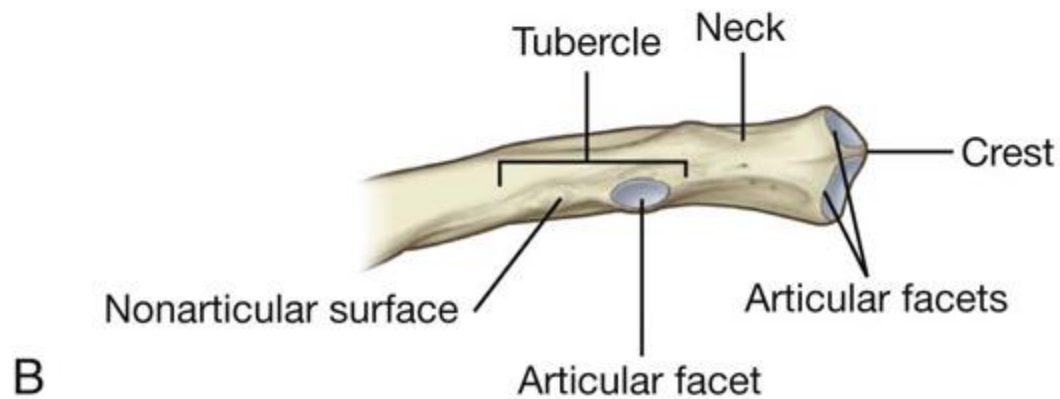
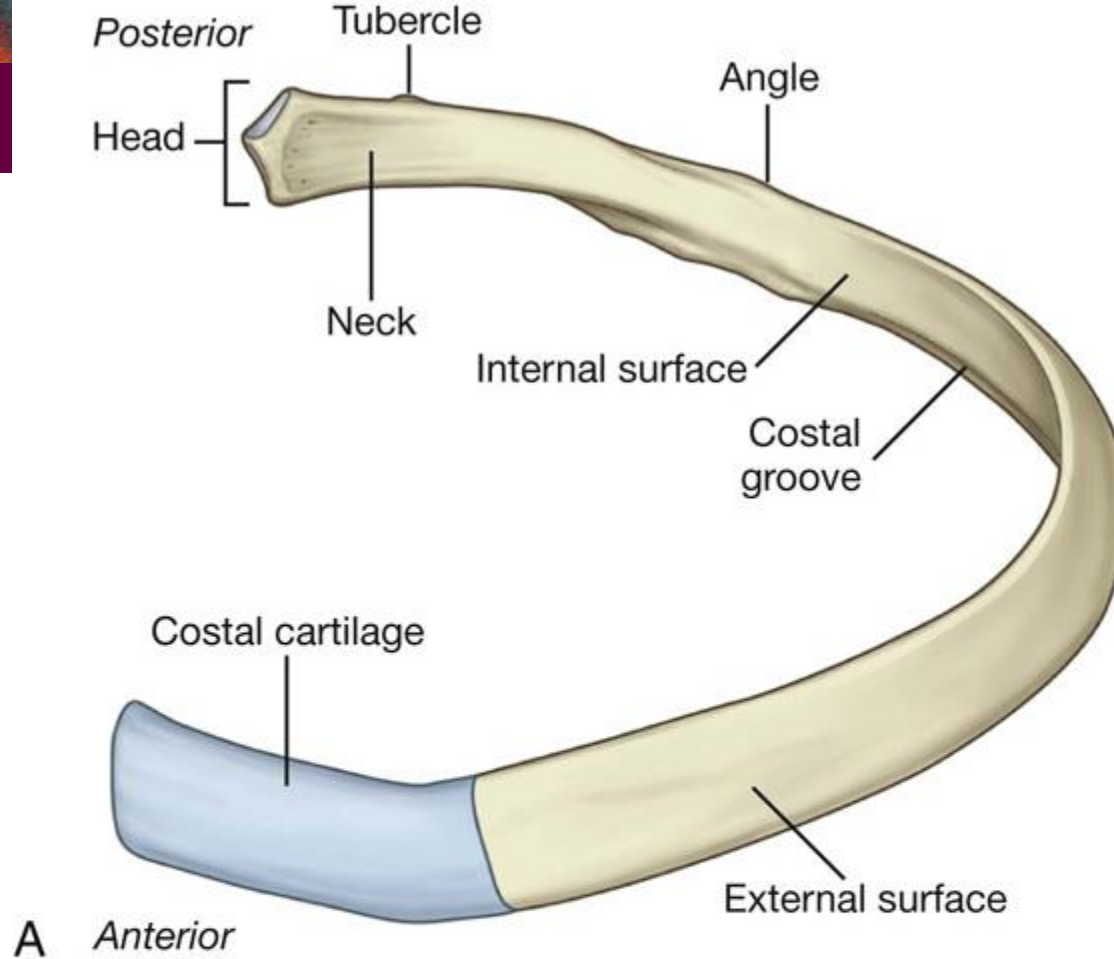
- **Manubrium** – superior part
 - ✓ Articulates with clavicles and 1st two pairs of ribs
- **Body** – bulk of sternum
 - ✓ Sides are notched at articulations for costal cartilage of ribs 2–7
- **Xiphoid process** – inferior end of sternum, may be bifid or perforated
 - ✓ Hyaline cartilage in youth
 - ✓ Ossifies around age 40

Ribs

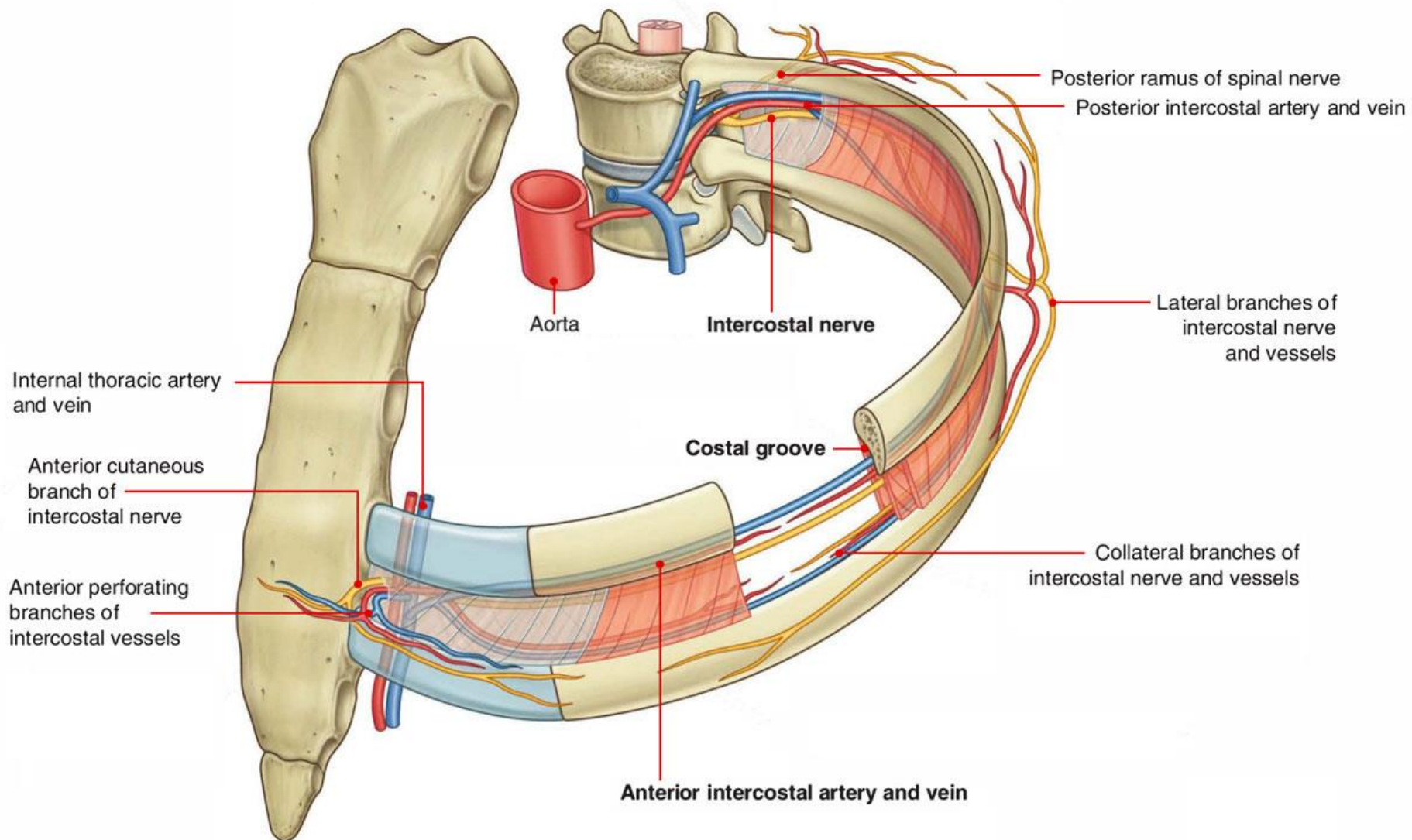
- 12 pairs. All ribs articulate with thoracic vertebrae posteriorly
- 3-9 typical; rest are atypical
 - **True ribs (vertebrosternal ribs)** – upper 7 pairs of ribs
 - Articulate directly with sternum anteriorly
 - **False ribs** – lower 5 pairs of ribs. Consists of
 - **vertebrochondral ribs:** ribs 8-10
 - are joined by a common cartilage to the costal cartilage of the seventh rib, which in turn is attached to the sternum
 - **vertebral/floating ribs:** ribs 11–12.
 - They do not attach to the sternum
- **Costal groove** at inferior border of inner surface
 - contain intercostal vein, arteries and nerve – **VAN**

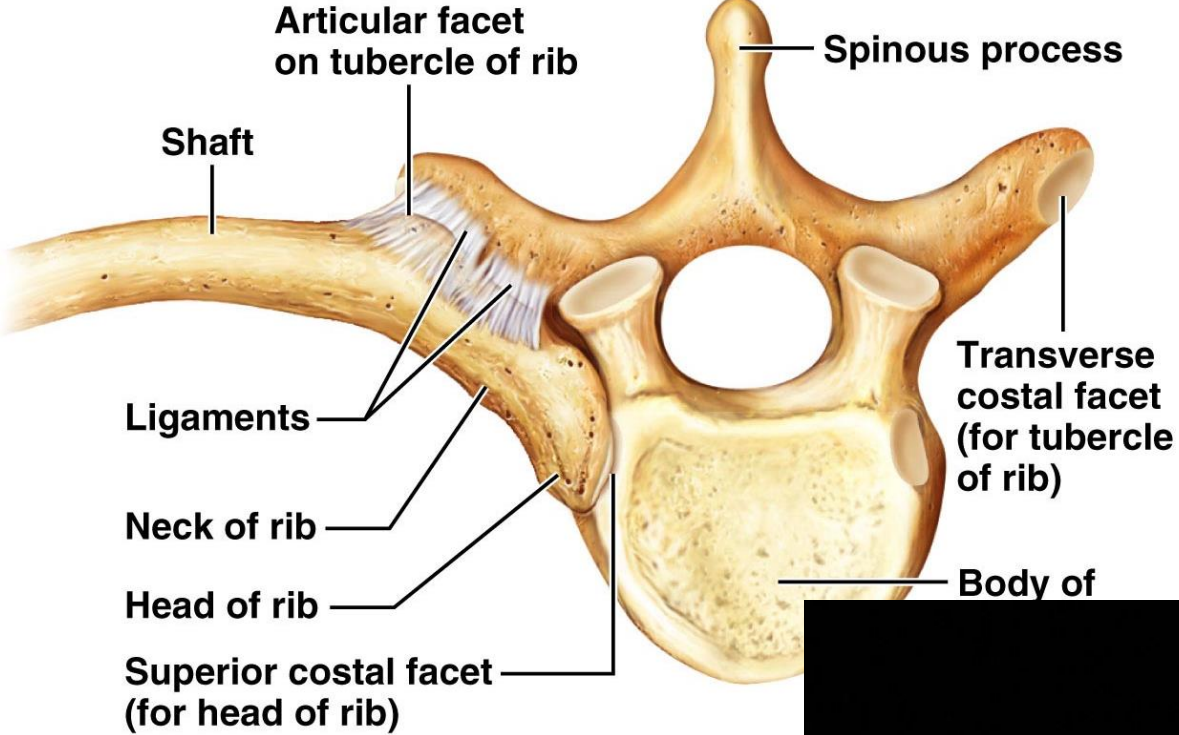


Ribs

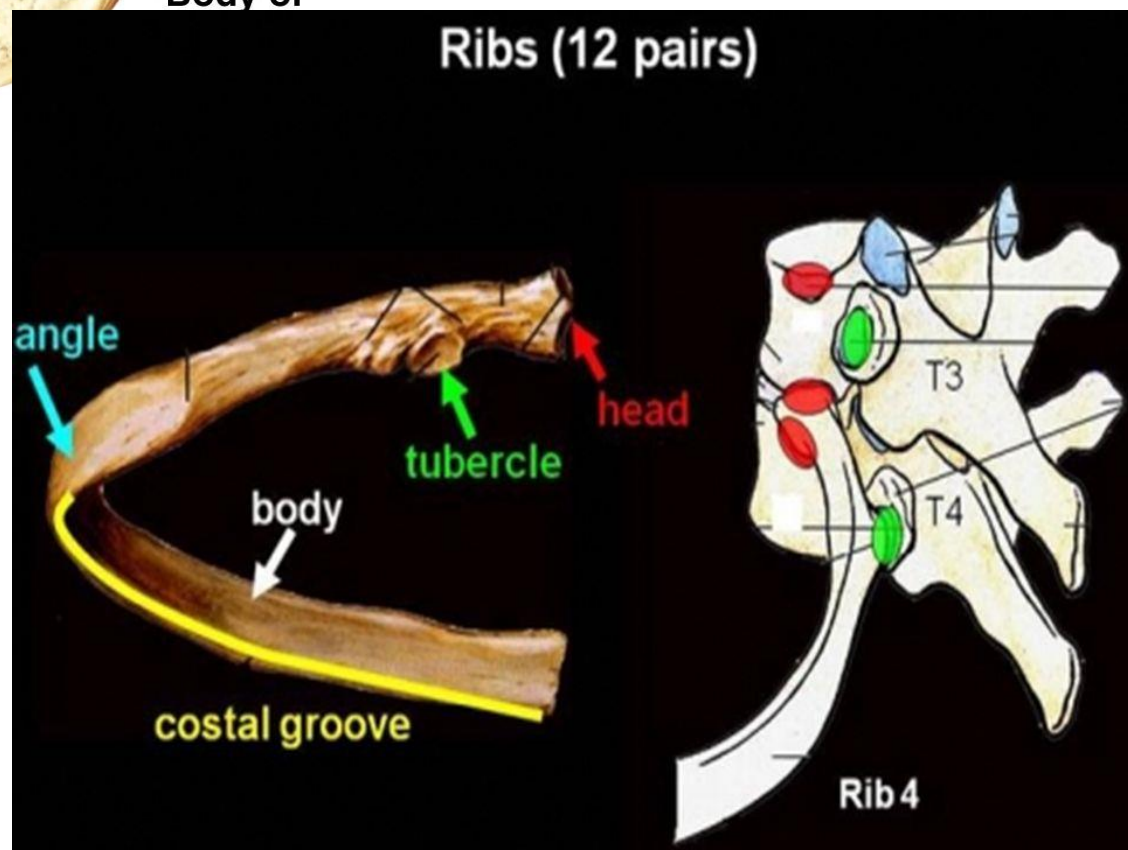


Ribs

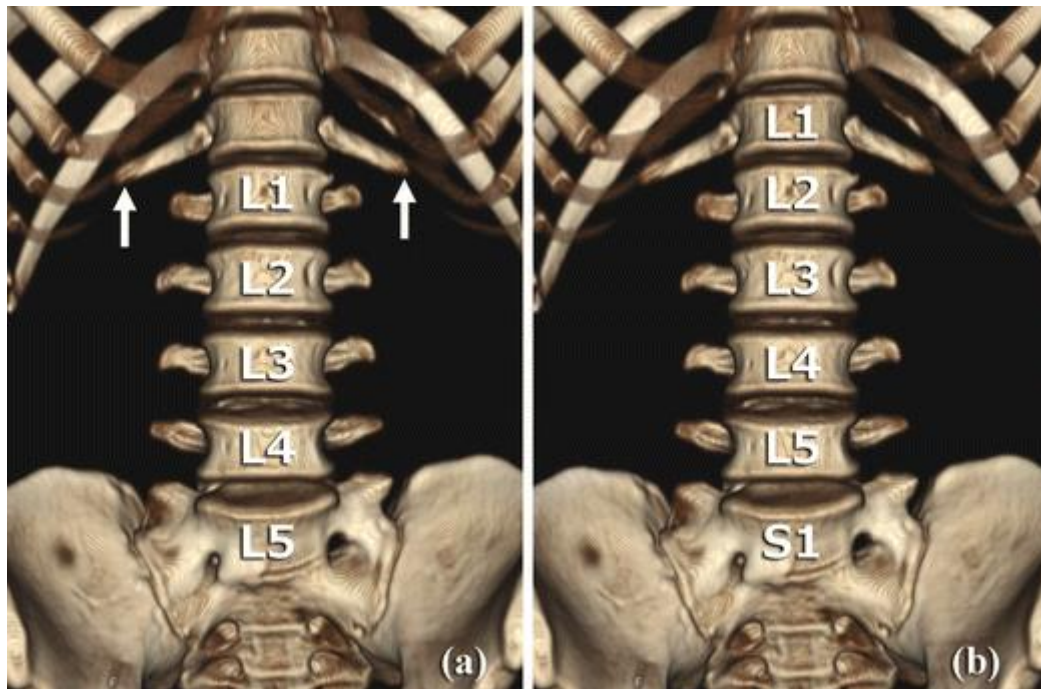




(b)

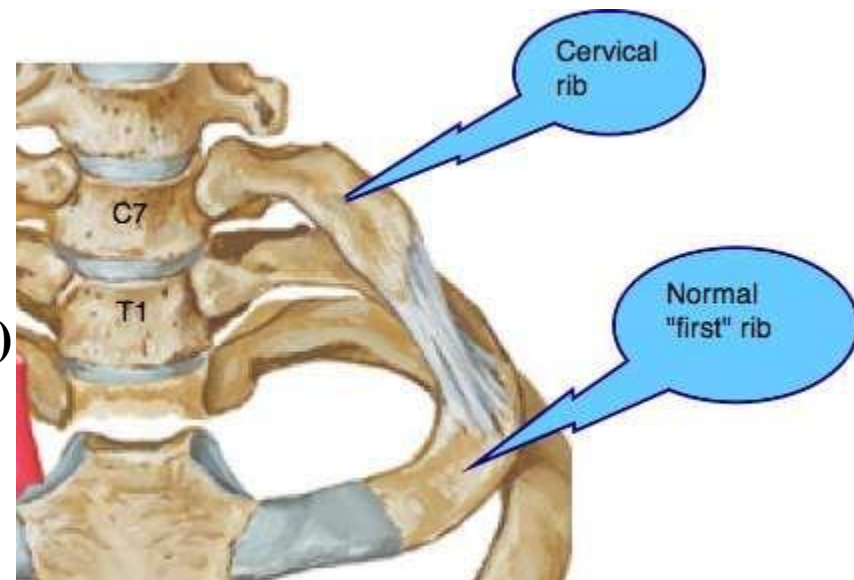


Ribs



Normal

Lumbar rib (Gorilla rib)

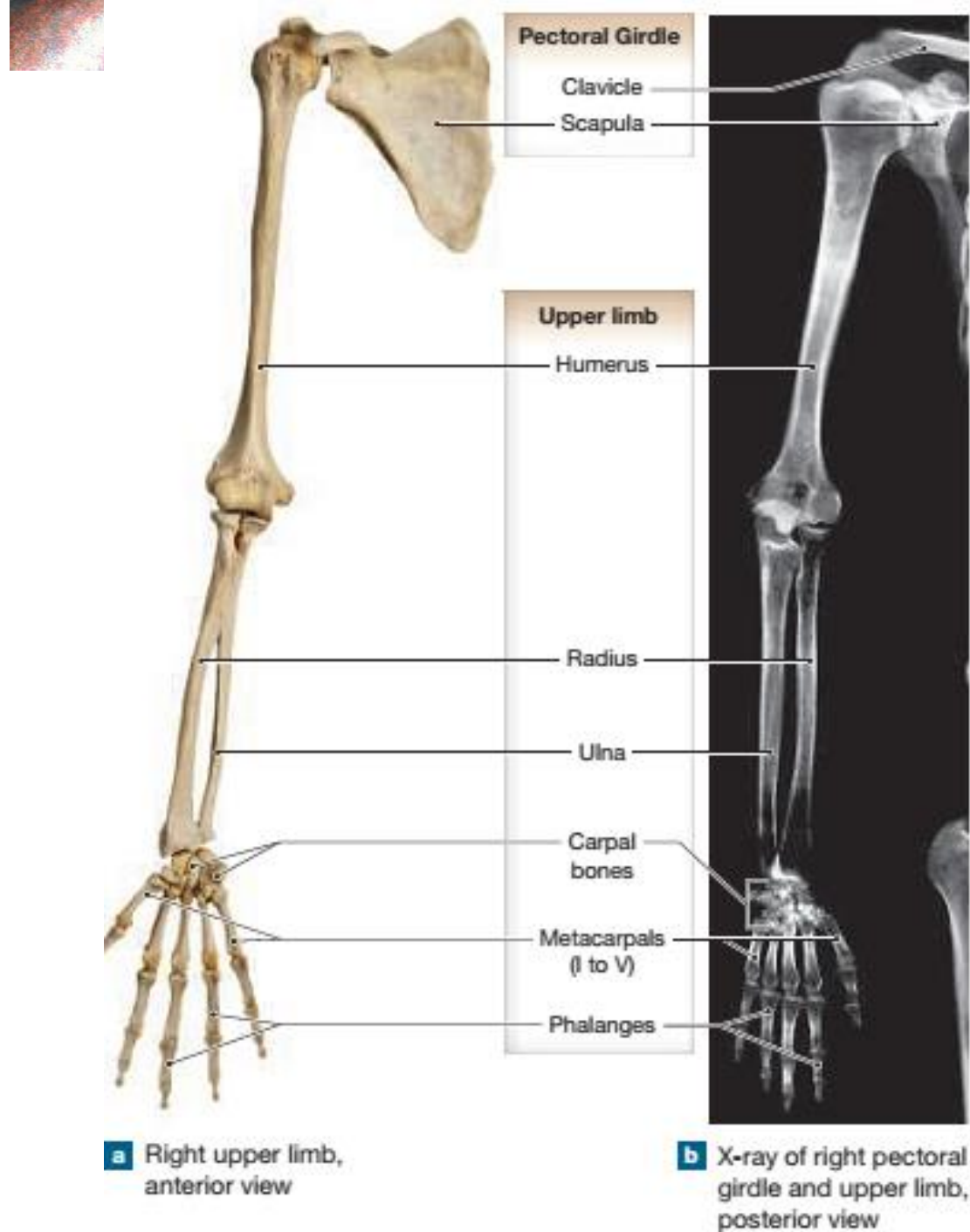
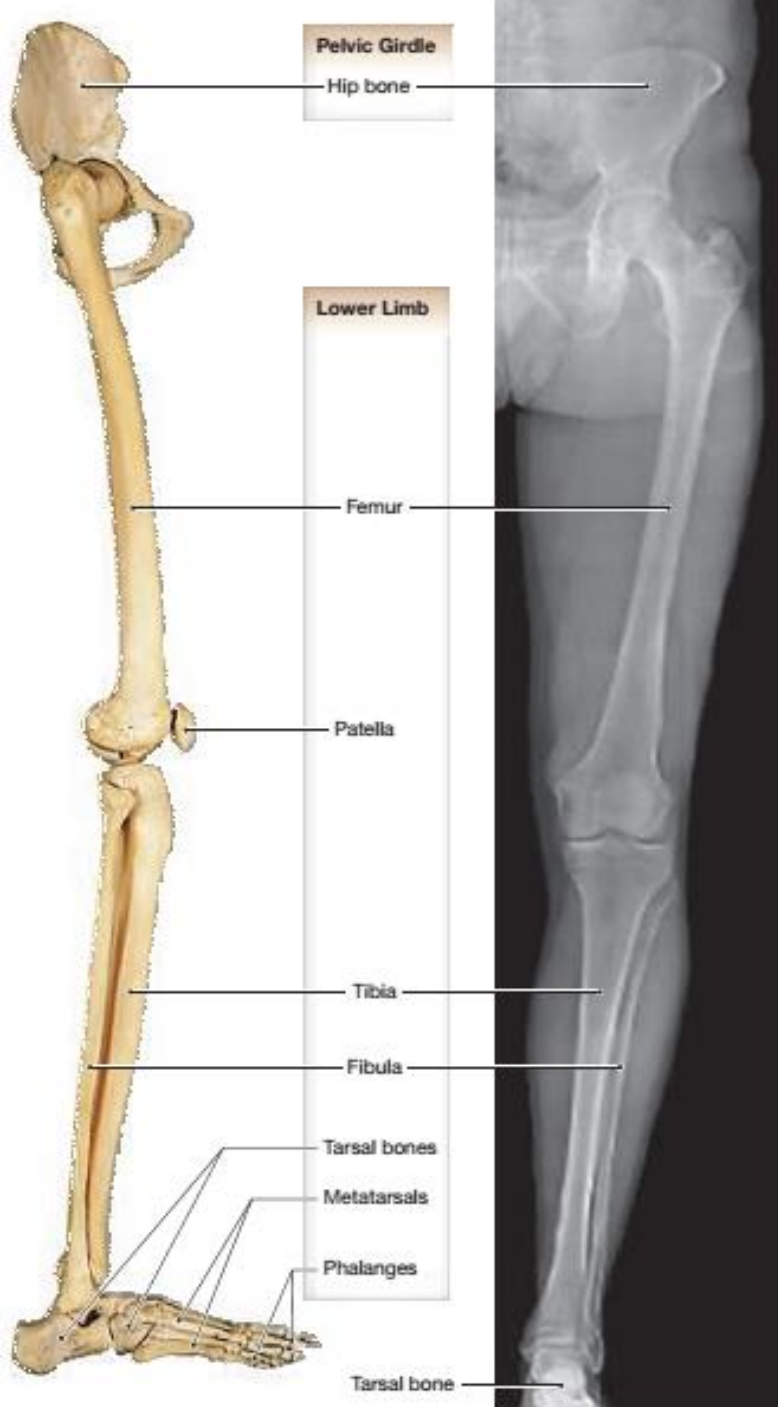


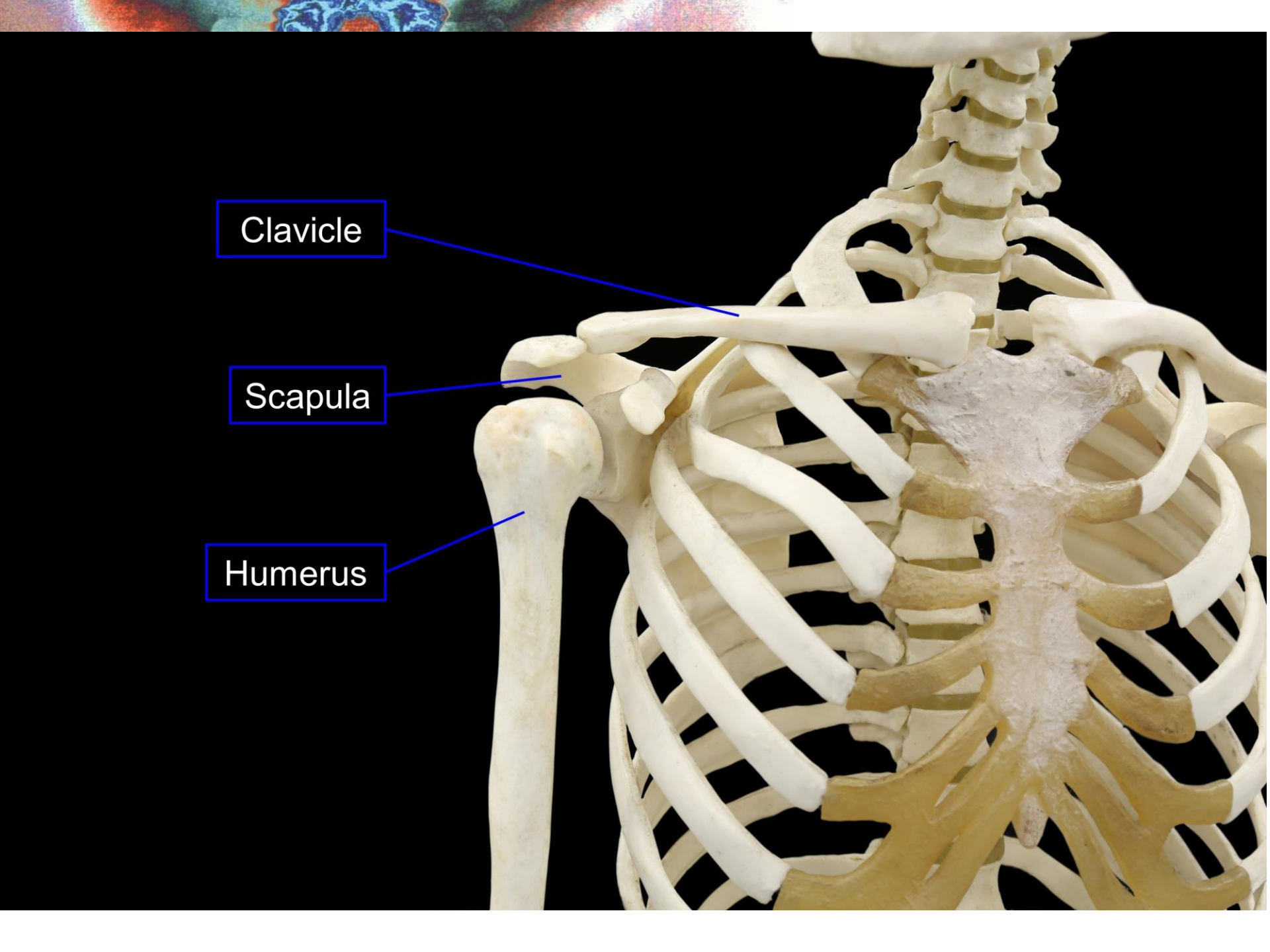
Cervical rib

*Thoracic outlet syndrome,
Flail chest (stove-in-chest)*



The Appendicular Skeleton



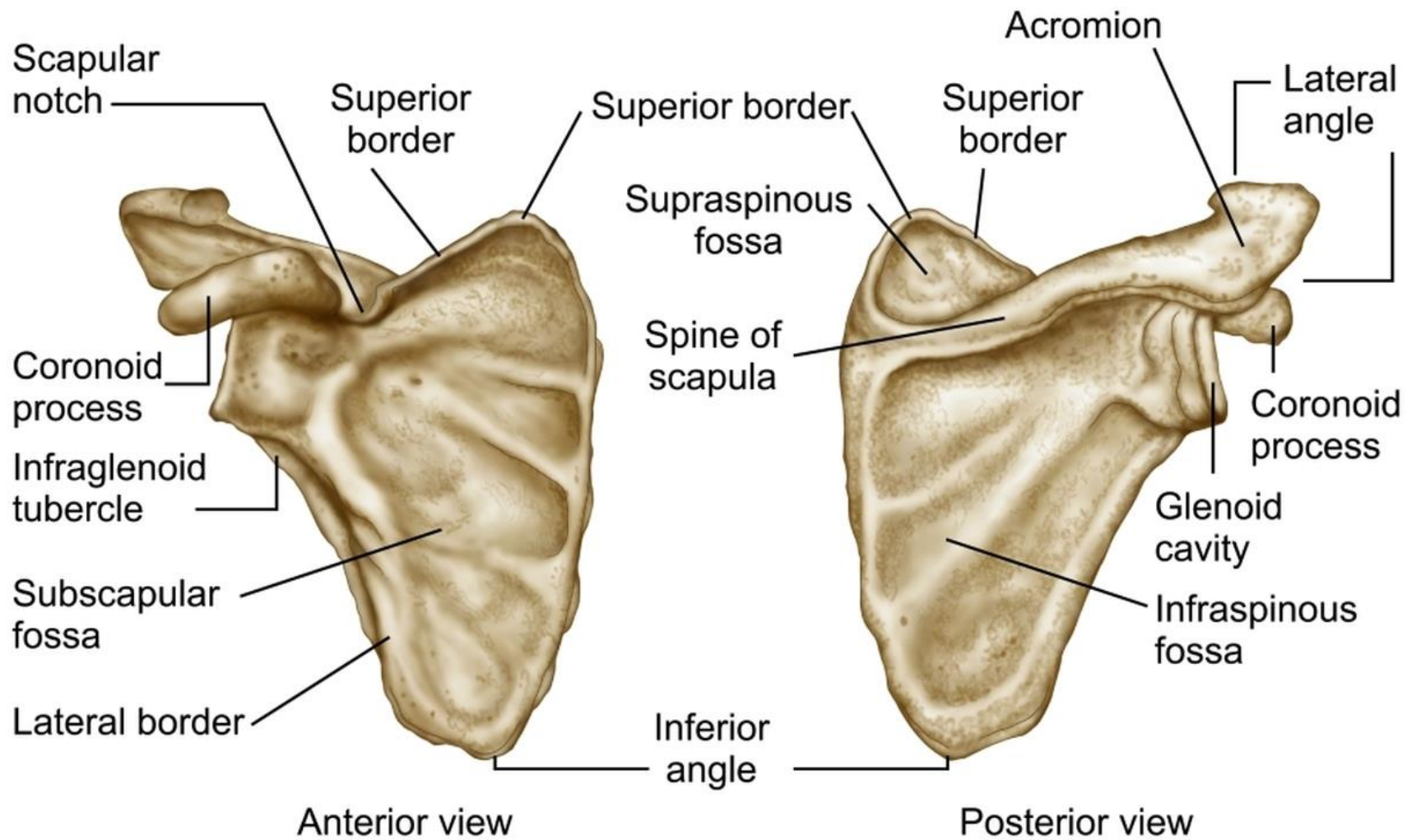


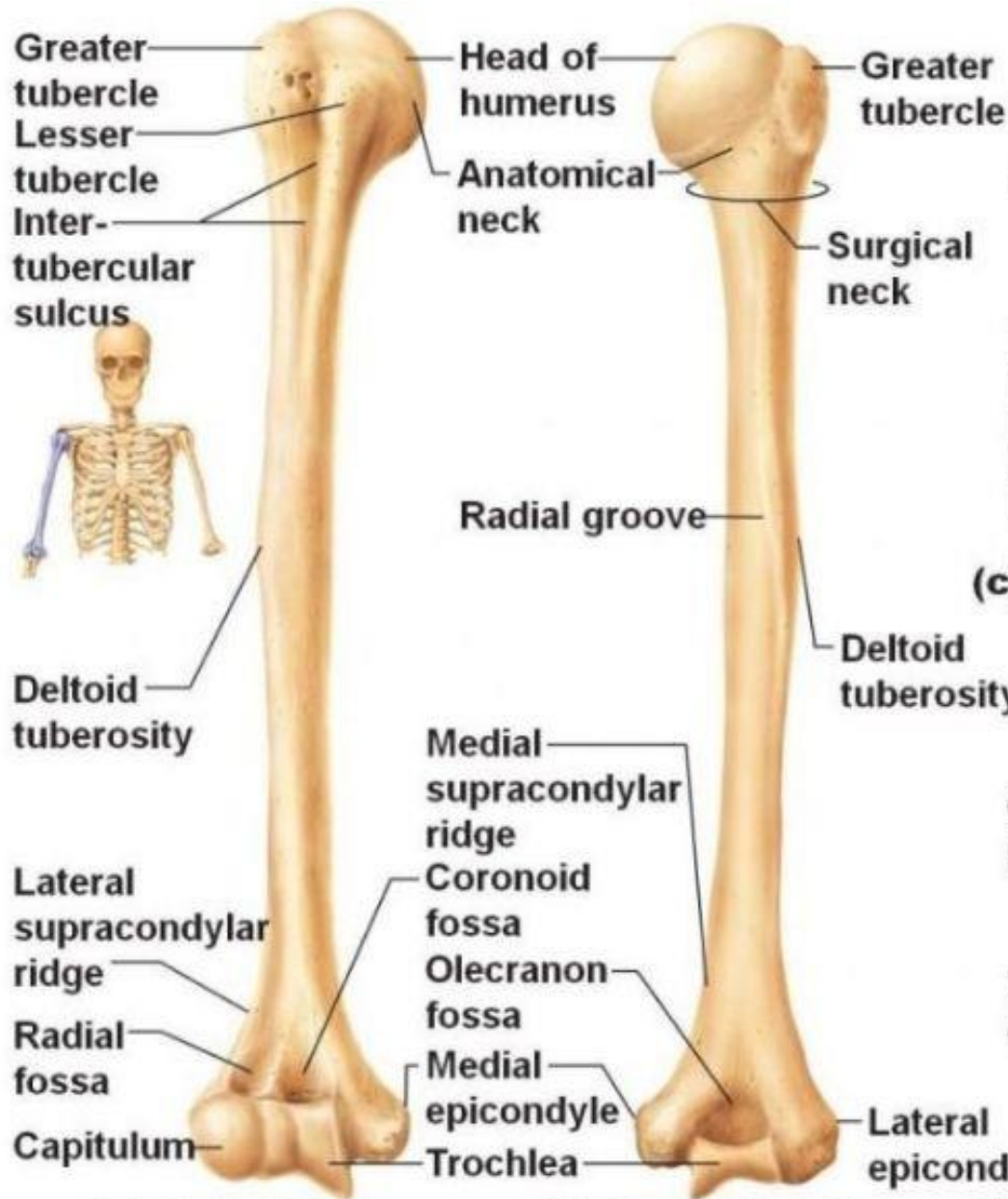
Clavicle

This anatomical diagram illustrates the upper human skeleton. The clavicle (collarbone) is shown as a long, slightly curved bone connecting the sternum to the scapula. The scapula (shoulder blade) is a triangular bone located on the back. The humerus (upper arm bone) is shown extending downwards from the shoulder joint. The rib cage and spine are also visible in the background.

Scapula

Humerus



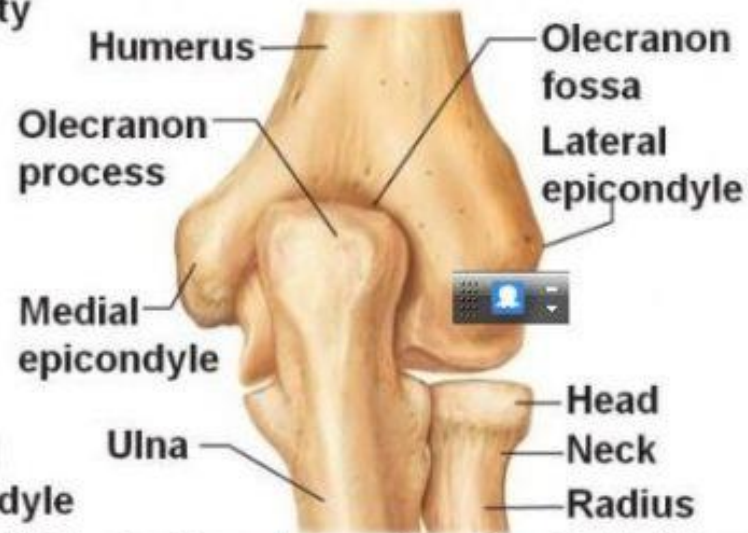


(a) Anterior view

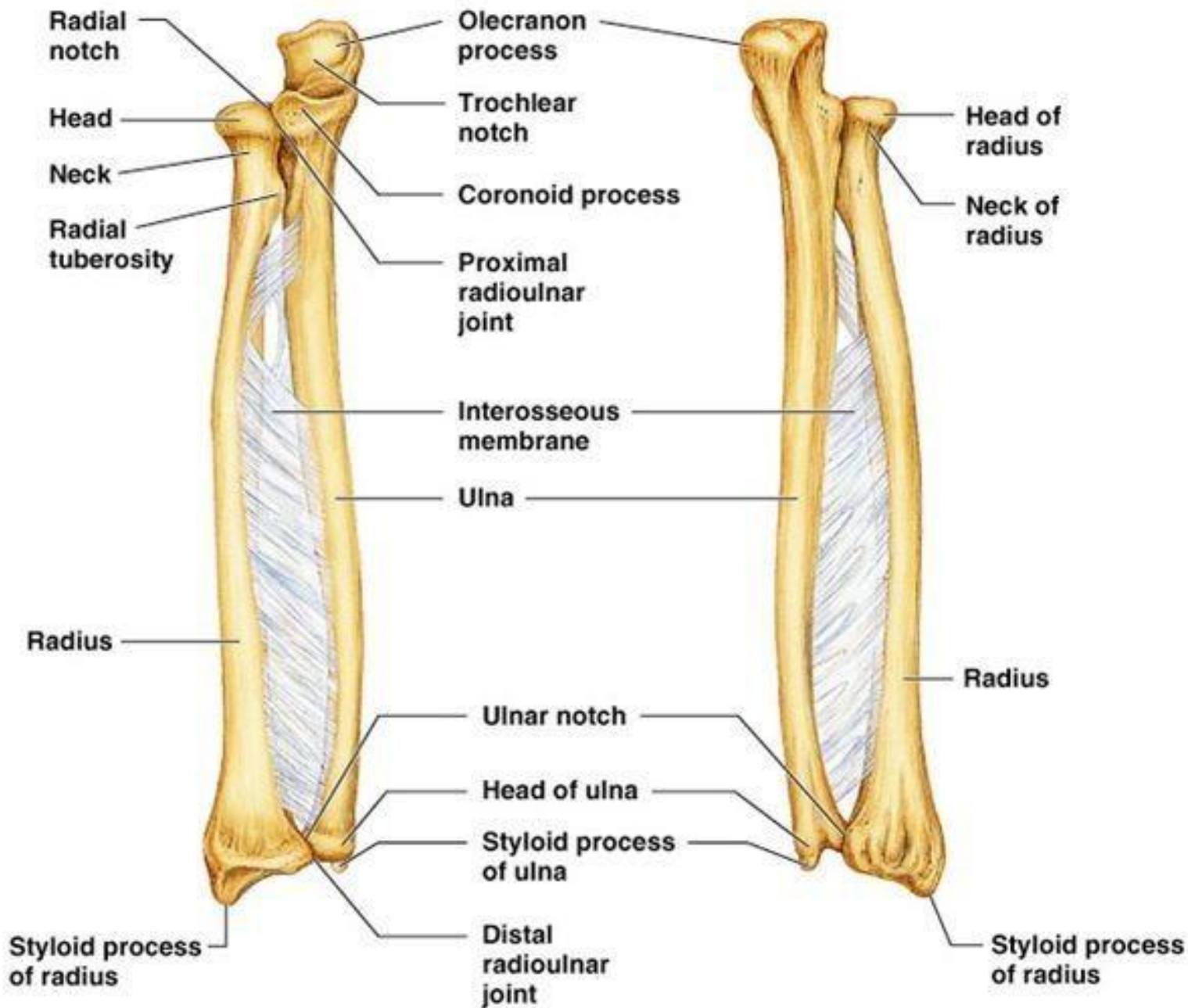
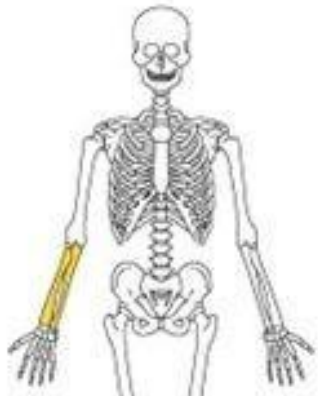
(b) Posterior view



(c) Anterior view at the elbow region



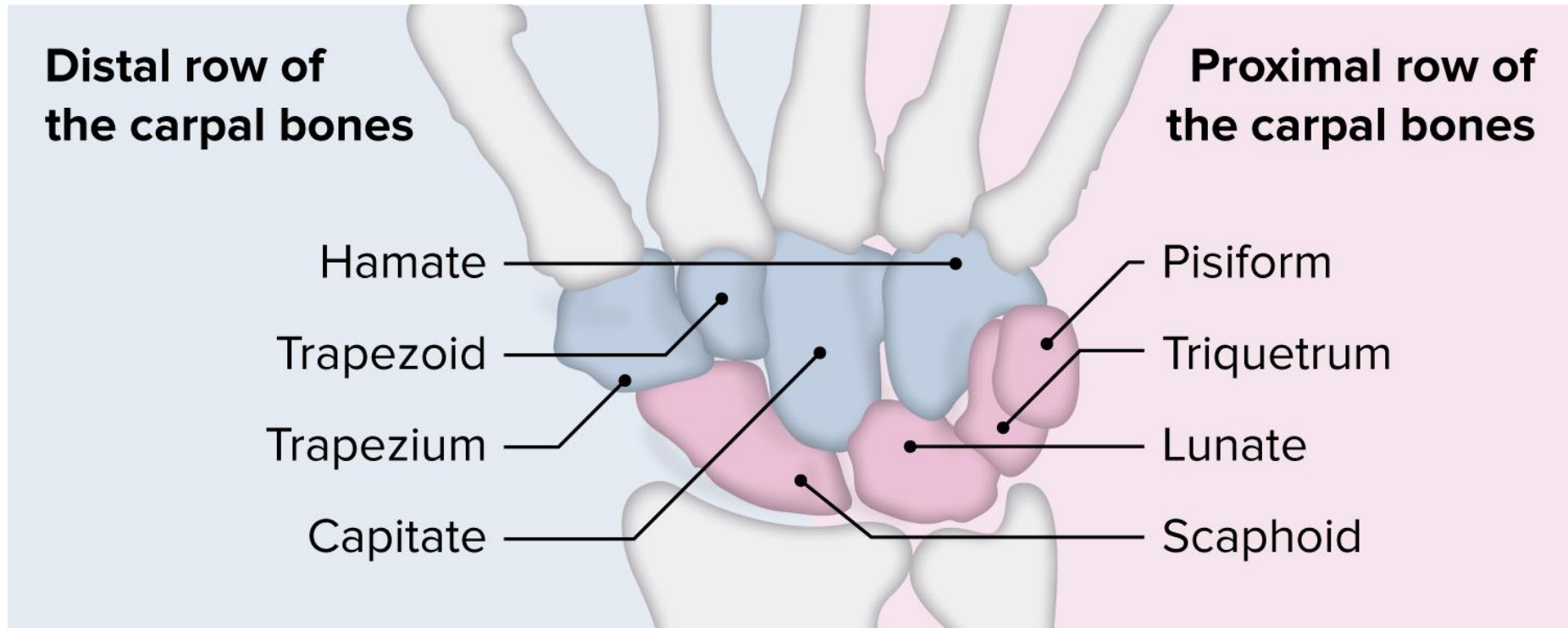
(d) Posterior view of extended elbow



(a) Anterior view

(b) Posterior view

Carpal bones

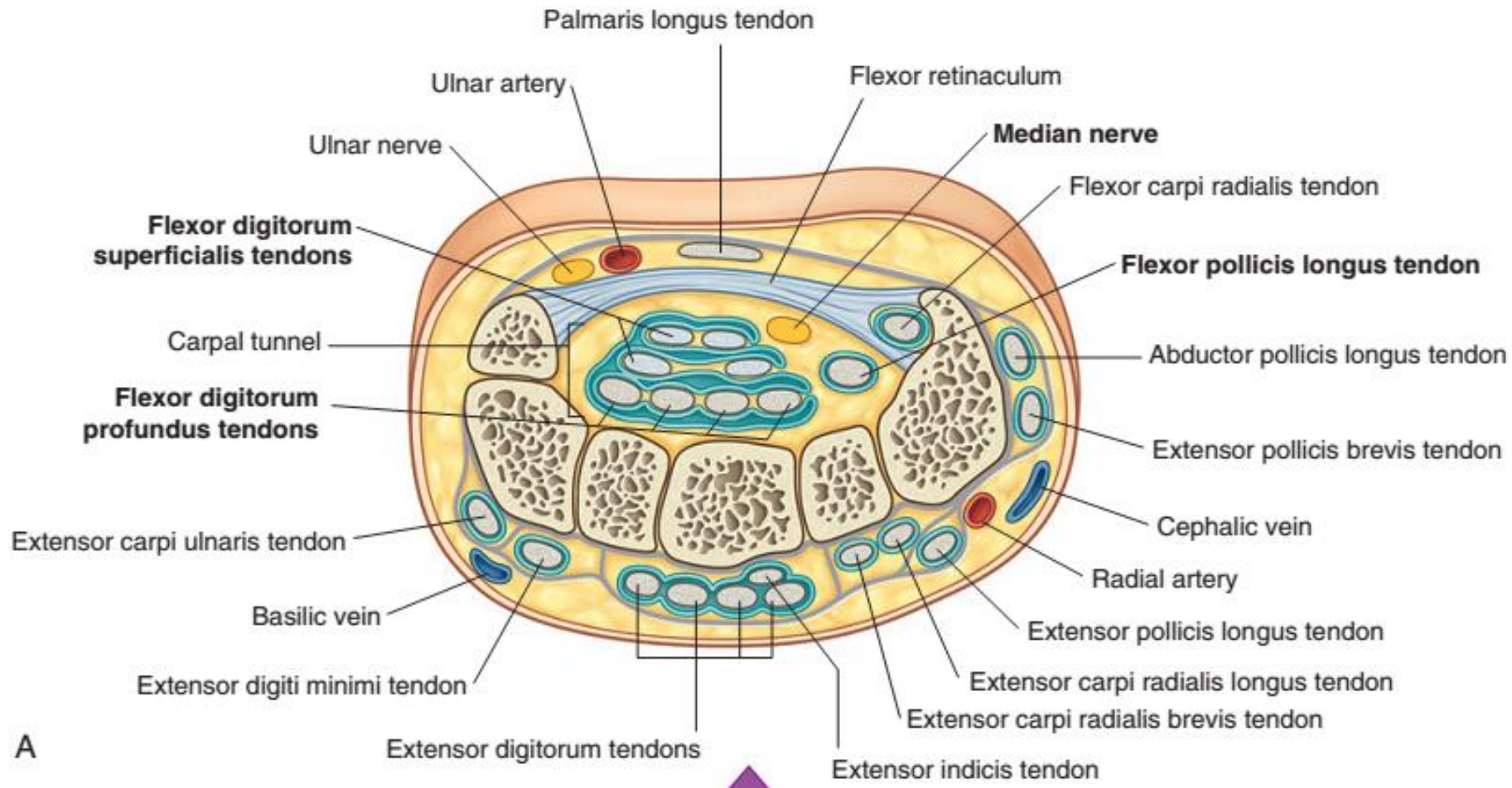


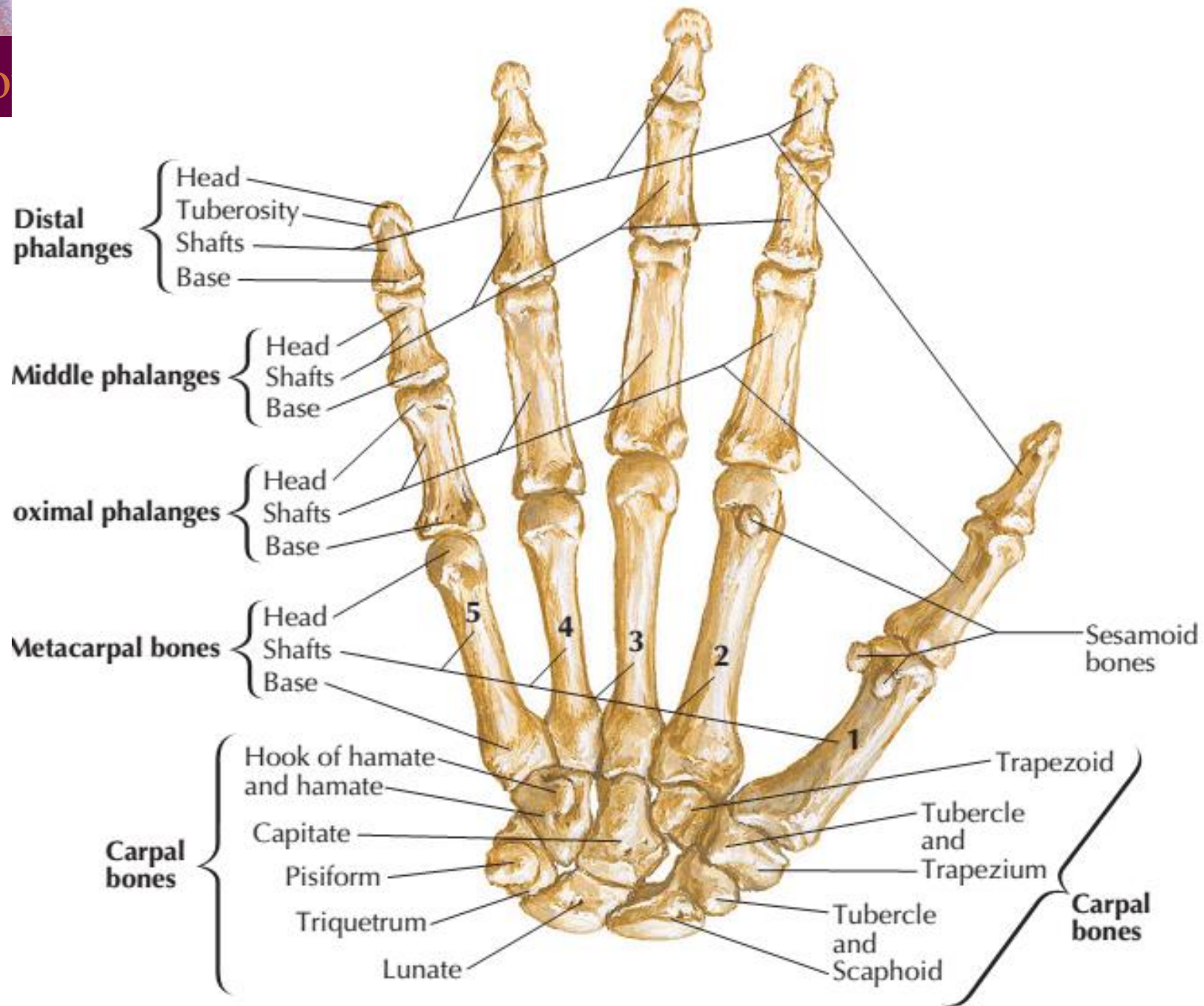
From lateral to medial

“Some Lovers Try Positions, That They Can’t Handle”

“So Luke Took Peter, To The Church Hall”

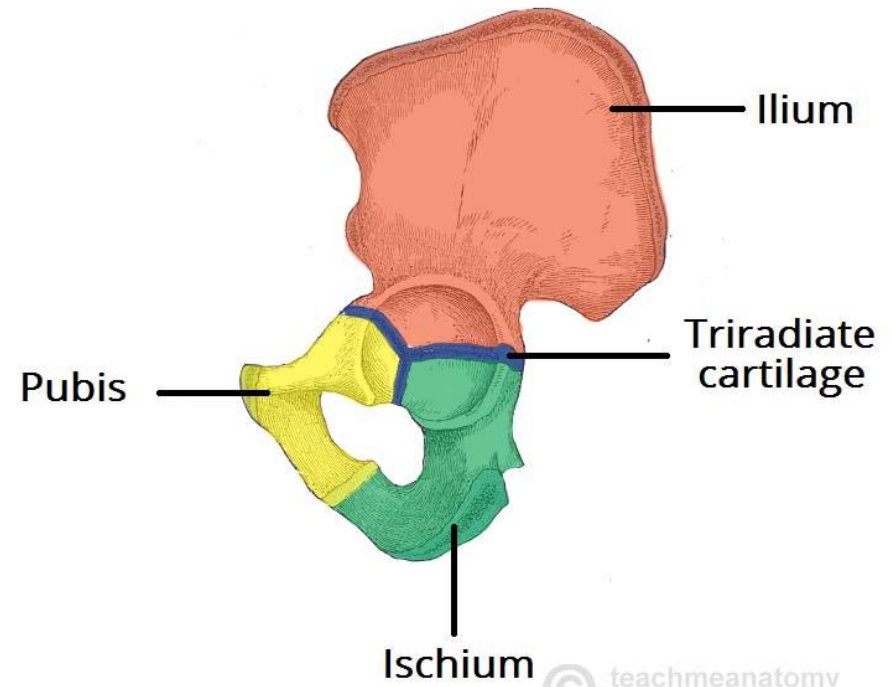
Carpal tunnel





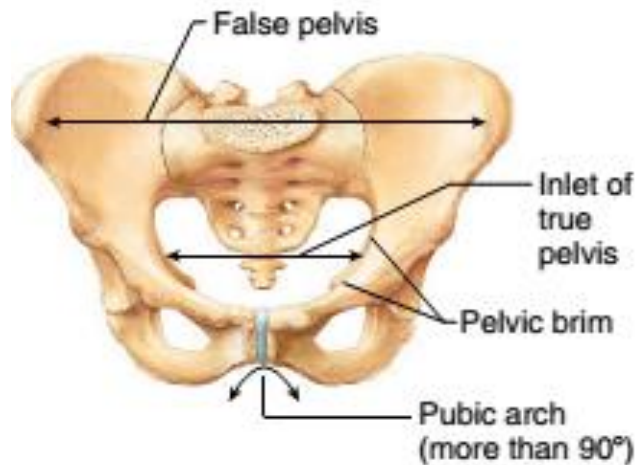
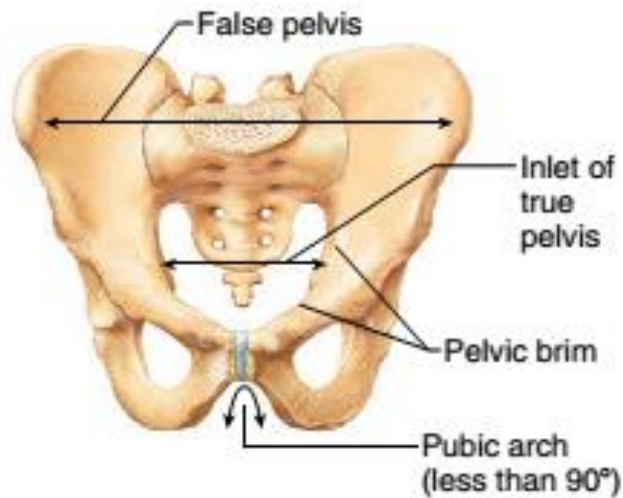
Hip bones (coxal bones or os coxae or innominate bones)

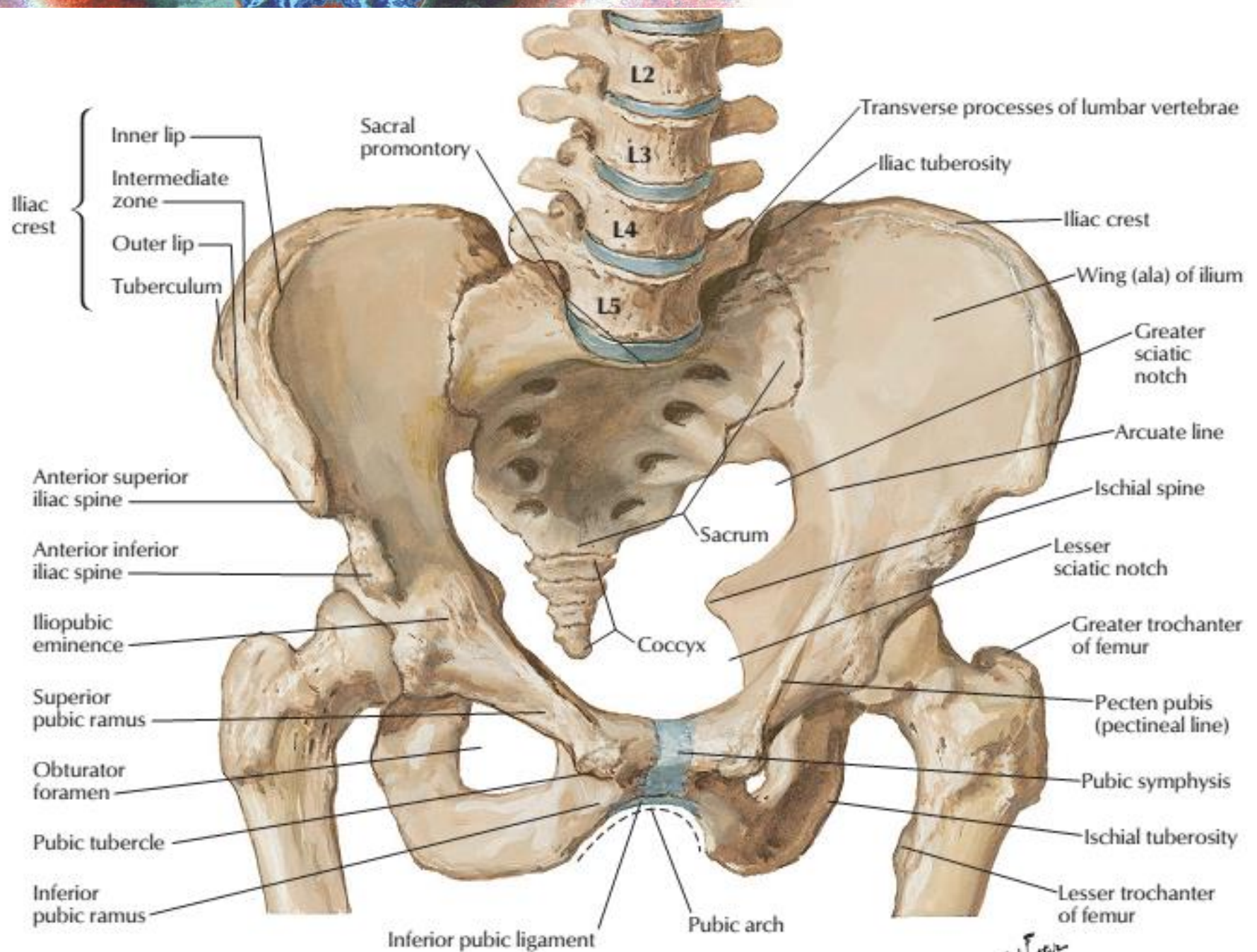
- Each hip bone consists of three parts: **ilium**, **ischium** and **pubis**. They fuse at the acetabular fossa (**acetabulum**).
- The 3 parts are separate bones in childhood
- The 2 hip bones articulate posteriorly with the sacrum, and joins anteriorly by pubic symphysis.
- Attaches the lower limbs to the trunk.



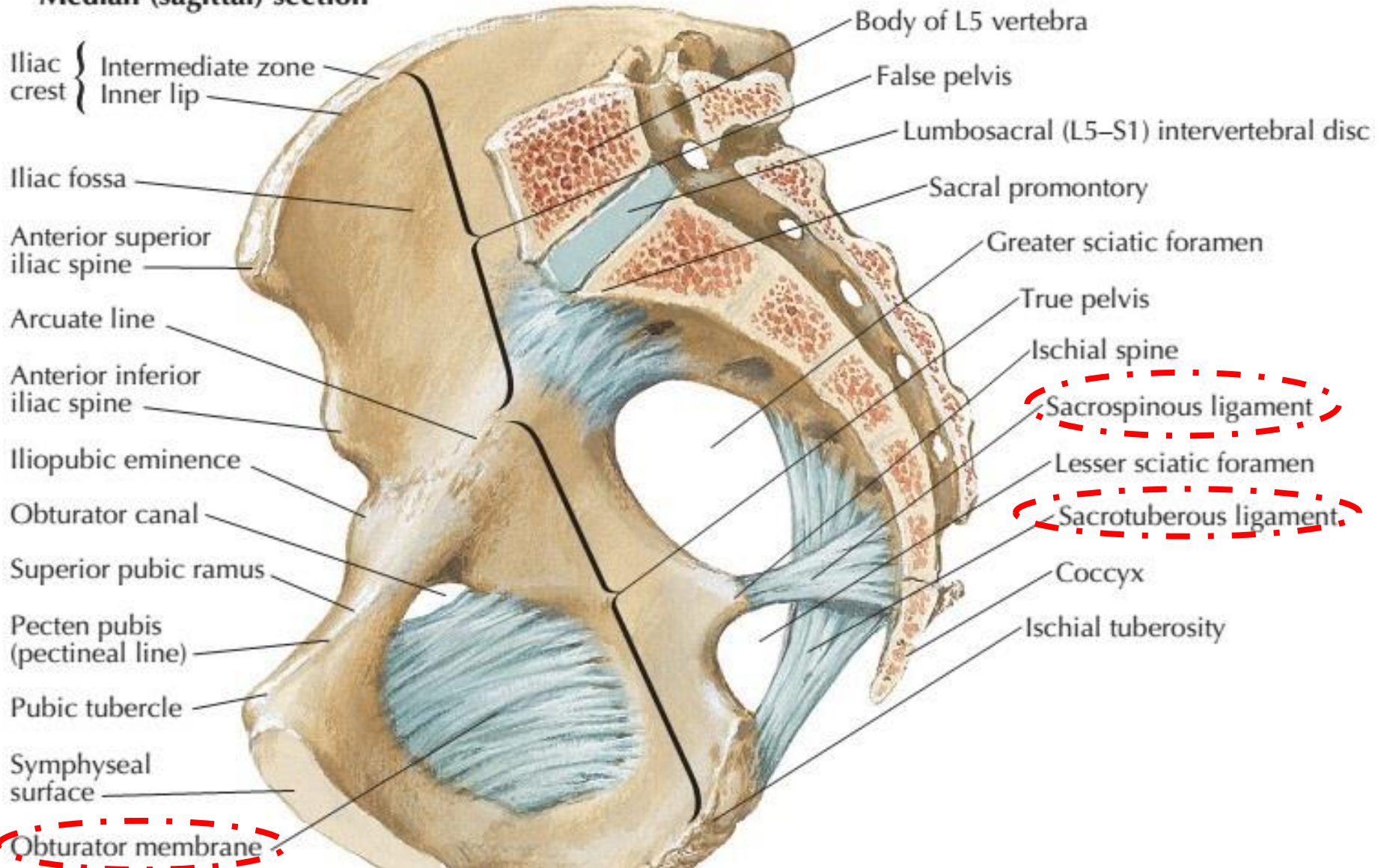
Bony Pelvis

- Consists of
 - 2 hip bones
 - 1 Sacrum
 - 1 Coccyx





Median (sagittal) section



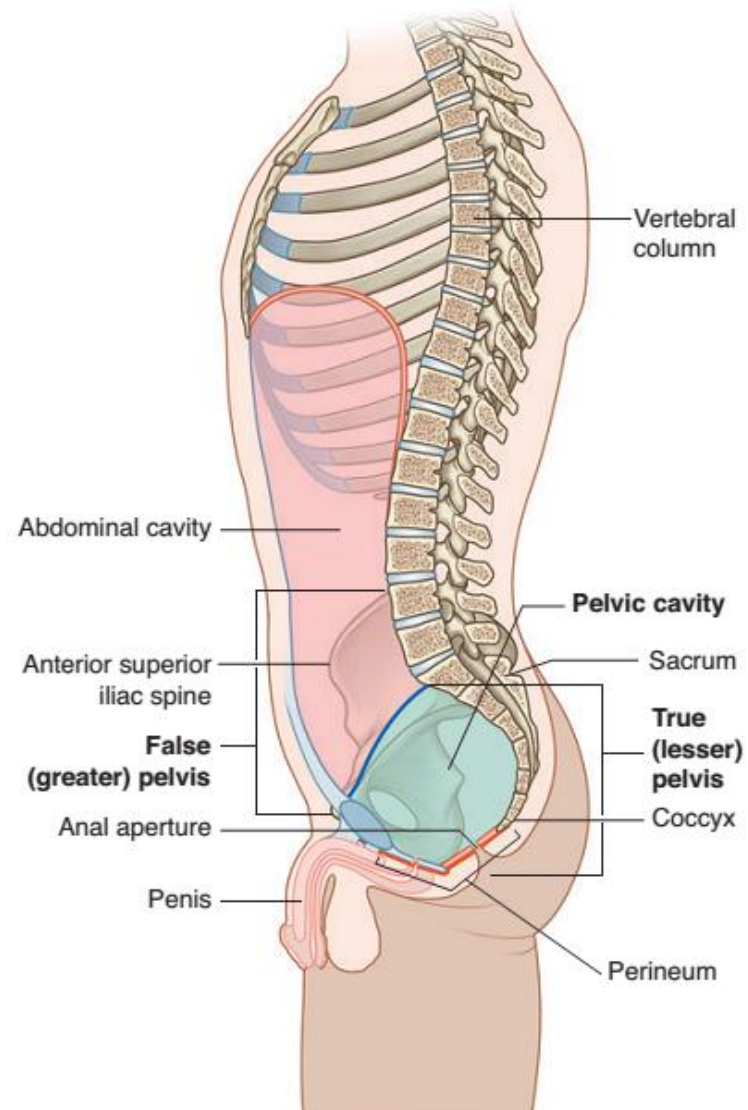
Pelvic brim/inlet divides pelvis into two parts

■ True pelvis or lesser pelvis

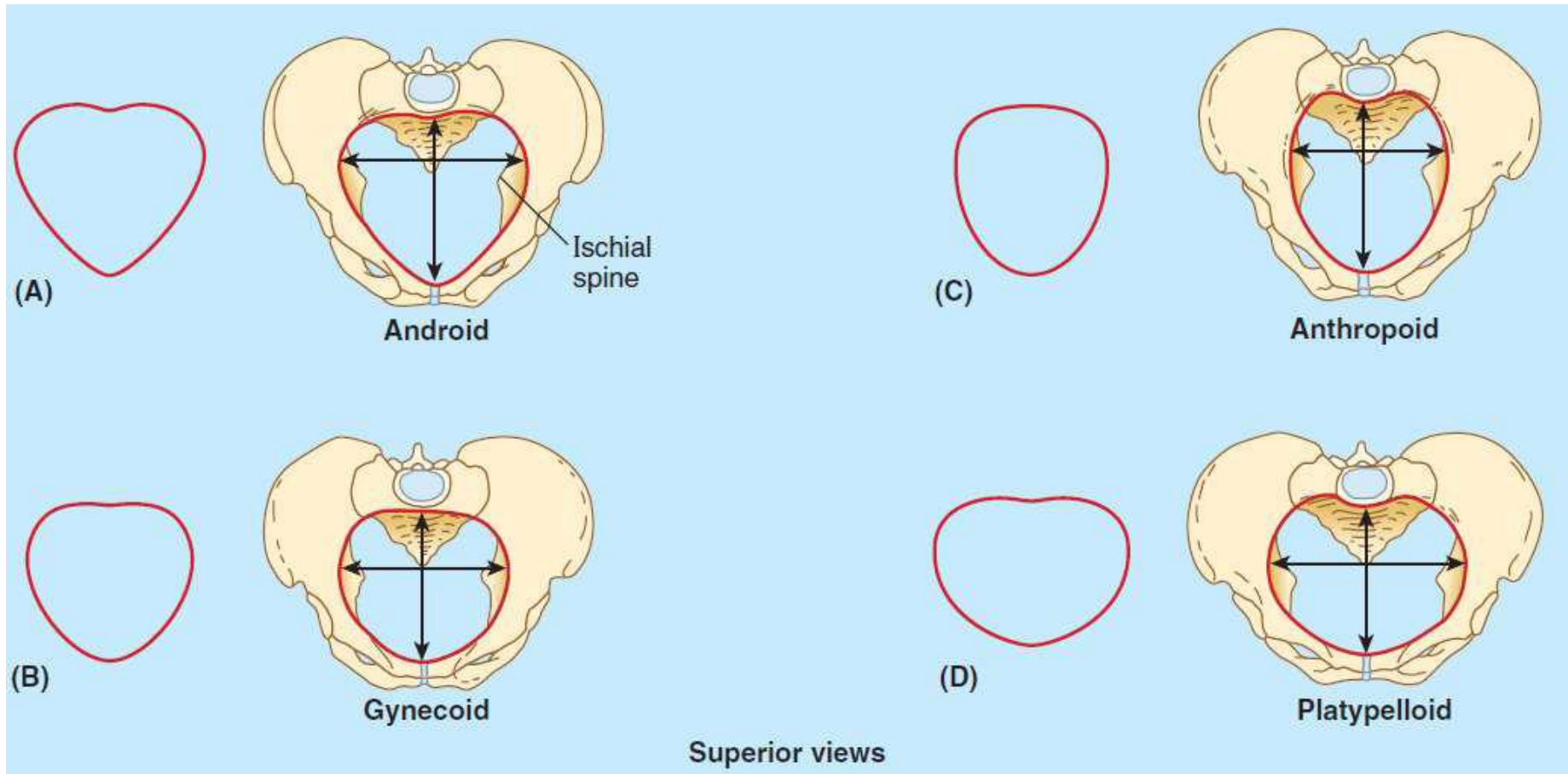
- lies between pelvic inlet and outlet
- related to the inferior parts of the pelvic bone sacrum, and coccyx ,
- contains the pelvic viscera
- divided into 3 regions: brim, cavity and outlet

■ False pelvis or greater pelvis

- lies above pelvic brim between the iliac fossae, lower part of anterior abdominal wall and lower lumbar vertebrae,
- considered part of the abdominal cavity — contains part of the ileum and sigmoid colon



Types of Bony Pelvis-according to Caldwell and Moloy





Difference Between Male & Female Bony Pelvis

Bony Pelvis

Male

Female

General structure

Thick and heavy

Thin and light

Greater pelvis (pelvis major)

Deep

Shallow

Lesser pelvis (pelvis minor)

Narrow and deep, tapering

Wide and shallow, cylindrical

Pelvic inlet (superior pelvic aperture)

Heartshaped, narrow

Oval and rounded; wide

Pelvic outlet (inferior pelvic aperture)

Comparatively small

Comparatively large

Pubic arch and subpubic angle

Narrow ($< 70^\circ$)

Wide ($> 80^\circ$)

Obturator foramen

Round

Oval

Acetabulum

Large

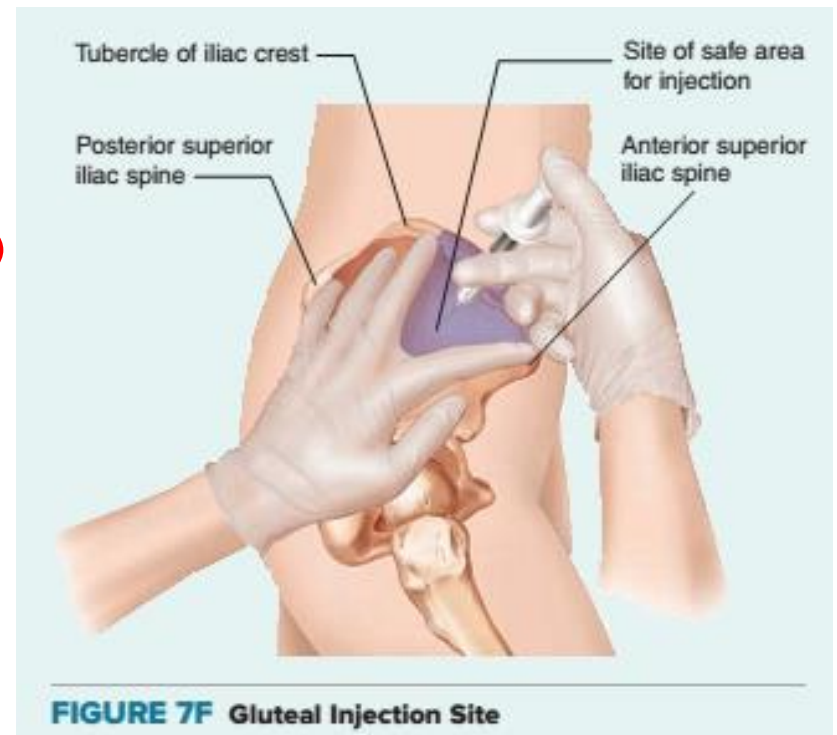
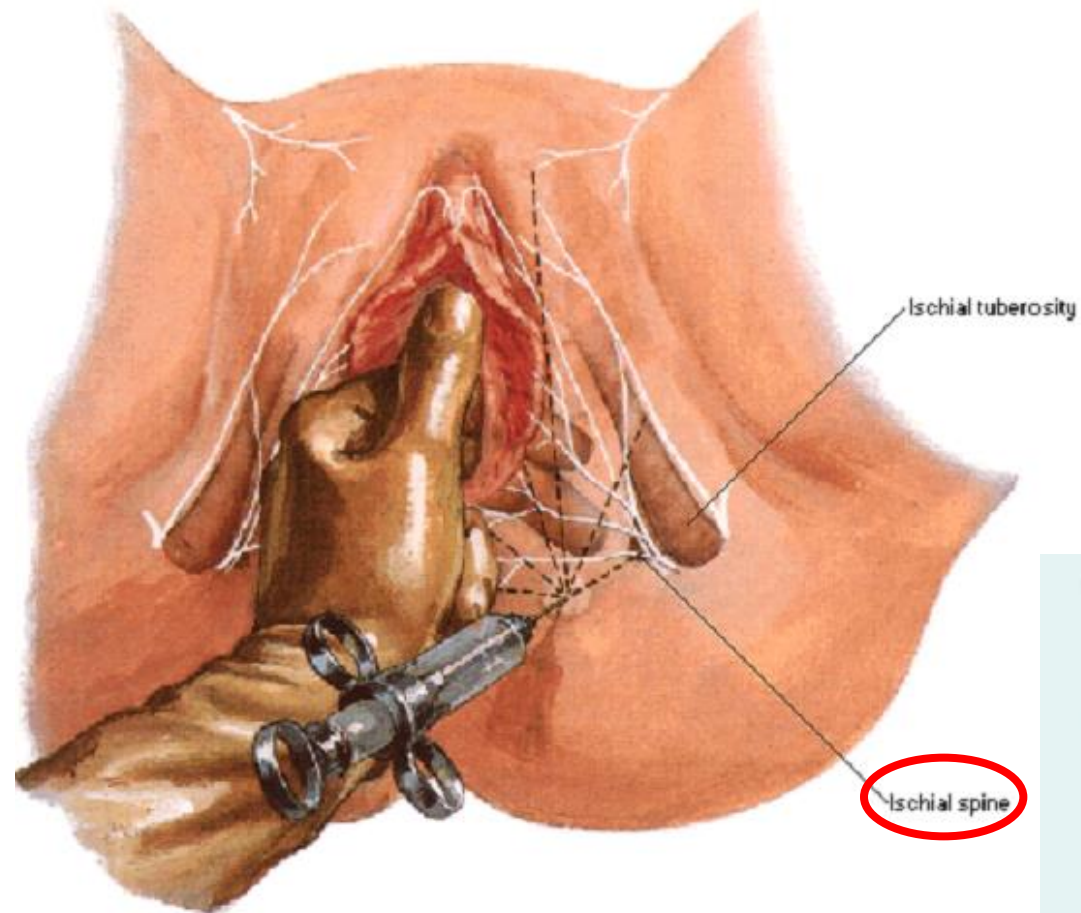
Small

Differences linked to function

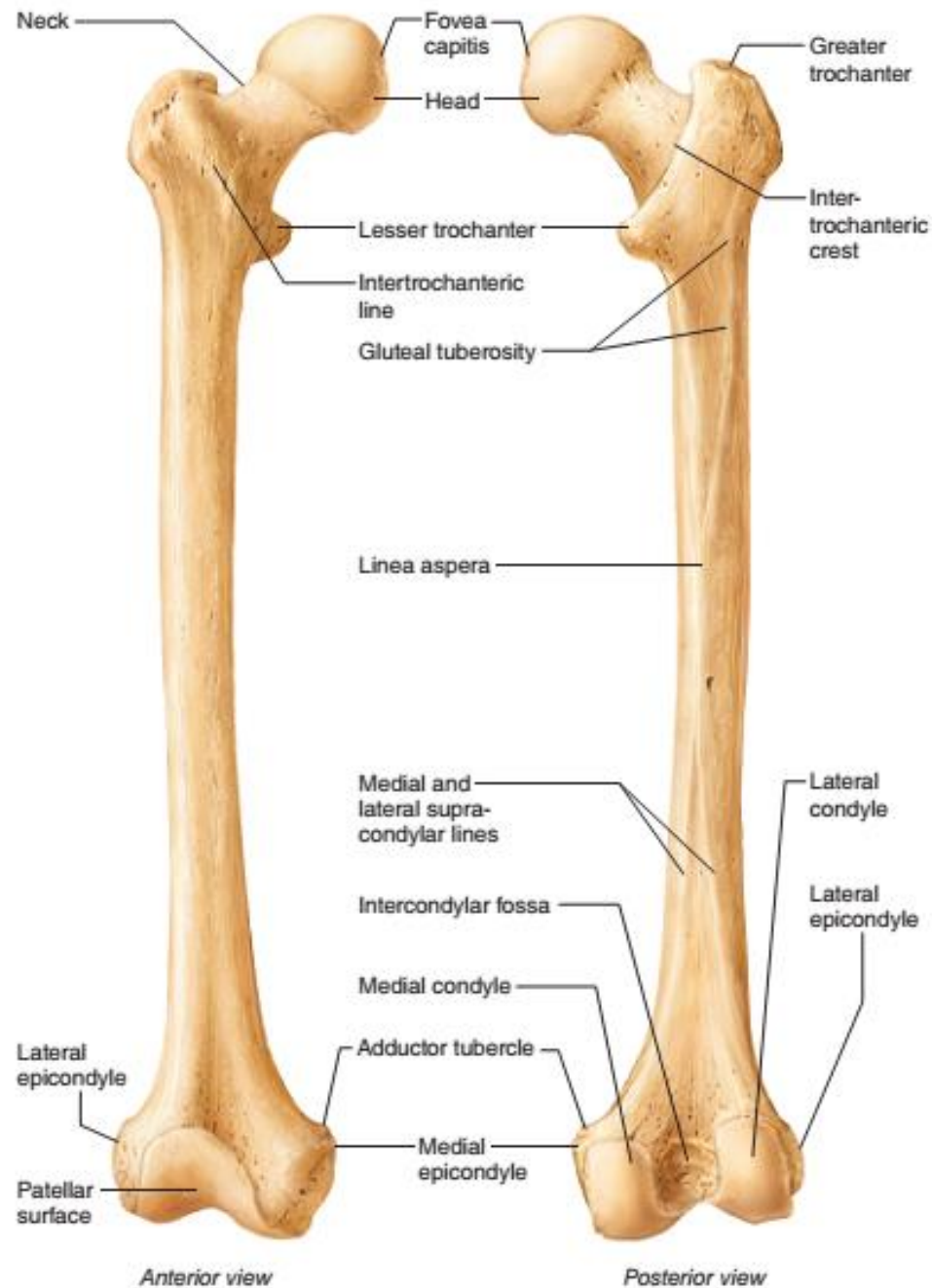
- Pregnancy and childbirth in females
- Heavier build and larger muscles of men

Significant in forensic medicine

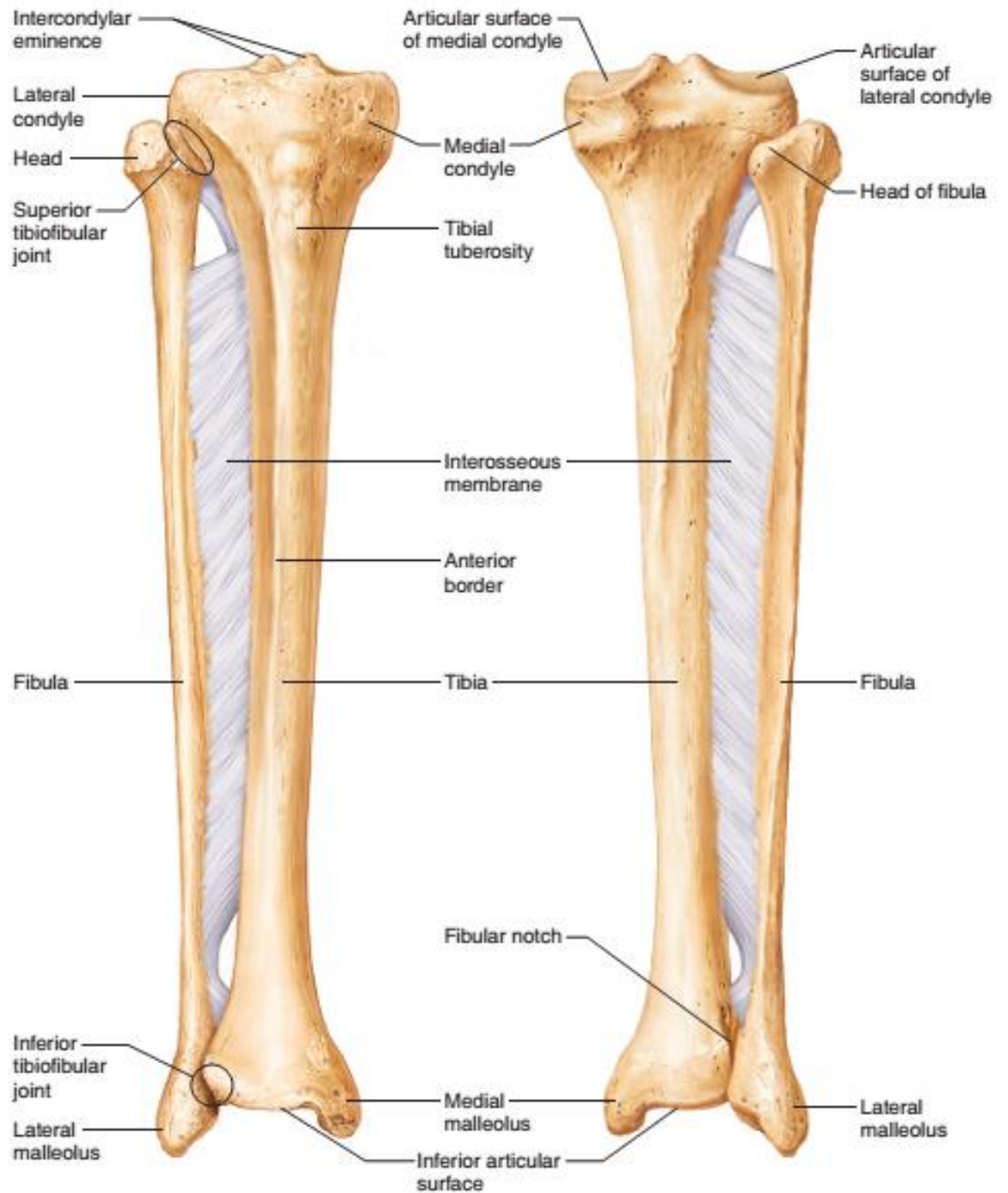
Primarily Pudendal Nerve



Femur

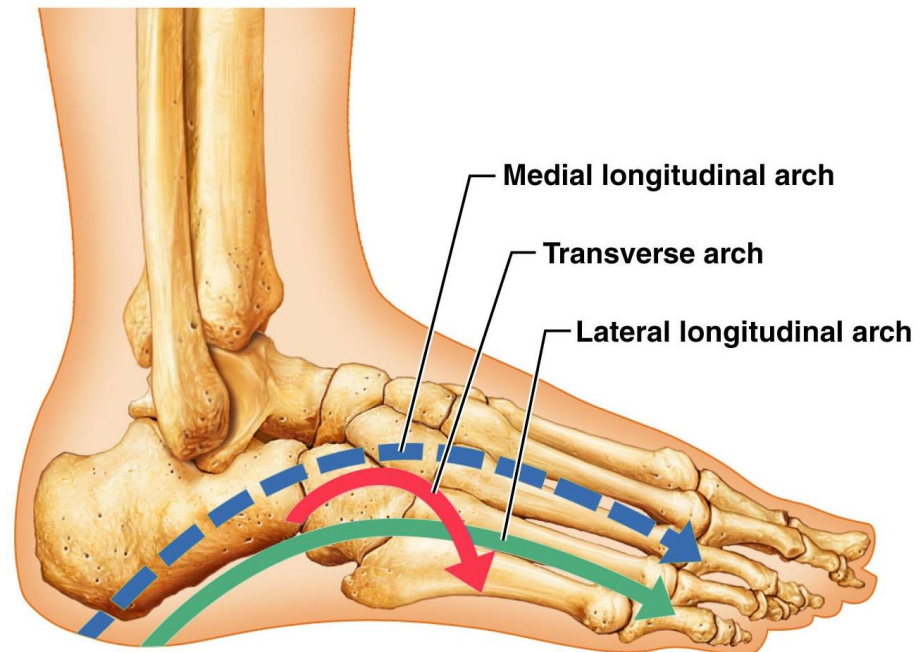


Tibia and Fibula

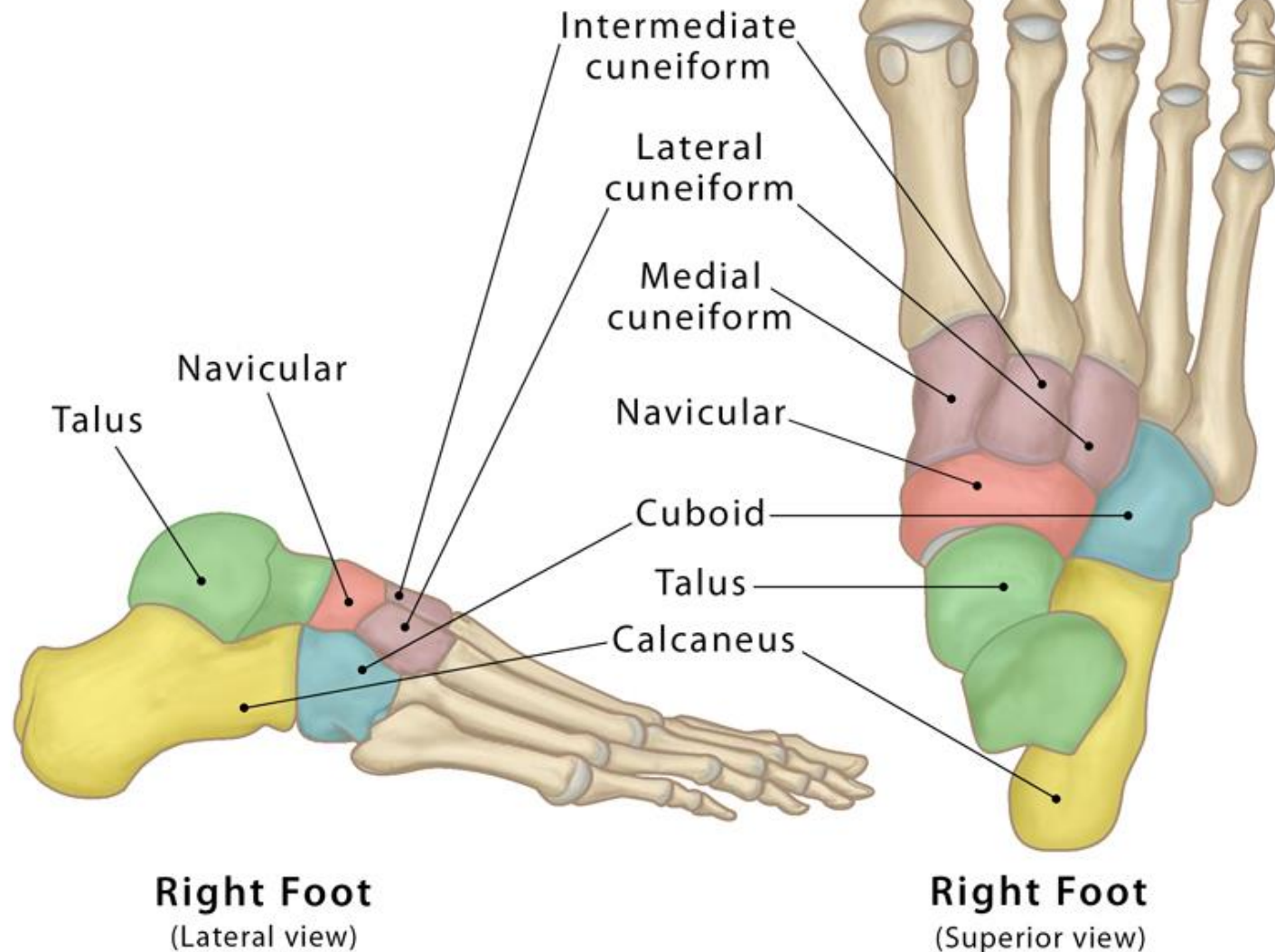


■ Fibula

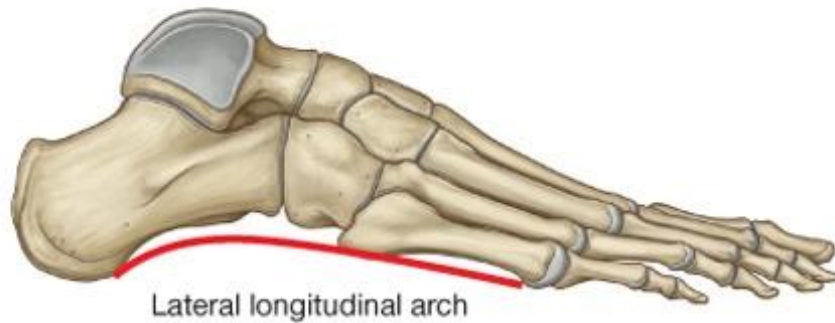
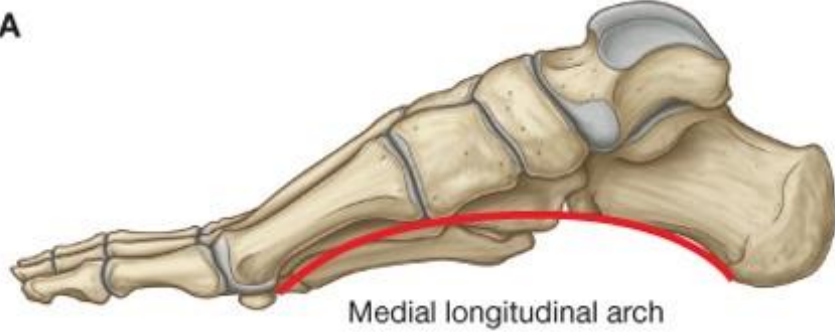
- Long pin-like bone, lies laterally
- Does not contribute to the knee joint; **helps stabilize the ankle joint via lateral malleolus**
- Articulates with the tibia and talus (ankle joint)



Tarsal Bones



A

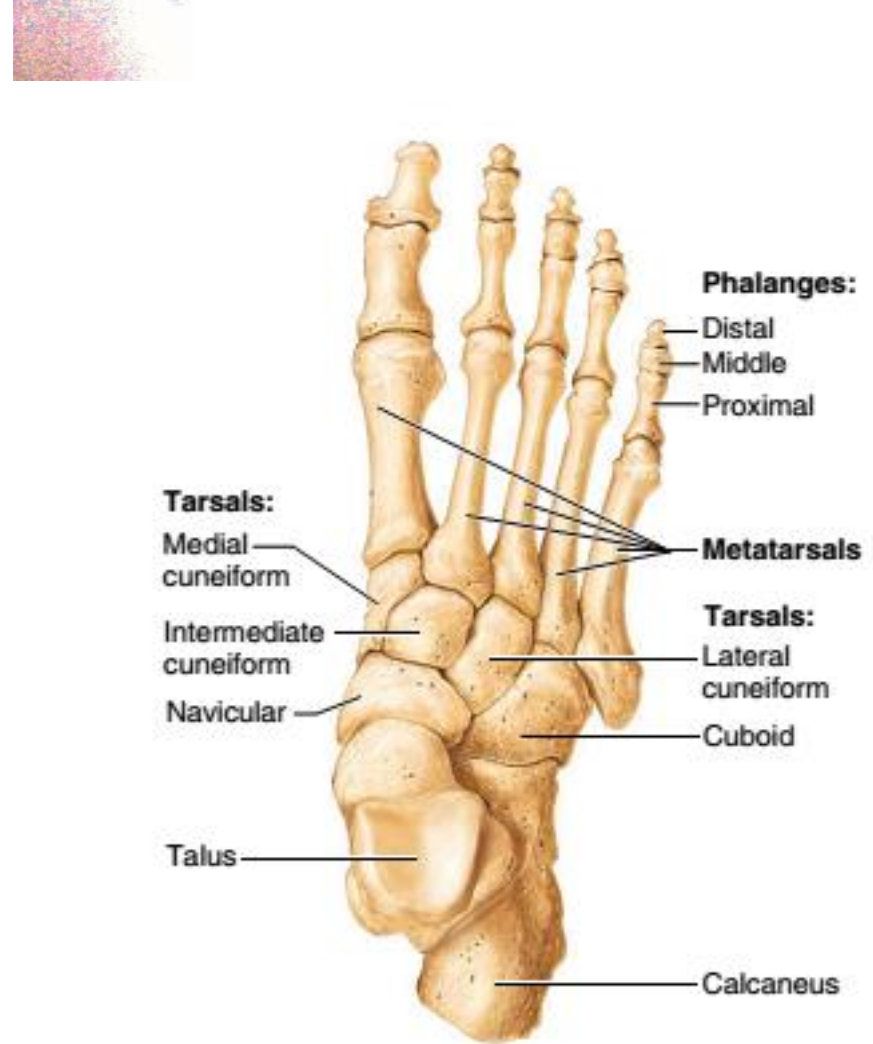


B



Transverse arch

Arches of the Foot



Pes planus (Flat foot)?

- **Talipes equinovarus (club foot):** commonest deformity of the foot. The foot is inverted, adducted, and plantar flexed.



Disorders of the Appendicular Skeleton



Ball of the foot



Bunion



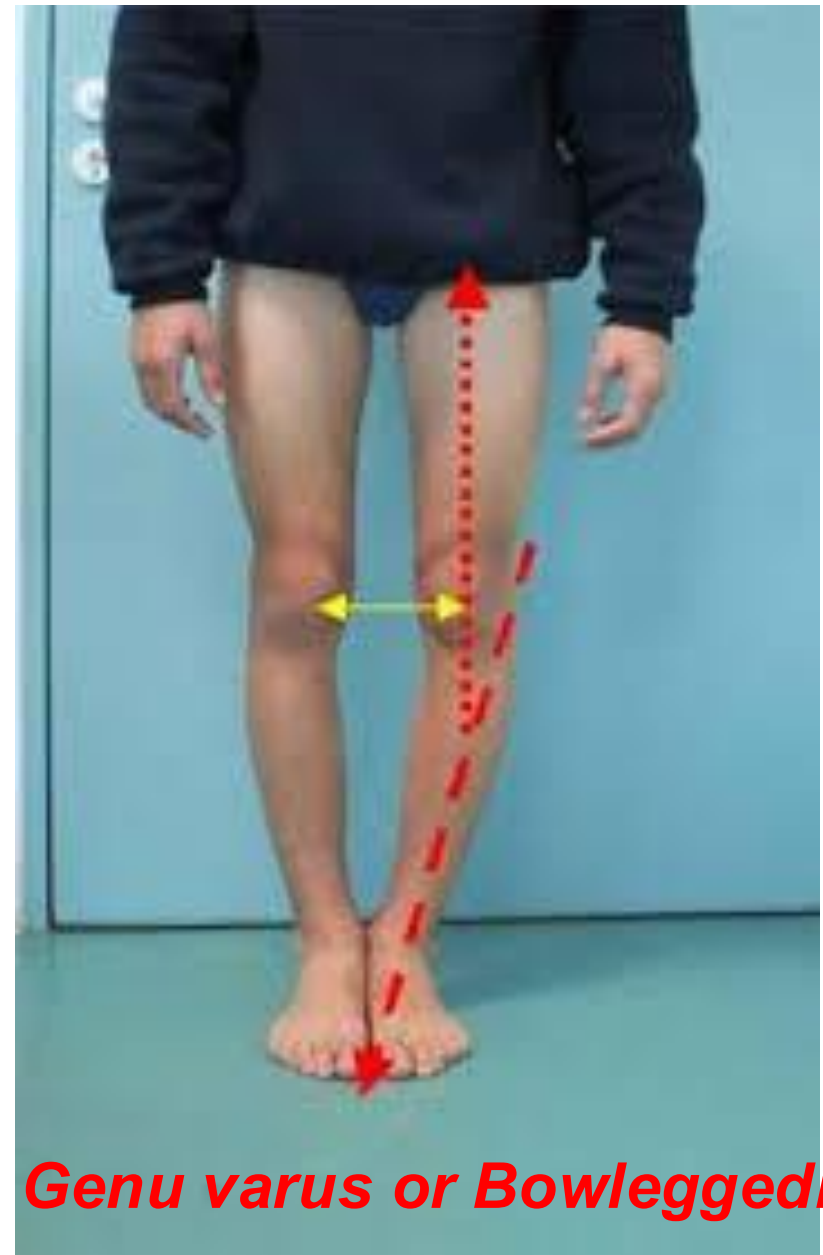
Hammer toe



effects of heels

don't forget to **SHARE** this

Disorders of the Appendicular Skeleton



Disorders of the Appendicular Skeleton

- *Hallux valgus/ bunions*

Hammer toe/claw-toe?

Lisfranc injury?

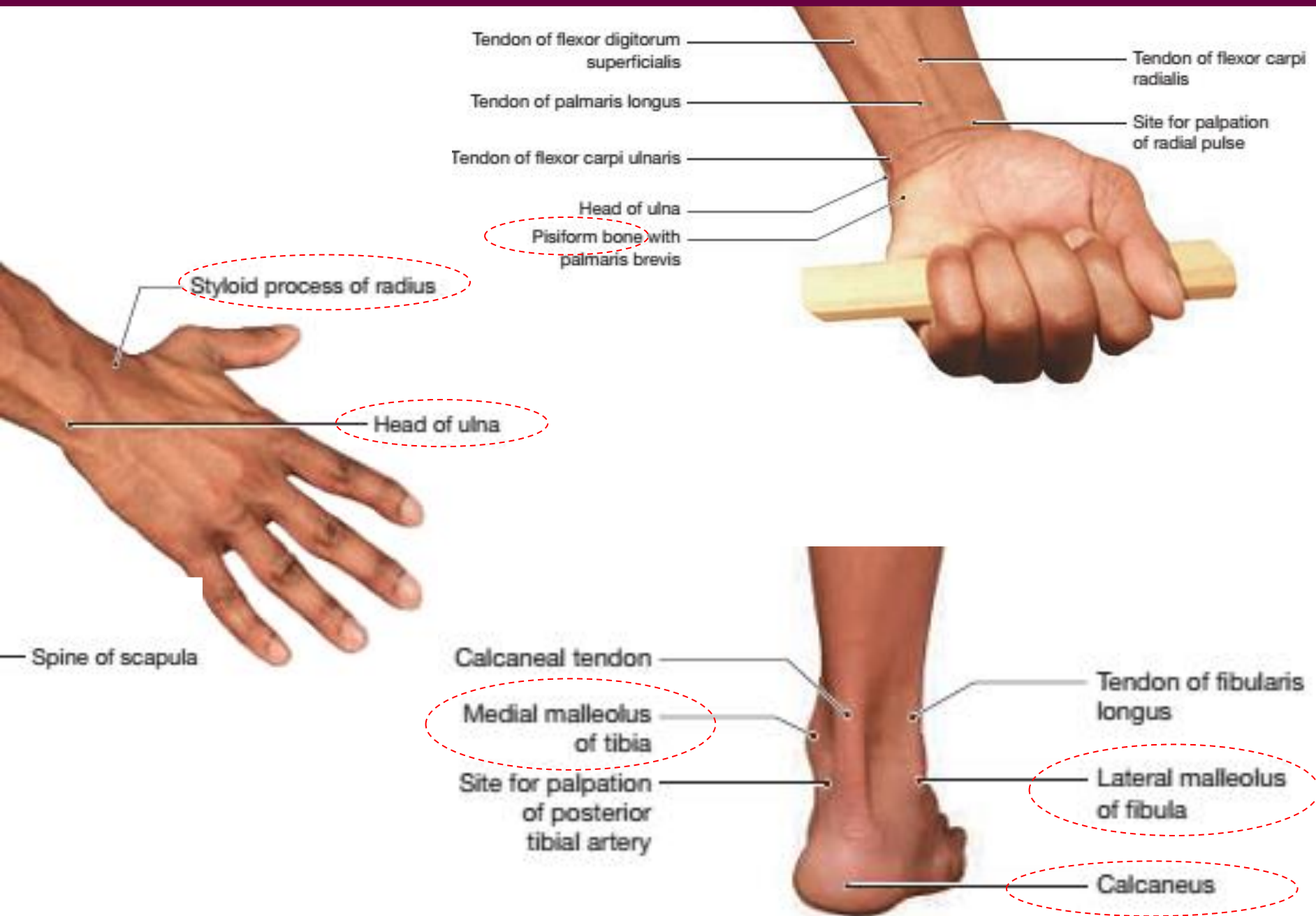


Bunion



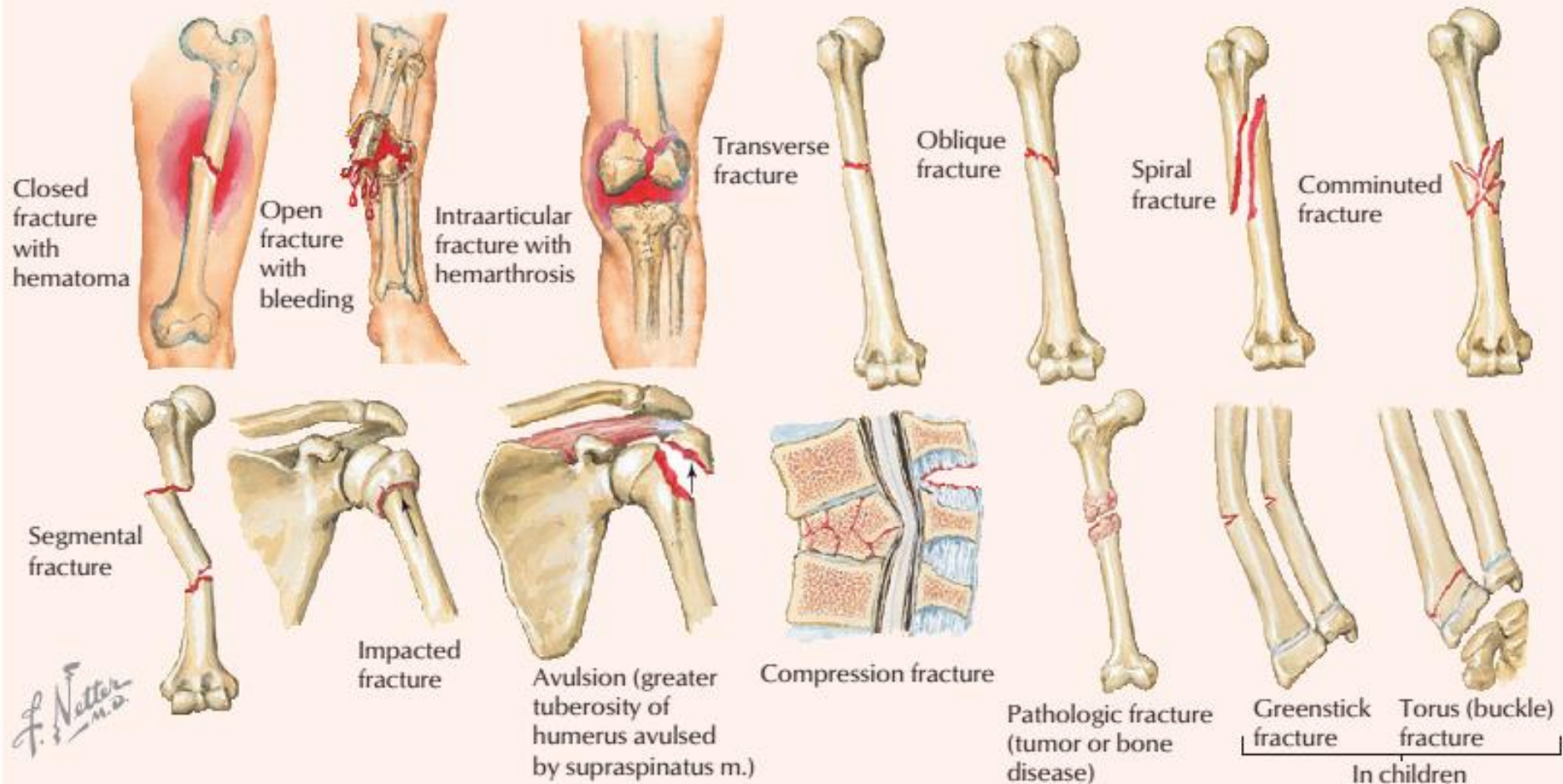
Hammer toe

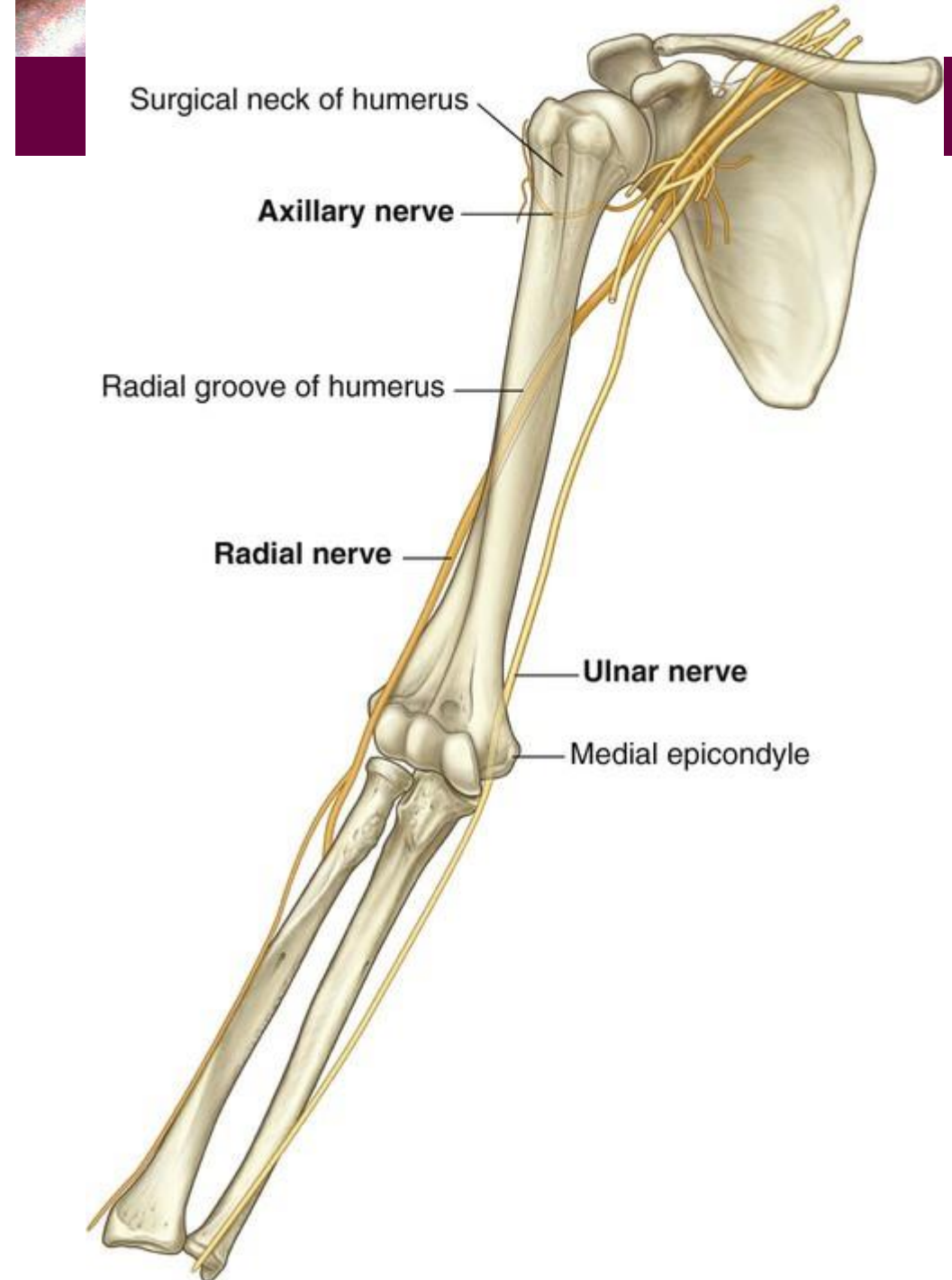
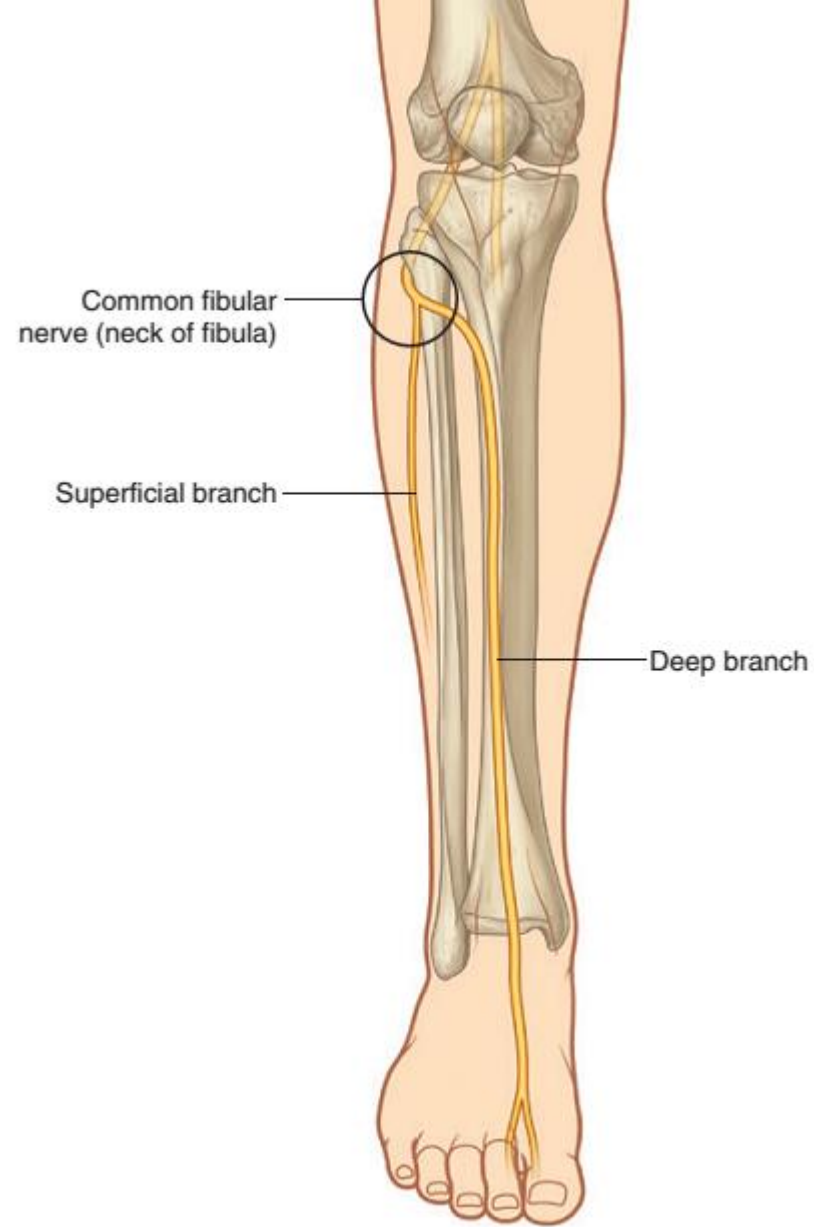
Surface Anatomy



Fractures

Fractures are classified as either **closed** (the skin is intact) or **open** (the skin is perforated; often referred to as a **compound fracture**). Additionally, the fracture may be classified with respect to its anatomical appearance (e.g., transverse, spiral, and so on).





- 
- Fracture at the surgical neck of the humerus may injure the **axillary nerve**.
 - A midshaft fracture of the humerus may injure the **radial nerve**.
 - The **ulnar nerve** passes posterior to the medial epicondyle
 - Fracture at the neck of fibula may injure the **common peroneal (fibular) nerve**.

Joints/articulations/arthroses

- unions or junctions between two or more bones or rigid parts of the skeleton.
- Allow rigid skeleton some flexibility
- Each joint has a normal range of motion, and bony surfaces, cartilages, ligaments, tendons, and muscles working together to keep movement within normal range.
- The anatomy of a joint determines its function and range of motion.

Arthrology

Arthritis

Arthroplasty?

Arthrocentesis?



Classifications of Joints

- Joints can be classified by function or structure
- *Functional classification*—based on amount of movement at the joint
 - ✓ **Synarthrosis (syn – together)**—**immovable joint**; eg. suture, gomphosis, synchondroses
 - ✓ **Amphiarthrosis**—**slightly movable joint**; eg. symphysis, syndesmosis
 - ✓ **Diarthroses**—**freely movable joint**; all synovial joints

Classifications of Joints

- *Structural classification* based on what separates/binds the articulating bones (fibers, cartilage or joint cavity)
- Structural classifications include
 - ✓ Fibrous
 - Bones are connected by a tough fibrous material. No joint cavity
 - ✓ Cartilaginous
 - Bones are united by cartilage. No joint cavity
 - ✓ Synovial
 - ✓ Presence of a joint cavity

Classifications of Joints

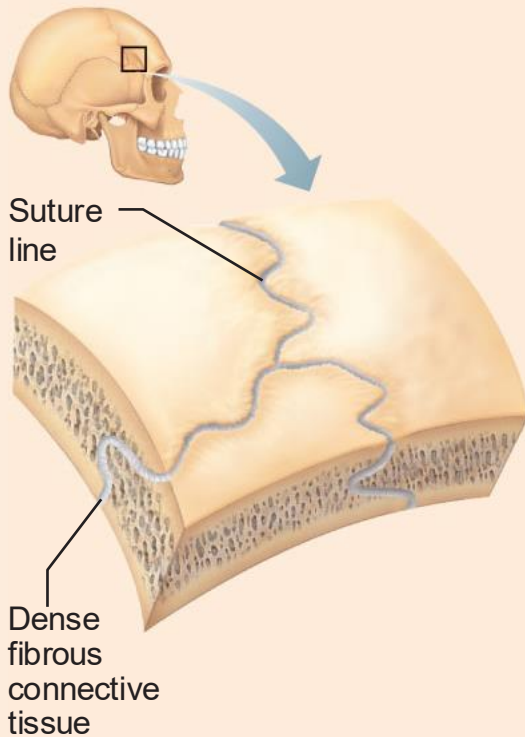
TABLE 9.1 Summary of Joint Classes

Structural Class	Structural Characteristics	Types		Mobility
Fibrous	Adjoining bones united by collagenic fibers	1. Suture (short fibers)		Immobile (synarthrosis)
		2. Syndesmosis (longer fibers)		Slightly mobile (amphiarthrosis) and immobile
		3. Gomphosis (periodontal ligament)		Immobile
Cartilaginous	Adjoining bones united by cartilage	1. Synchondrosis (hyaline cartilage)		Immobile
		2. Symphysis (fibrocartilage)		Slightly movable
Synovial	Adjoining bones separated by a joint cavity, covered with articular cartilage, and enclosed within an articular capsule lined with synovial membrane	1. Plane 2. Hinge 3. Pivot	4. Condylloid 5. Saddle 6. Ball and socket	Freely movable (diarthrosis); movements depend on design of joint

Fibrous Joints

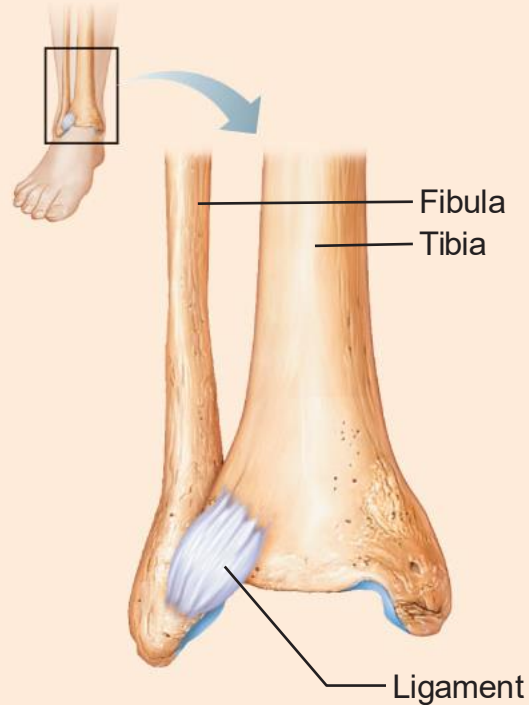
(a) Suture

Joint held together with very short, interconnecting fibers, and bone edges interlock. Found only in the skull.



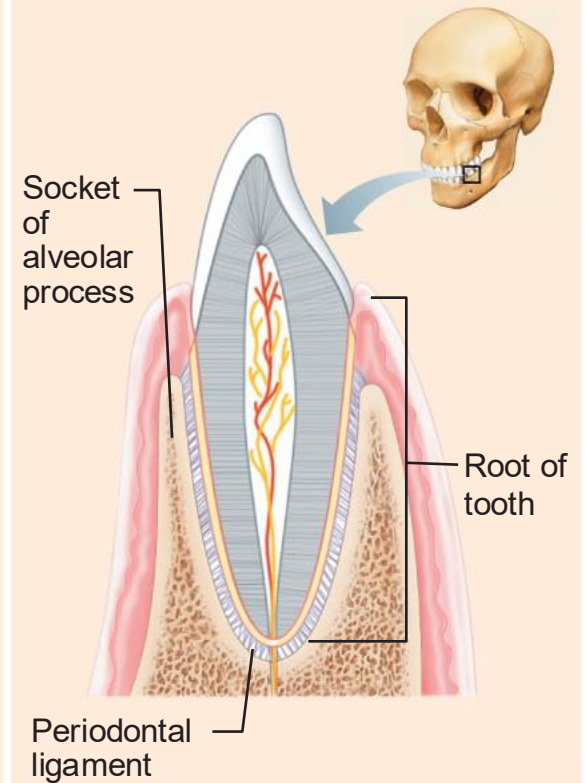
(b) Syndesmosis

Joint held together by a ligament. Fibrous tissue can vary in length but is longer than in sutures.



(c) Gomphosis

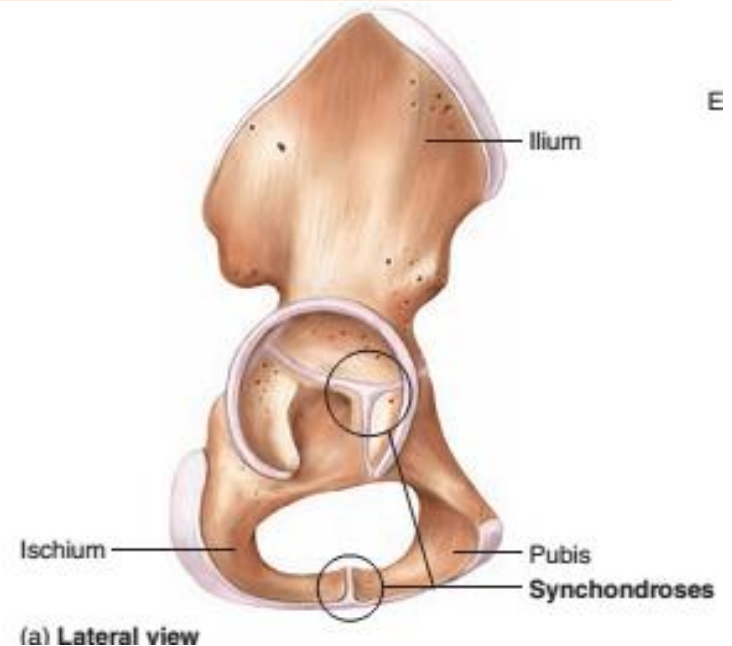
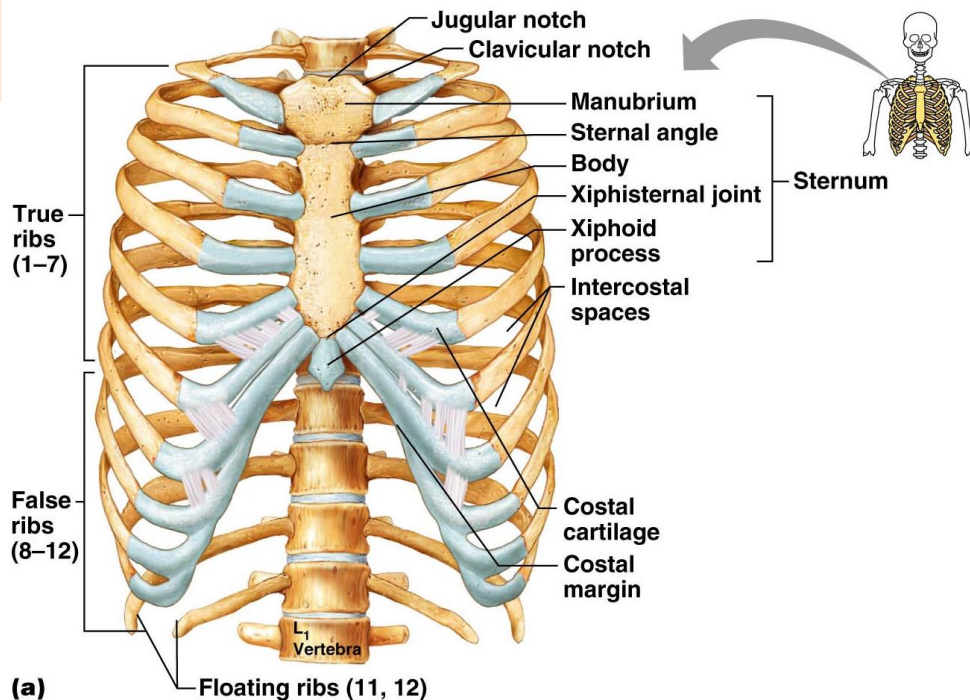
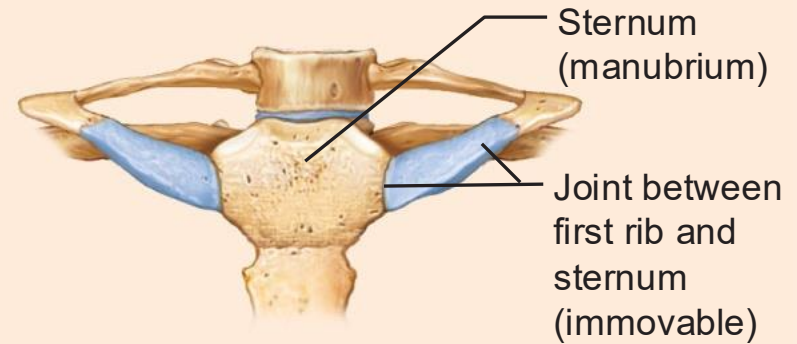
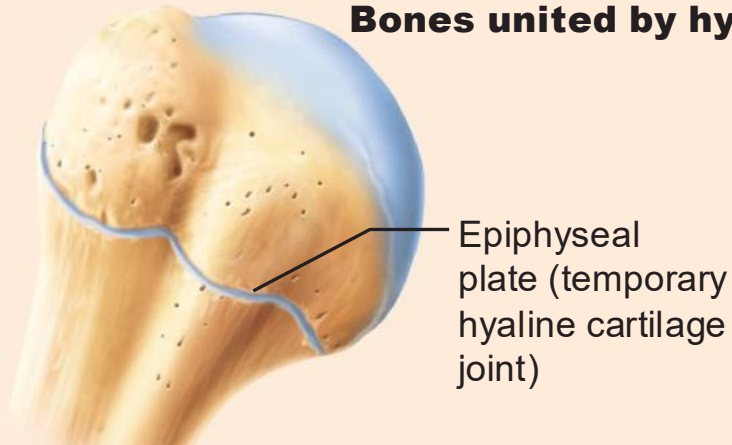
Peg-in-socket fibrous joint. Periodontal ligament holds tooth in socket.



Examples of synchondroses/Primary Cartilaginous Joints

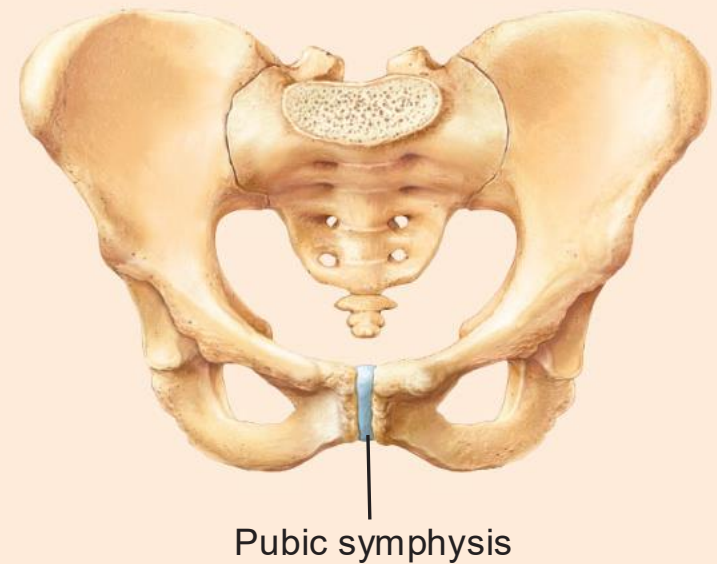
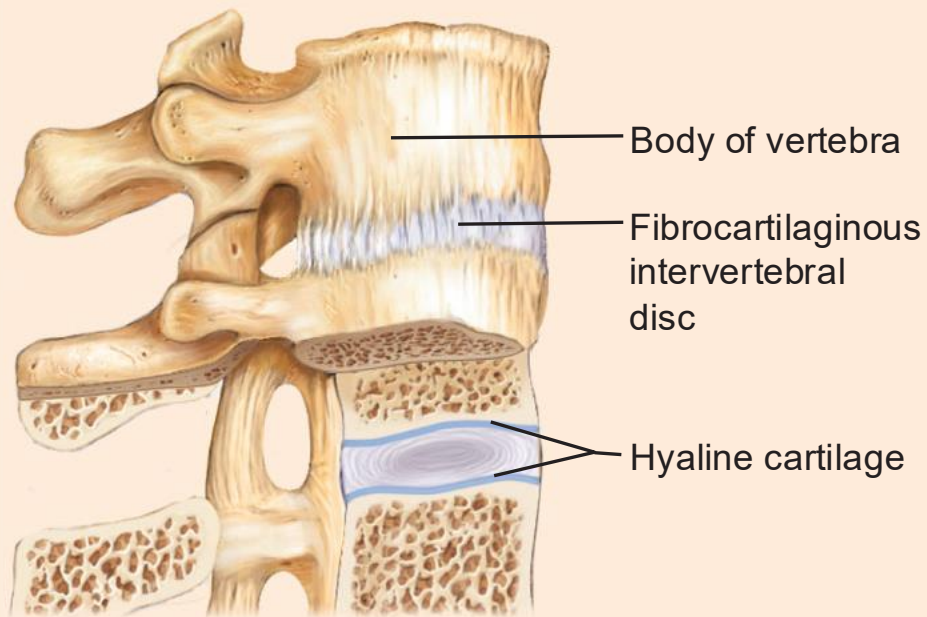
(a) Synchondroses

Bones united by hyaline cartilage



Examples of symphyses/Secondary Cartilaginous Joints

(b) Symphyses Bones united by fibrocartilage





Synovial Joints

- Most movable type of joint
- All are **diarthroses**
- Each contains a fluid-filled joint cavity
- Certain synovial joints contain a disc of fibrocartilage, called an *articular disc or a meniscus* (“*crescent*”).
 - Articular discs occur in the temporomandibular (jaw) joint, sternoclavicular joint, knee joint.



General Structure of Synovial Joints

- **Articular cartilages**
 - Cover ends of opposing bones. Thickness decreases with age
 - Act as shock absorbers and help reduce friction within the joint
- **Joint cavity** (synovial cavity) filled with synovial fluid
- **Articular (joint) capsule** —encloses a joint cavity
 - **(Outer) fibrous capsule**
 - dense irregular connective tissue, which strengthens the joint
 - **(Inner) synovial membrane**
 - loose connective tissue, lines joint capsule and covers parts of the bones within the joint not covered by articular cartilage;
 - functions to make synovial fluid

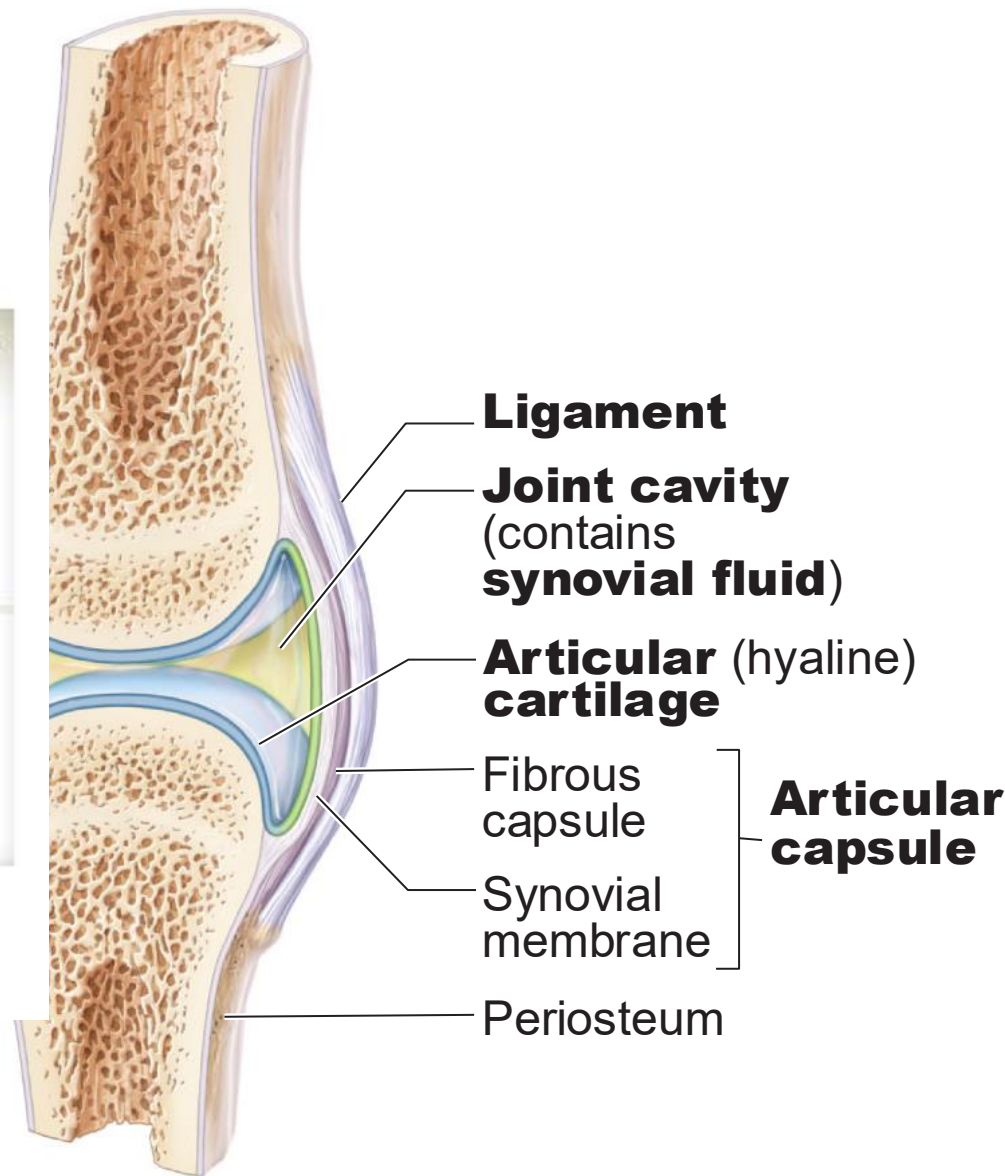
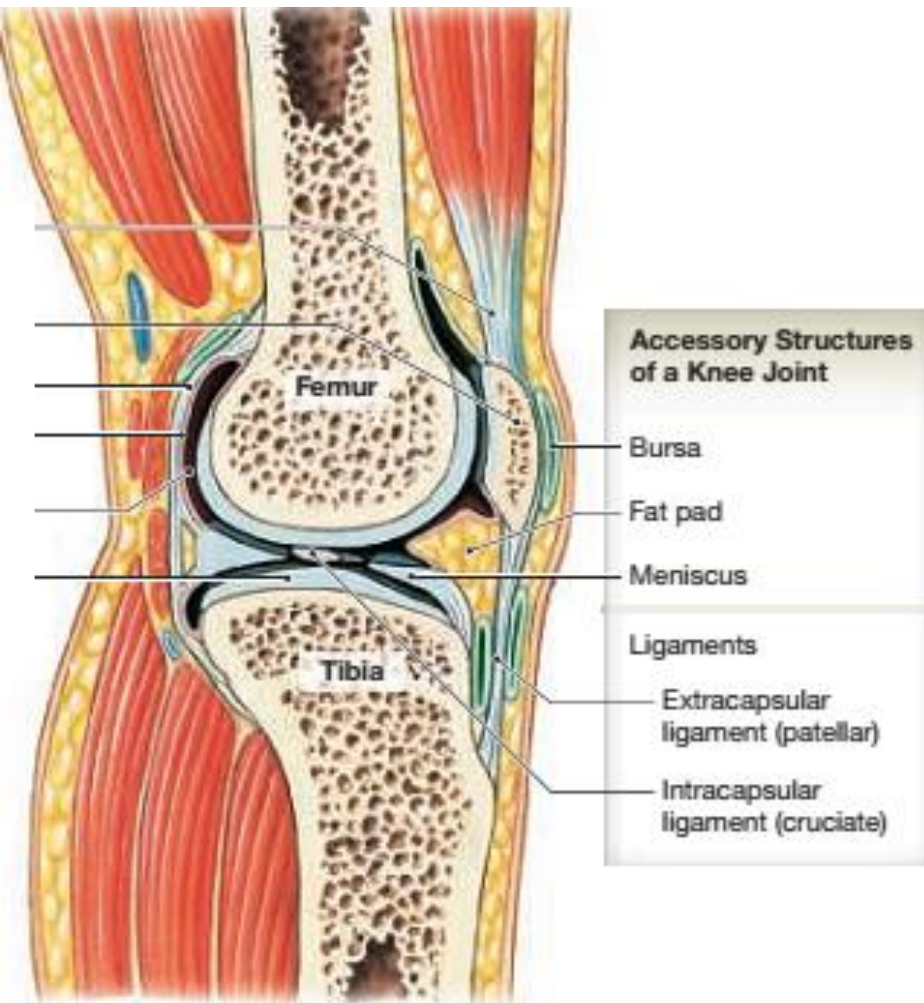
General Structure of Synovial Joints

- **Synovial fluid** – colourless, viscous fluid
 - ✓ provides nutrients for the structures within the joint cavity;
 - ✓ contains phagocytes, which remove microbes and cellular debris;
 - ✓ acts as a lubricant; maintains joint stability;
 - ✓ cushions joints that are subjected to compression;
 - ✓ prevents the ends of the bones from being separated, as does a little water between two glass surfaces

- **Sensory nerves and blood vessels**
 - Nerves and blood vessels crossing a joint usually supply the capsule and the muscles that move it.

- **Accessory structures**
 - fat pads, menisci, ligaments, bursae, tendons

General Structure of Synovial Joints



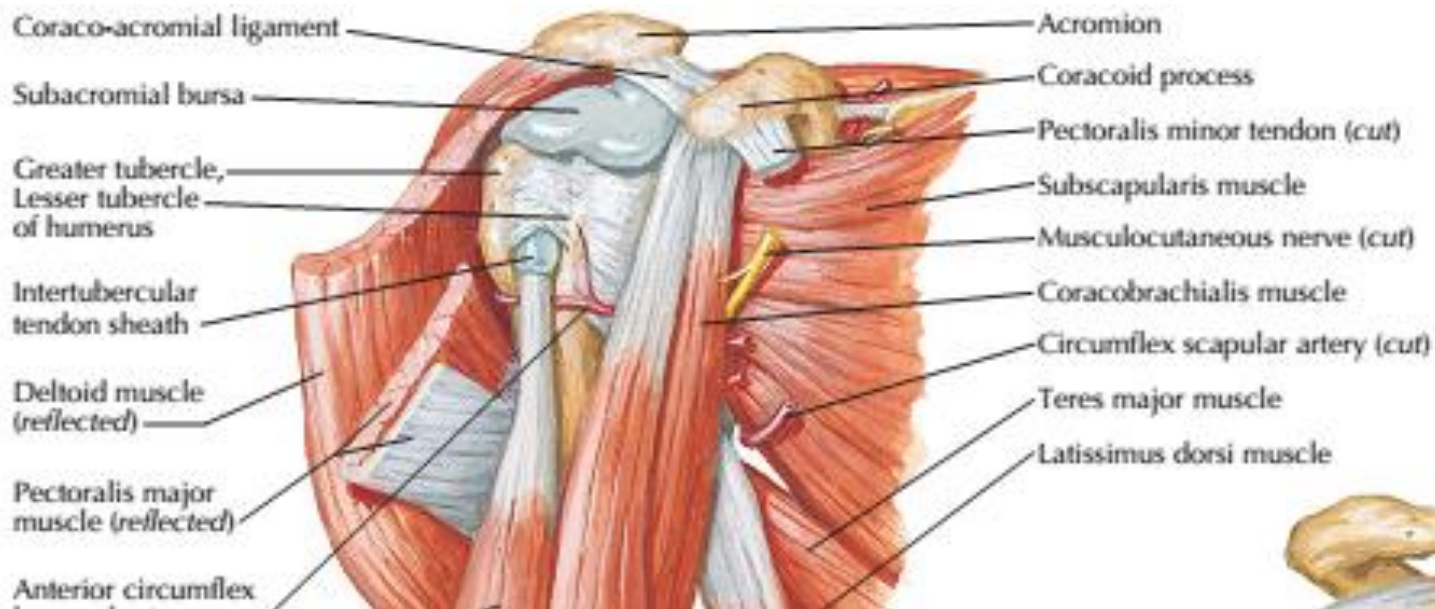
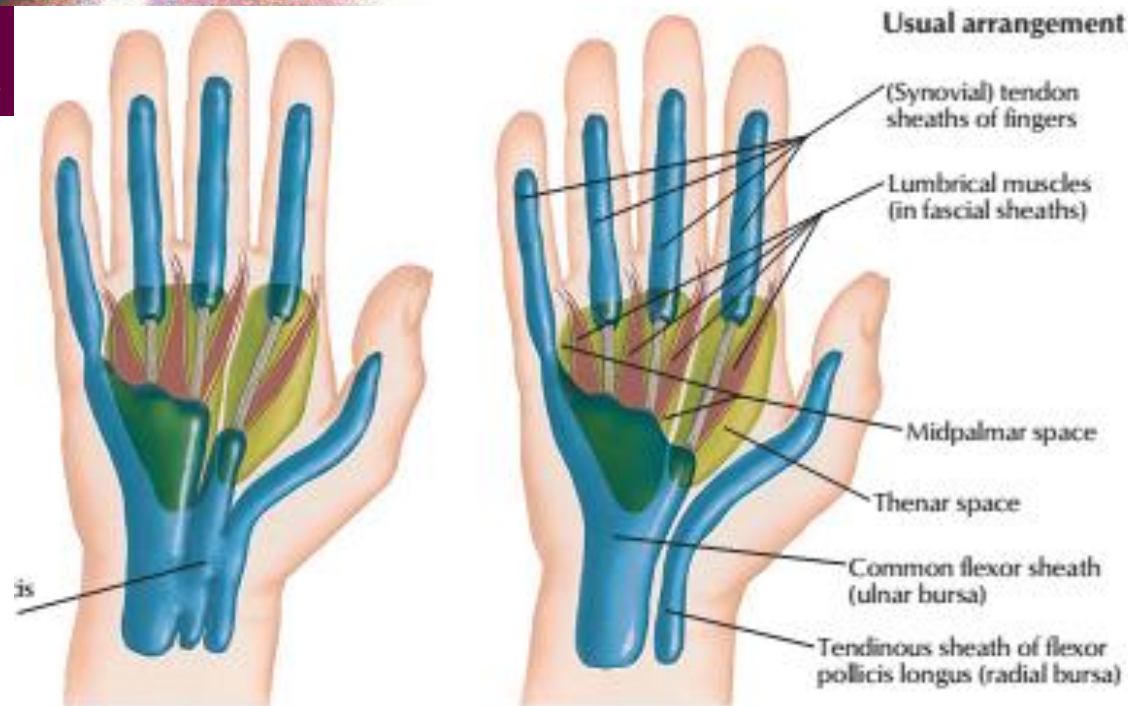
(a) A typical synovial joint



Bursa and Tendon sheath

- Bursae and tendon sheaths contain synovial fluid and often are associated with synovial joints
- **Bursa** (“purse”) – A synovial fluid-filled sac. Occur where ligaments, muscles, skin, tendons, or bones overlie each other and rub together. Reduce friction, cushion certain muscles, facilitate movements of tendons or muscles over bone or ligament. **What is bursitis?**
- **Tendon sheath** - an elongated bursa that wraps around tendons of certain muscles, occurs only on tendons that are subjected to friction.

Bursa and Tendon sheath





Bursitis

Synovial Joints

- **Factors Influencing the Stability of Synovial Joints**
 - the shapes of the articulating surfaces preventing movement in specific directions,
 - the number and positioning of ligaments
 - **muscle tone** – tension in the relaxed muscle. Important in maintaining posture

Distinguish between:

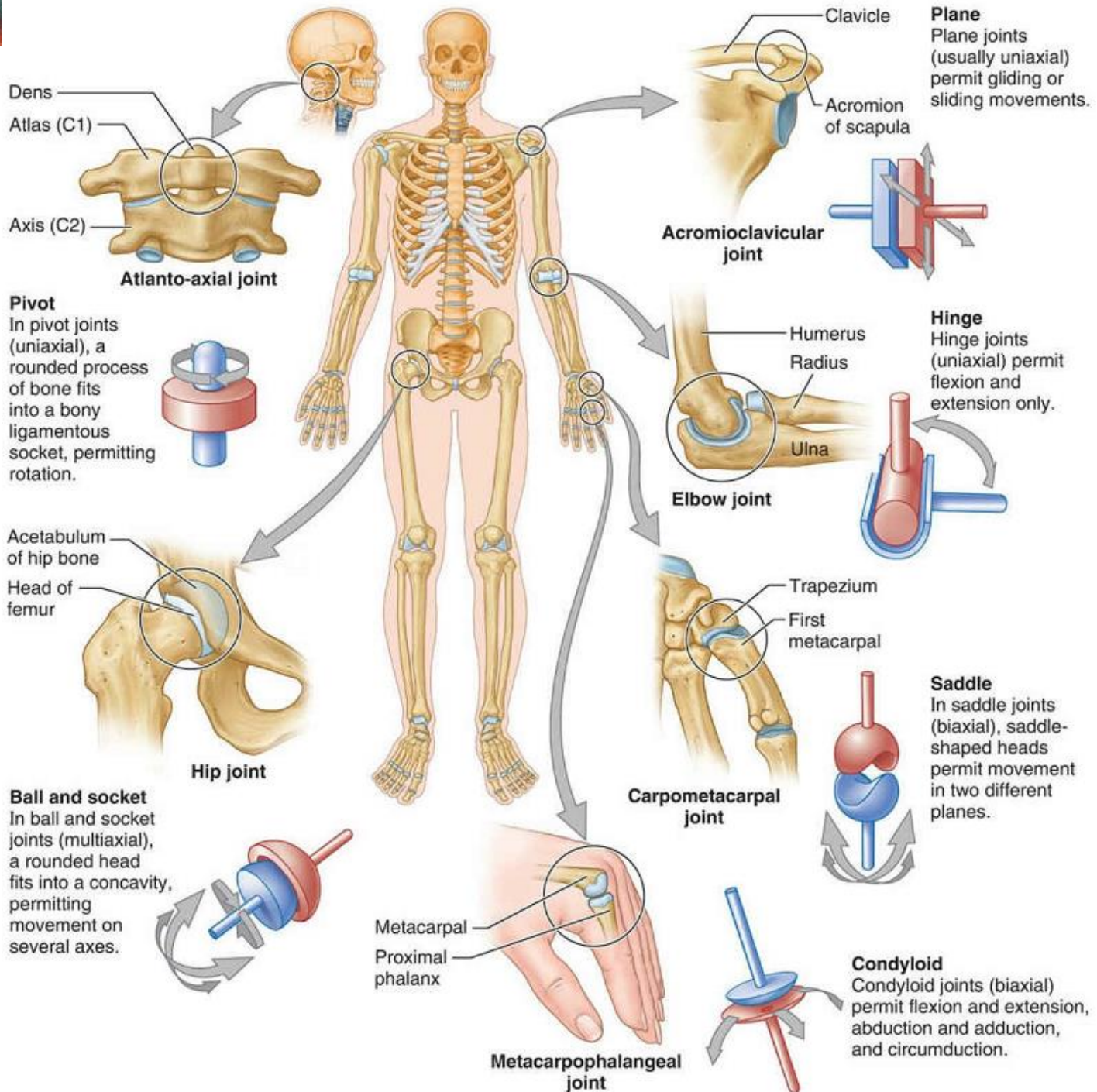
- Strains or Sprains
- Dislocation or luxation and subluxation

Synovial Joints

- Synovial joints are classified according to the shape of the articulating surfaces and/or the type of movement (uni-, bi-, or multiaxial movement) they permit
- **Plane:** allow gliding or sliding movements
 - Eg: Intertarsal and intercarpal joints, 2-7th sternocostal, acromioclavicular, zygapohyseal joints
- **Hinge** (ginglymus): uniaxial (monoaxial) joints for flexion and extension
 - Eg: Elbow, knee, ankle, and interphalangeal joints
- **Pivot** (trachoid): uniaxial joints for rotation
 - Eg: proximal radio-ulnar joint, atlanto-axial joint

Synovial Joints

- **Saddle:** biaxial joints for flexion, extension, abduction, adduction, and circumduction
 - Examples: 1st carpometacarpal joint (allows opposition of the thumb), sternoclavicular, calcane-ocuboid
- **Condylod (ellipsoid):** biaxial joints for flexion, extension, abduction, adduction, and slight circumduction
 - Examples: Metacarpophalangeal joints (knuckle joint), Metatarsophalangeal joints, temporomandibular (TMJ) joint, atlanto-occipital joint, radiocarpal joint (wrist)
- **Ball-and-socket (spheroid):** multiaxial (polyaxial or triaxial) joints for flexion, extension, abduction, adduction, medial and lateral rotation, and circumduction
 - Examples: Shoulder and hip joints



Types of Joints

TABLE 7.4

Types of Joints

Type of Joint	Description	Possible Movements	Examples
Fibrous	Articulating bones are fastened together by a thin layer of dense connective tissue.	None or slight twisting	Suture between bones of skull, joint between the distal ends of tibia and fibula
Cartilaginous	Articulating bones are connected by hyaline cartilage or fibrocartilage.	Limited movement, as when back is bent or twisted	Joints between the bodies of vertebrae, pubic symphysis
Synovial	Articulating ends of bones are surrounded by a joint capsule of ligaments and synovial membranes; ends of articulating bones are covered by hyaline cartilage and separated by synovial fluid.	Allow free movement (see the following list, "Types of Joint Movements")	
1. Ball-and-socket	Ball-shaped head of one bone articulates with cup-shaped cavity of another.	Movements in all planes, including rotation	Shoulder, hip
2. Condylar	Oval-shaped condyle of one bone articulates with elliptical cavity of another.	Variety of movements in two planes, but no rotation	Joints between the metacarpals and phalanges
3. Plane	Articulating surfaces are nearly flat or slightly curved.	Sliding or twisting	Joints between various bones of wrist and ankle, sacroiliac joints, joints between ribs 2–7 and sternum
4. Hinge	Convex surface of one bone articulates with concave surface of another.	Flexion and extension	Elbow, joints of phalanges
5. Pivot	Cylindrical surface of one bone articulates with ring of bone and ligament.	Rotation around a central axis	Joint between the atlas and dens of the axis
6. Saddle	Articulating surfaces have both concave and convex regions; the surface of one bone fits the complementary surface of another.	Variety of movements, mainly in two planes	Joint between the carpal and metacarpal of thumb